

SIEMENS

SINAMICS/SIMOTION

SINAMICS/SIMOTION DCC editor description

Programming and Operating Manual

Preface

Introduction

1

DCC editor functionality

2

DCC for SINAMICS

3

DCC for SIMOTION

4

Appendix

A

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

with a safety alert symbol, indicates that minor personal injury can result if proper precautions are not taken.

CAUTION

without a safety alert symbol, indicates that property damage can result if proper precautions are not taken.

NOTICE

indicates that an unintended result or situation can occur if the corresponding information is not taken into account.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation for the specific task, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING

Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be adhered to. The information in the relevant documentation must be observed.

Trademarks

All names identified by ® are registered trademarks of the Siemens AG. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

Preface

SIMOTION Documentation

An overview of the SIMOTION documentation can be found in a separate list of references.

This documentation is included as electronic documentation with the supplied SIMOTION SCOUT.

The SIMOTION documentation consists of 9 documentation packages containing approximately 80 SIMOTION documents and documents on related systems (e.g. SINAMICS).

The following documentation packages are available for SIMOTION V4.1 SP4:

- SIMOTION Engineering System
- SIMOTION System and Function Descriptions
- SIMOTION Service and Diagnostics
- SIMOTION Programming
- SIMOTION Programming - References
- SIMOTION C
- SIMOTION P350
- SIMOTION D4xx
- SIMOTION Supplementary Documentation

SINAMICS documentation

The SINAMICS documentation is organized into 2 parts:

- General documentation/catalogs
- Manufacturer/service documentation

A current overview of the documentation in the available languages can be found on the Internet:

<http://www.siemens.com/motioncontrol>

Select the menu items "Support" --> "Technical Documentation" --> "Overview of Publications."

The Internet version of DOConCD (DOConWEB) is available on the Internet:

<http://www.automation.siemens.com/doconweb>

Information on the range of training courses and FAQs (Frequently Asked Questions) are available on the Internet:

<http://www.siemens.com/motioncontrol>

Follow the menu item "Support".

Further documentation for the DCC editor

- SINAMICS/SIMOTION Function Manual, *Description of the standard DCC blocks*

Hotline and Internet addresses

Siemens Internet address

The latest information about SIMOTION products, product support, and FAQs can be found on the Internet at:

- General information:
 - <http://www.siemens.de/simotion> (German)
 - <http://www.siemens.com/simotion> (international)
- Downloading documentation
Further links for downloading files from Service & Support.
<http://support.automation.siemens.com/WW/view/en/10805436>
- Individually compiling documentation on the basis of Siemens contents with the My Documentation Manager (MDM), refer to <http://www.siemens.com/mdm>

My Documentation Manager provides you with a range of features for creating your own documentation.
- FAQs
You can find information on FAQs (frequently asked questions) by clicking <http://support.automation.siemens.com/WW/view/en/10805436/133000>.

Additional support

We also offer introductory courses to help you familiarize yourself with SIMOTION.

For more information, please contact your regional Training Center or the main Training Center in 90027 Nuremberg, Germany.

Information about training courses on offer can be found at:

www.sitrain.com

Technical support

If you have any technical questions, please contact our hotline:

	Europe / Africa
Phone	+49 180 5050 222 (subject to charge)
Fax	+49 180 5050 223
€0.14/min from German wire-line network, mobile phone prices may differ.	
Internet	http://www.siemens.com/automation/support-request

	Americas
Phone	+1 423 262 2522
Fax	+1 423 262 2200
E-mail	mailto:techsupport.sea@siemens.com

	Asia / Pacific
Phone	+86 1064 757575
Fax	+86 1064 747474
E-mail	mailto:support.asia.automation@siemens.com

Note

Country-specific telephone numbers for technical support are provided under the following Internet address:

<http://www.automation.siemens.com/partner>

Questions about this documentation

If you have any questions (suggestions, corrections) regarding this documentation, please fax or e-mail us at:

Fax	+49 9131- 98 2176
E-mail	mailto:docu.motioncontrol@siemens.com

Table of contents

	Preface	3
1	Introduction.....	11
2	DCC editor functionality	17
2.1	Overview	17
2.2	Requirement.....	19
2.3	Establish the project requirements	20
2.3.1	Create a project.....	20
2.3.2	Inserting a device into a project	21
2.3.3	Inserting the DCC chart in a project.....	23
2.3.4	Inserting a new chart (subchart)	27
2.3.5	Inserting new chart partitions	27
2.3.6	View and representation	28
2.4	Library handling	29
2.4.1	Importing block libraries	29
2.4.2	Updating the block library	30
2.4.3	Changing the block library language	32
2.4.4	Removing block libraries from the configuration.....	34
2.4.5	Exchanging the basic library version for installed libraries	35
2.5	Handling blocks.....	36
2.5.1	Introduction	36
2.5.2	Inserting blocks in the DCC editor	36
2.5.3	Inserting text.....	38
2.5.4	Specifying execution properties	40
2.5.5	Editing block connections	41
2.5.5.1	General	41
2.5.5.2	Block connection properties	41
2.5.6	Interconnecting blocks	44
2.5.7	Data type abbreviation in the DCC for connection and transformer blocks.....	45
2.5.8	Interconnection to chart connections	46
2.5.9	Interconnection to global operands in DCC SIMOTION	47
2.5.10	Deleting blocks.....	47
2.6	Compiling	48
2.6.1	Consistency check without code generation.....	48
2.6.2	Compiling the DCC in the DCC editor	49
2.6.3	Error log	51
2.7	Editing configurations further	52
2.7.1	Editing programs further: overview	52
2.7.2	Changing the block library	52
2.7.3	Copying of charts or chart sections	53
2.7.4	Search in the project from STARTER/SCOUT	55
2.7.5	Replace in the project from STARTER/SCOUT	56
2.8	Test mode	57
2.8.1	Test modes	57
2.8.2	Monitoring in laboratory mode	58

2.8.3	Monitoring in process mode	58
2.8.4	Logging on/logging off connections for testing	59
2.8.5	Activating/deactivating connection monitoring	59
2.8.6	Activating test mode	60
2.8.7	Monitoring test mode	61
2.8.8	Enabling the value and trend display during a test	62
2.8.9	Editing DCCs in test mode	63
2.8.10	Deactivating test mode	64
2.8.11	Changing online during test mode	64
2.8.11.1	Preliminary remark	64
2.8.11.2	Changing values at block inputs online	65
2.8.11.3	Deleting an interconnection online	65
2.8.11.4	Establishing an interconnection online	66
2.8.11.5	Moving interconnections online	66
2.8.11.6	Inserting a block online	67
2.8.11.7	Deleting a block online	67
2.8.11.8	Inserting comments in the chart	68
2.8.11.9	Changing comments in the chart	68
2.8.11.10	Moving blocks in the chart	68
2.8.11.11	Consistency of the charts in test mode	68
2.9	Reference data	70
2.9.1	Chart reference data	70
2.9.2	List of block types	71
2.9.3	Cross References List Execution Groups (only for DCC-SIMOTION)	71
2.9.4	List of operand cross references	72
2.10	Know-how Protection	73
2.10.1	Information on know-how Protection	73
2.11	Startup behavior	77
2.12	Software upgrade and module exchange	78
2.13	Version information	79
3	DCC for SINAMICS	81
3.1	Overview	81
3.1.1	Introduction	81
3.1.2	Software requirements	81
3.1.3	SINAMICS system integration	82
3.1.3.1	Applications and features	82
3.1.3.2	Execution groups in the DCC editor	83
3.1.3.3	Fixed execution groups	85
3.1.3.4	Free execution groups	95
3.1.3.5	Execution sequence order, creating new execution groups	96
3.1.3.6	Creating customer-specific parameters ("declare")	98
3.1.3.7	Interconnection with SINAMICS parameters	102
3.1.3.8	Copy charts within a project	104
3.1.3.9	Copy charts between projects	104
3.1.4	Computing time load, memory requirement and assignment of the HW sampling times	105
3.1.4.1	Computing time load for the SINAMICS V2.5 and V2.6 software versions	105
3.1.4.2	Computing time load as of software version V4.3	107
3.1.4.3	Memory requirement for software version SINAMICS V2.6	110
3.1.4.4	Memory requirement as of software version SINAMICS V4.3	112
3.1.4.5	Number of possible different hardware sampling times	112
3.2	Working with DCC SINAMICS	114

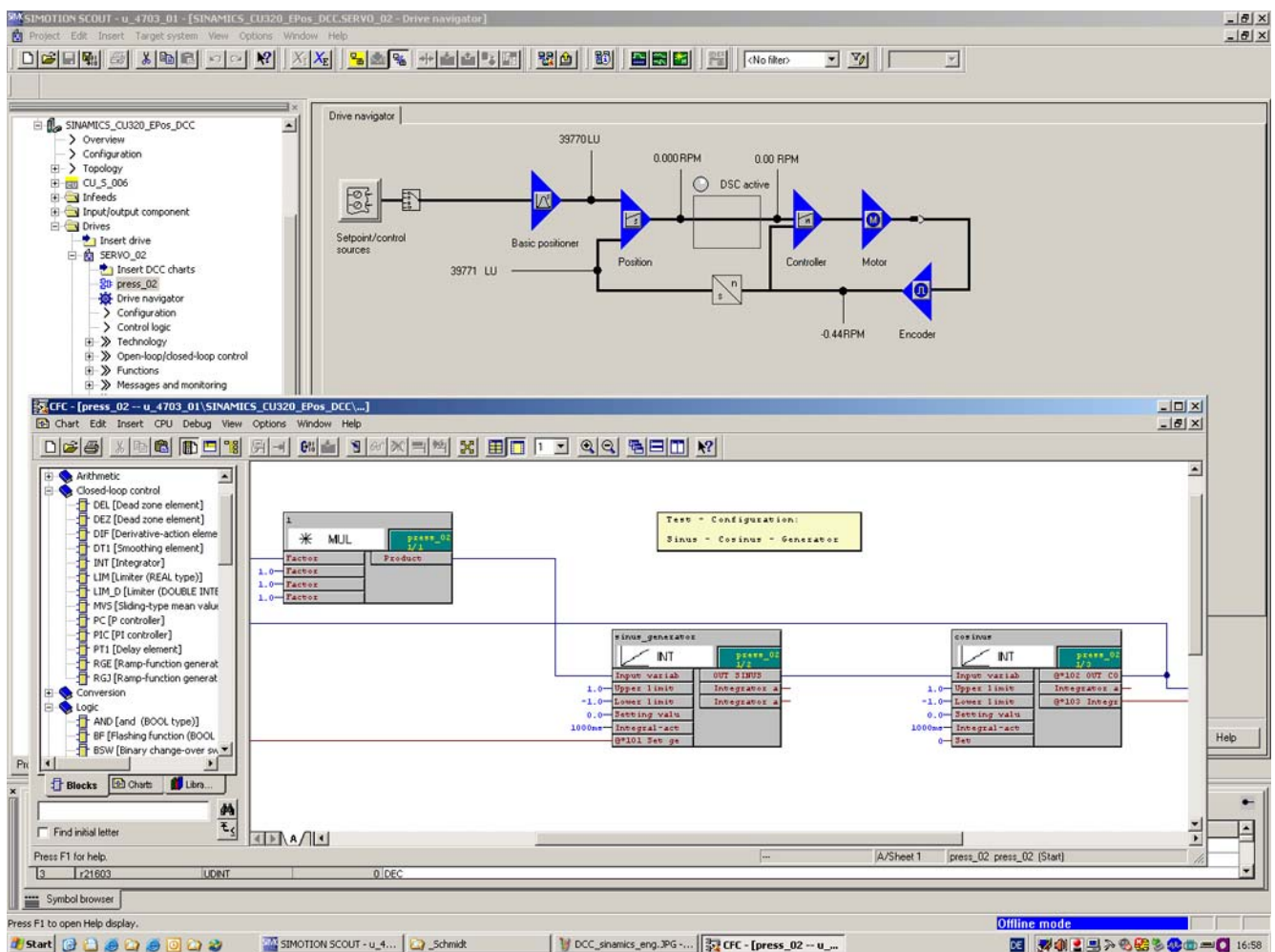
3.2.1	Preliminary remarks on configuration	114
3.2.2	Creating a new project	115
3.2.3	Inserting a DCC	116
3.2.4	Inserting blocks	118
3.2.5	Interconnecting blocks	119
3.2.6	Parameterizing block connections in the chart	119
3.2.7	Publishing block connections as parameters	121
3.2.8	Compiling the DCC chart in the DCC editor	123
3.2.9	Setting execution sequence within an execution group	124
3.2.10	Setting sampling time for an execution group	124
3.2.11	Loading the DCC technology option onto the CF card of the drive device	125
3.2.12	Downloading compiled DCC chart into the drive	127
3.2.13	Displaying values of block connections online	127
3.2.14	Interconnection to the BICO parameters of the basic system in DCC SINAMICS	129
3.2.15	BICO interconnections and reading back parameters	130
3.2.16	Record with the trace signals from the DCC chart	131
3.2.17	Archiving a project	133
3.2.18	Creating documentation	133
3.3	Connecting the DCC to the drive	135
3.3.1	Overview	135
3.3.2	Calculating a DCC chart with per-unit variables	135
3.3.3	Calculating a DCC chart with absolute variables	137
3.3.3.1	Example 2.1 (interconnecting input value)	137
3.3.3.2	Example 2.2 (interconnecting output value)	139
3.3.4	Interconnecting DCC signals with communication interfaces IF1 and IF2	141
3.3.4.1	Preliminary remark	141
3.3.4.2	Interconnecting received process data with DCC	141
3.3.4.3	Interconnecting sent process data with DCC	141
3.4	DCC SINAMICS specifications	142
3.4.1	Rules for assigning names in the DCC editor	142
3.4.2	Field/name lengths and conventions	143
3.4.3	Representation of the dynamic value display	143
4	DCC for SIMOTION	145
4.1	Overview	145
4.1.1	Introduction	145
4.1.2	Software requirements	145
4.1.3	SIMOTION system integration	146
4.1.3.1	Execution level, execution group and execution sequence	146
4.1.3.2	HMI variables (publishing of variables and @ variables)	149
4.1.3.3	Interconnecting with SIMOTION variables	151
4.1.4	Creating block libraries	154
4.1.4.1	Fundamentals	154
4.1.4.2	Creating comments for DCB libraries you have created yourself	157
4.1.4.3	Special features of the 1:N interconnection of chart connection inputs	158
4.1.4.4	Reliance of the generated block on the underlying block libraries	159
4.1.4.5	Creating an installable DCB library from DCC libraries	160
4.1.4.6	Creating the online help for block libraries	161
4.1.4.7	Installation and uninstallation of DCB libraries	164
4.1.5	DCC and SIMOTION trace	165
4.2	Working with DCC SIMOTION	166
4.2.1	Preliminary remarks on configuration	166
4.2.2	Creating a project	167
4.2.3	Inserting a chart	167

4.2.4	Inserting blocks	168
4.2.5	Interconnecting blocks	169
4.2.6	Parameterizing block connections in the chart	169
4.2.7	Default connection values for delta downloads.....	171
4.2.8	Compiling the DCC in the DCC editor.....	172
4.2.9	Loading the compiled DCC	174
4.2.10	Starting the CPU	174
4.2.11	Enable attribute, execution groups	174
4.2.12	Creating documentation	174
4.3	DCC SIMOTION specifications	176
4.3.1	Rules for assigning names in the DCC editor	176
4.3.2	Field/name lengths and conventions	177
4.3.3	Representation of the dynamic value display	178
4.4	Faults and warnings	179
4.4.1	Notes on error message display	179
A	Appendix.....	181
A.1	List of abbreviations	181
A.2	Glossary	182
	Index.....	185

Introduction

Drive Control Chart (DCC) for SINAMICS and SIMOTION means graphic configuration and expansion of the device functionality by means of freely available control, arithmetic and logic blocks

Drive Control Chart (DCC) expands the facility for the simplest possible configuring of technological functions both for the SIMOTION motion control system as well as for the SINAMICS drive system. This opens up a new dimension for users for adapting the specified systems to the specific functions of their machines. DCC has no restriction with regard to the number of usable functions; this is only limited by the performance capability of the target platform.



DCC comprises the DCC editor and the DCB library (block library with standard DCC blocks).

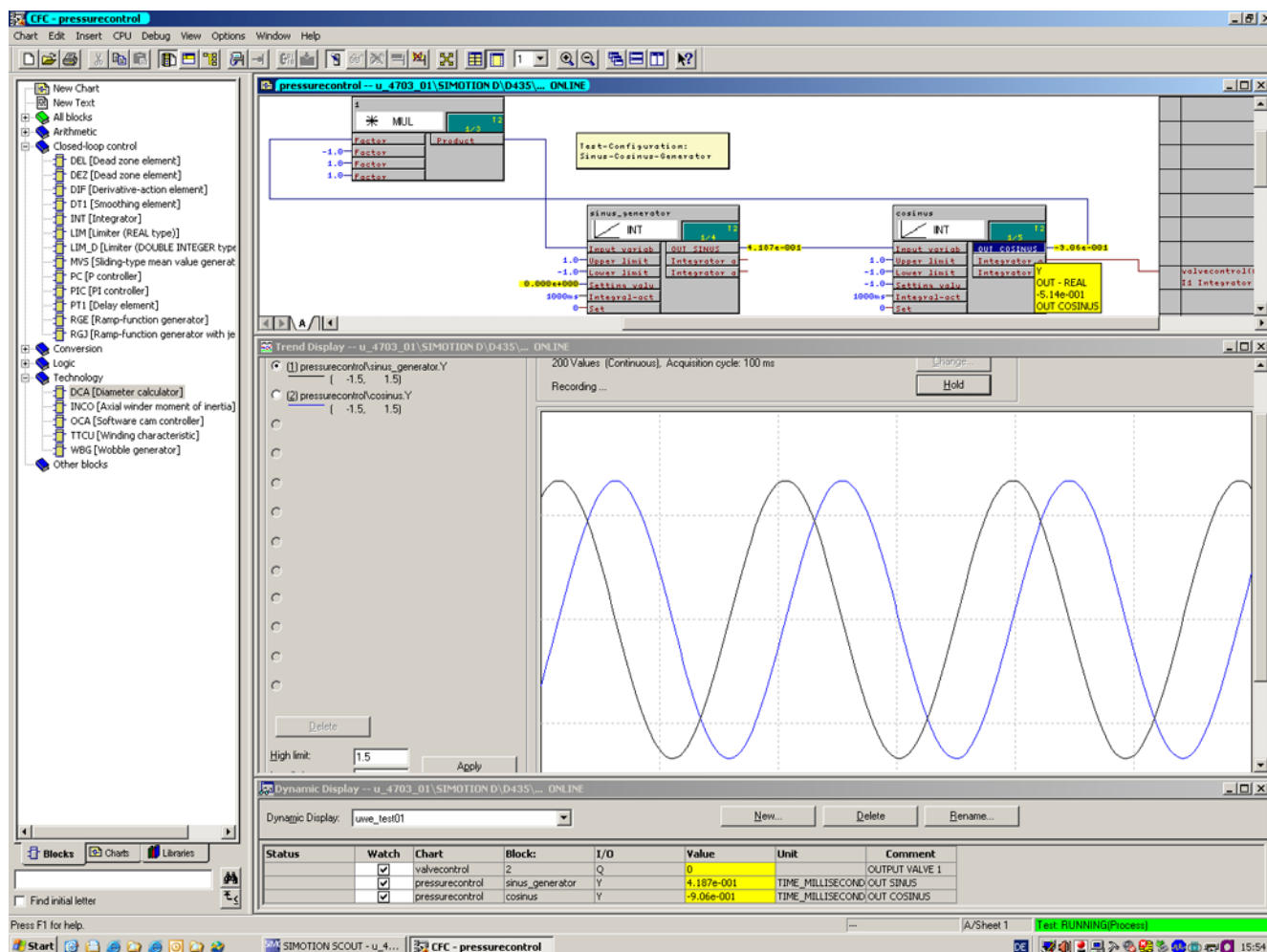
The user-friendly **DCC editor** enables easy graphic configuration and a clear representation of control loop structures as well as a high degree of reusability of existing charts. The open-loop and closed-loop control functionality is defined by using multi-instance-capable blocks (**Drive Control Blocks, DCBs**) from a pre-defined library (**DCB library**) that are selected and graphically linked by dragging and dropping. Test and diagnostic functions allow verification of program behavior or the identification of causes in the event of errors.

The block library contains a large selection of control, arithmetic and logic blocks as well as extensive open-loop and closed-loop control functions.

All commonly used logic functions are available for selection (AND, XOR, On/Off delay, RS flipflop, counters, etc.) for the logic operation, evaluation and acquisition of binary signals. Diverse arithmetic functions such as absolute-value generation, dividers and minimum/maximum analysis are available for monitoring and evaluating numeric variables. In addition to the drive control, axial winder functions, PI controllers, ramp-function generators or sweep generators can be configured simply and without problem.

Almost unlimited programming of control structures is possible in conjunction with the SIMOTION motion control system. These can then be combined with other program sections to form an overall program.

Drive Control Chart for SINAMICS drives also provides a convenient basis for resolving drive-level open-loop and closed-loop control tasks directly in the converter. This results in further adaptability of SINAMICS for the tasks set. On-site processing in the drive supports modular machine concepts and results in increased overall machine performance.



- With SINAMICS, DCC can be activated simultaneously on several drive objects (DOs) on a drive unit. With SIMOTION, several DCCs can be created in a program container.
- Block library with management, arithmetic, control, logic and complex blocks.
- Graphic component connection editor with various editing, macro, help, search, comparison and print functions.
- Simple configuration of axial winder functions, PI controller, ramp-function generator or sweep generator.
- Execution environment for SIMOTION with sampling times that can be selected and mixed, and consistent data transfer between sampling times.
- Execution environment for SINAMICS with the option of embedding technology in SINAMICS using BICO technology so that applications can be set via configured parameters. Up to 10 different sampling times can be configured.

- Diagnostics environment with signal display, diagnostic and trace functions.
- Scalability with different performance features and quantity structures in DCC SIMOTION and DCC SINAMICS.

Note

The DCC editor is a programming system based on CFC (Continuous Function Chart). The following sections describe the principles of DCC operation that have not already been covered in the manual titled "CFC Manual for S7".

Note

Further references for the DCC editor:

SINAMICS/SIMOTION Function Manual, *Description of the standard DCC blocks*

Differences between DCC SIMOTION and DCC SINAMICS

DCC SIMOTION and DCC SINAMICS differ to some extent in their mode of operation. The basic differences are listed below:

Table 1- 1 Differences between DCC SIMOTION and DCC SINAMICS

	SIMOTION	SINAMICS
System integration	Via interconnection to variables of the basic system	Via adjustable parameter or interconnection via BICO parameter to the basic system
Execution system	Any number of execution groups that can be assigned five time slices (depending on the system cycle clocks)	Maximum of 10 execution groups that can be assigned 10 different sampling times
Consistency in the data transfer	Consistent data transfer also across the time slices	Consistency in the data transfer must be established by the user, when required, by means of standard blocks (SAH_X)
Use of libraries	Additional libraries can be used (Page 154)	No additional libraries can be used
Scope of standard blocks	See SINAMICS/SIMOTION Function Manual, <i>Description of the standard DCC blocks</i>	See SINAMICS/SIMOTION Function Manual, <i>Description of the standard DCC blocks</i>

Availability of DCC features

Table 1- 2 Availability of DCC features

Feature	Available from version
Monitoring and tracing of published parameters	SINAMICS 2.5 / SIMOTION 4.1
Online changes in test mode and tracing of unpublished block connections	SINAMICS 2.6 / SIMOTION 4.1
Read back of BICO interconnections from the target device	SINAMICS 2.5 / DCC 2.0.3 / STARTER/SCOUT 4.1.3
Creation of installable libraries	SINAMICS 2.5 / DCC 2.0.3 / STARTER/SCOUT 4.1.3
Installation of block libraries in the case of an opened SCOUT	SCOUT V4.1.2
Online insertion and deletion of block instances	SINAMICS 2.6 / SIMOTION 4.1.2
Online changes, insertion and deletion of interconnections	SINAMICS 2.6 / SIMOTION 4.1.2
Display of reference data for DCC charts	SCOUT/STARTER 4.1.3
Search and replace for DCC sheet bar interconnections	SCOUT/STARTER 4.1.2
Compiling without DCC license	SCOUT/STARTER 4.1.3
Creating C-block libraries automated from DCC libraries	DCC 2.0.3
Fixed execution group "BEFORE actual position value"	SINAMICS 4.3
Available HW sampling times for CU320 = 25	SINAMICS 4.3

DCC editor functionality

2.1 Overview

The product offers a modular, scalable technology option, which has chiefly been developed for drive-related, continuous open-loop and closed-loop control engineering tasks.

The DCC technology option for SIMOTION controllers and SINAMICS drives can be configured graphically using the Drive Control Chart editor (referred to below as DCC editor), which is based on CFC. The following figure illustrates the data flow of the configuration data when configuring with the DCC technology option:

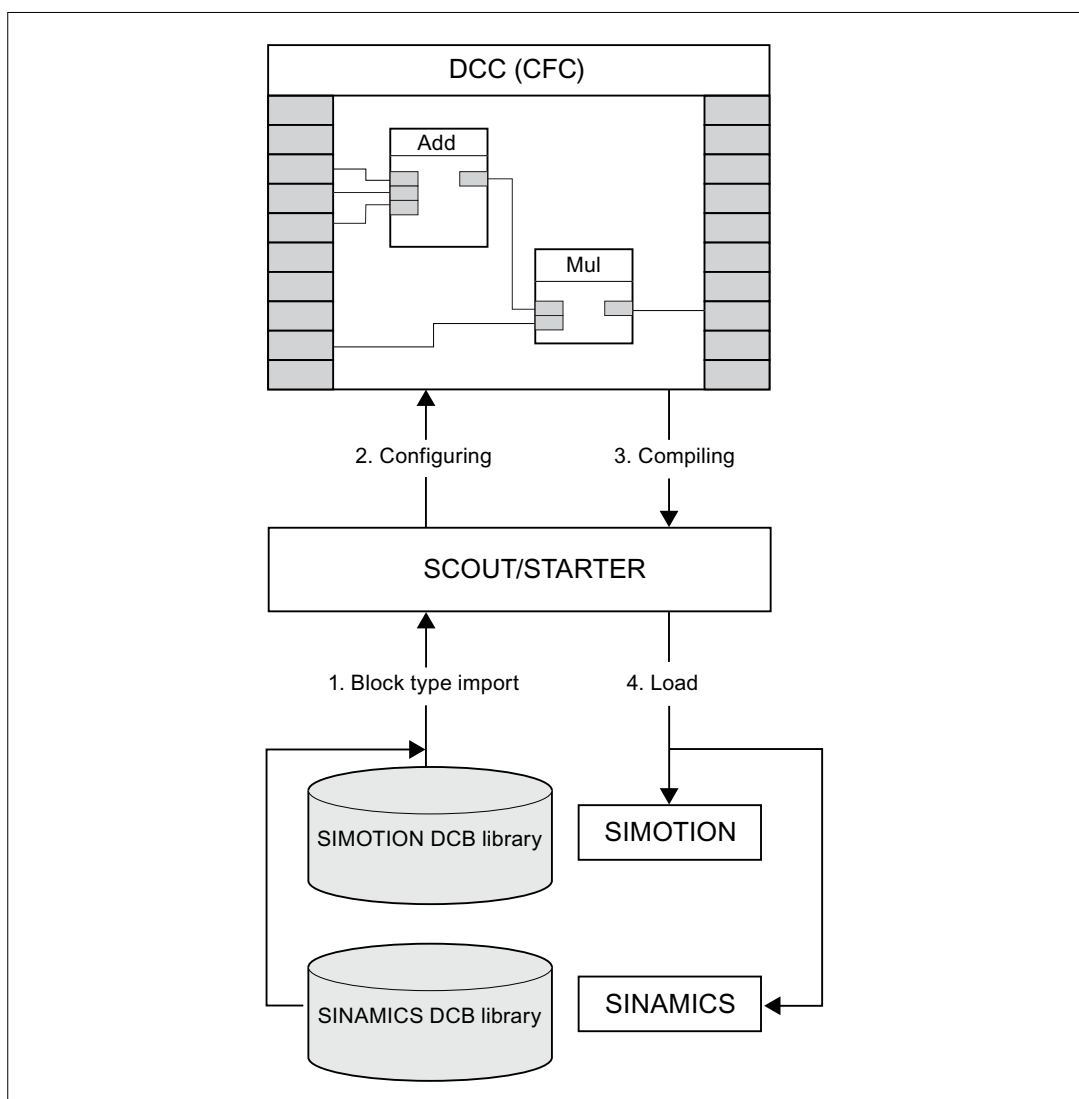


Figure 2-1 Flow of configuration data

1. When a new chart is created, the block types are taken from the device-specific block library and inserted in the DCC editor's block manager.
2. The DCC editor is used to create charts, in which you can insert, parameterize and interconnect blocks.
3. When you compile the charts, an intermediate code is created.
4. This is downloaded to the device or drive unit using STARTER/SCOUT.

2.2 Requirement

This description refers to the following device SW versions:

- DCC 2.0.5
- SIMOTION P, C, D as of Version 4.1.5
- SINAMICS Integrated of the SIMOTION D410 as of Version 2.5
- SINAMICS Integrated of the SIMOTION D as of Version 2.6
- SINAMICS S120, S150, SM150, G130, G150, GM150, GL150 as of Version 4.3
- SINAMICS DCM as of Version 1.1
Except for the load level, the SINAMICS DC MASTER as of Version 1.1 supports the same function scope as SINAMICS S120 as of Version 4.3.
Detailed information about SINAMICS DC MASTER as of Version 1.1 is contained in the *SINAMICS DC MASTER 6RA80 DC Converters* operating instructions.
- At least CFC 7.0.1.3 must be installed as of DCC 2.0.3.

As well as the SIMOTION SCOUT and STARTER engineering tools as of V4.1.3

Note

Configuration should not be carried out in SIMATIC Manager, but directly using SCOUT/STARTER. SIMATIC Manager uses a type of syntax for naming new charts that causes an error when they are compiled in SIMOTION SCOUT/STARTER.

If errors occur when charts are being compiled in SIMOTION SCOUT/STARTER, it is possible to generate a comprehensive error report by compiling them again in the DCC editor. To do this, activate *Display all messages with 'Save and compile all'* on the *Compiler* tab under the *Settings* menu command in SIMOTION SCOUT. A corresponding error report is automatically created in STARTER.

Note

The DCC editor is installed automatically with the SIMOTION SCOUT/STARTER engineering system. An appropriate program license is required for the DCC editor. This is on the USB stick supplied with the DCC SIMOTION or DCC SINAMICS product.

Note

It is only possible to print DCCs in the DCC editor, but not in SIMOTION SCOUT/STARTER.

Note

Only context-sensitive help is available for the DCC blocks, i.e. the descriptions cannot be accessed from the main help page.

Note

If the DCC charts are opened with a different CFC version than the one with which they were created, this may lead to inconsistencies within the project. The DCC chart must be recompiled and reloaded. DCC charts that have been edited with CFC 7.1 can no longer be edited with CFC 7.0.1.3.

2.3 Establish the project requirements

2.3.1 Create a project

You must create a new project in the SIMOTION SCOUT/STARTER engineering system before using the DCC editor.

Procedure

Proceed as follows to create a project:

1. Start the SIMOTION SCOUT/STARTER engineering system.
2. Execute the **Project > New** function from the menu bar.
3. In the **New Project** window, enter the name of the project in the **Name** field.

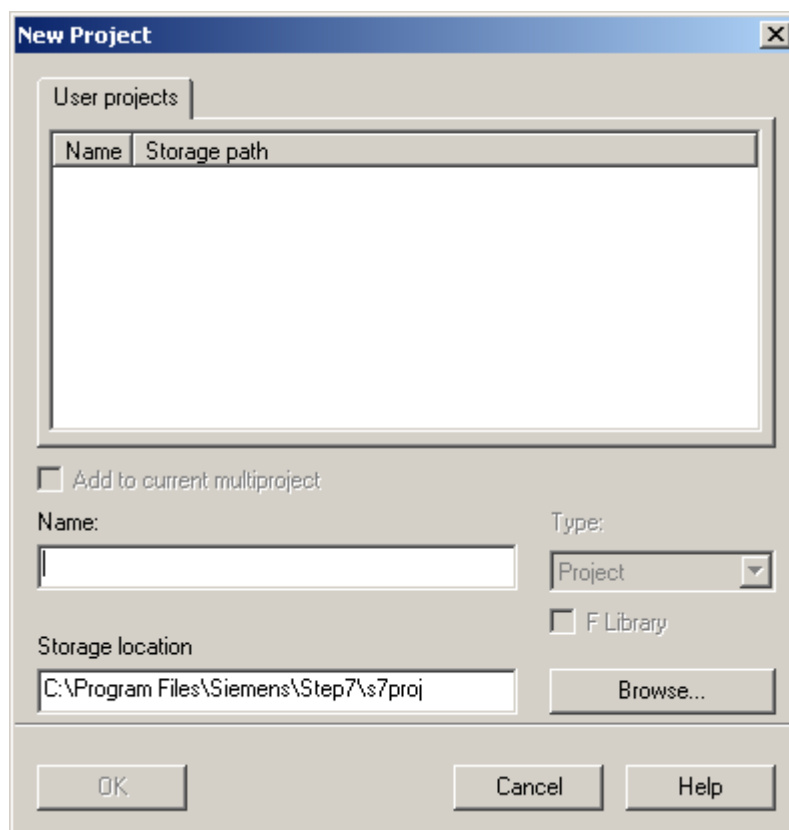


Figure 2-2 Window - New Project

4. Click **OK** to close the window.

The new project is created and then automatically opened.

Note**Convention for assigning names for projects**

The project name may contain a maximum of 24 characters. The folder name is generated using the first eight characters of the project name. It is therefore important to ensure that the first eight characters of the project name are unambiguous.

2.3.2 Inserting a device into a project

SIMOTION

Proceed as follows to insert a device into a project:

1. Open an existing SIMOTION SCOUT project if a project is not already open.
2. Execute the **Create new device** command.

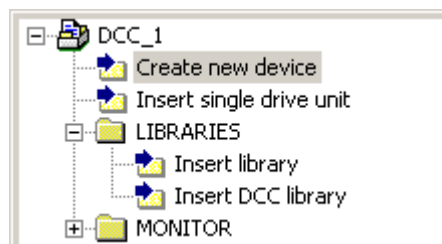


Figure 2-3 SIMOTION: Create new device

3. In the **Create new device** window, select the required device and close the window with **OK**.

Note

For further information on the **Open HW Config** switch, please refer to the documentation for SIMOTION SCOUT.

All requirements for creating a DCC in the project have now been met.

SINAMICS

1. Open an existing project or create a new project in which you want to insert a SINAMICS drive unit (e.g. SINAMICS S120 CU 320). Note that the SINAMICS S110 (CU 305 module) does not support DCC.

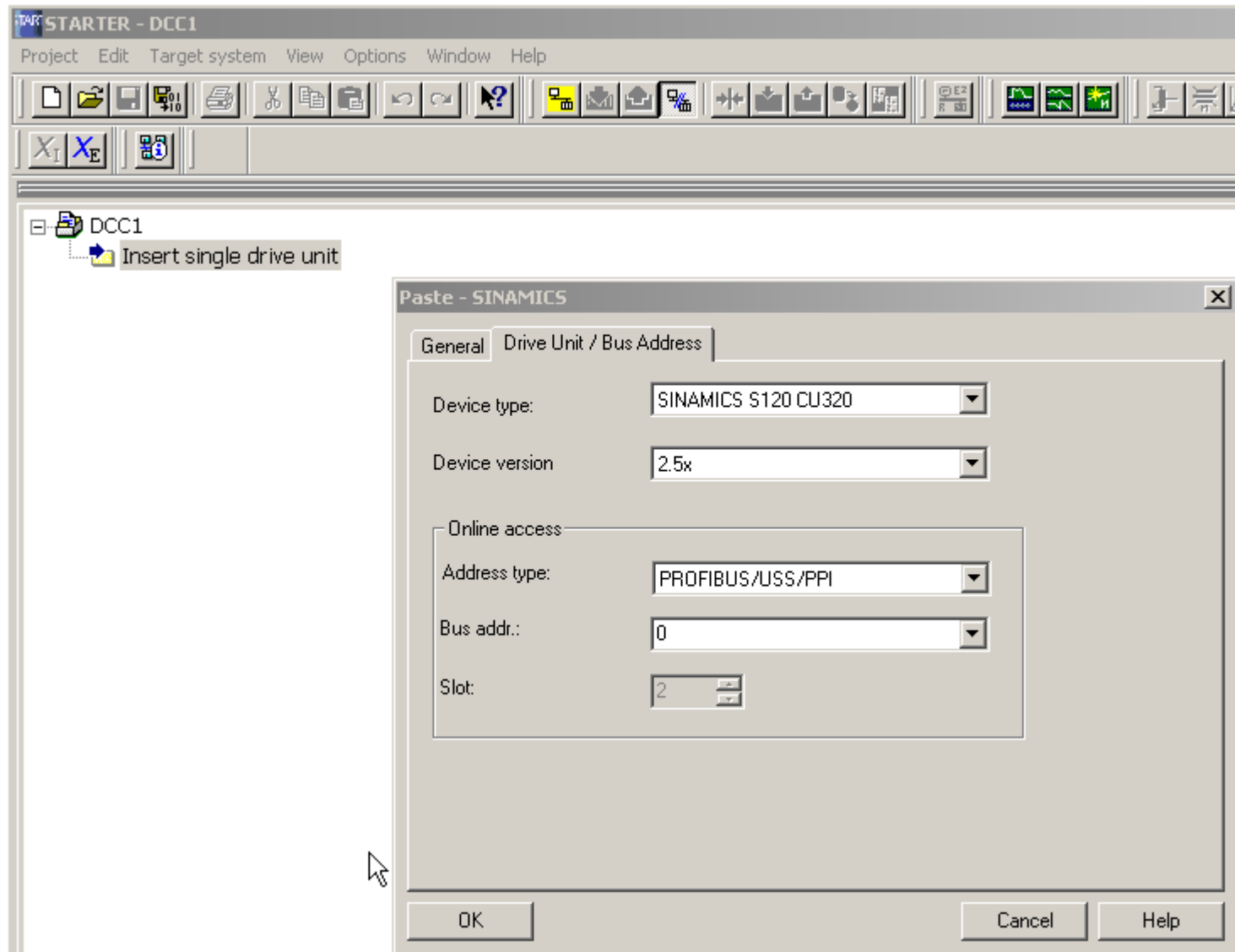


Figure 2-4 Inserting a device into a project

2.3.3 Inserting the DCC chart in a project

You can now insert a DCC into the existing project.

Procedure

1. Select a device from the project.
2. **SIMOTION:**

Execute the **Insert DCC** function from the **PROGRAM** subitem of the device.

SINAMICS, STARTER:

Execute the **Insert DCC** function on the desired drive object. There may only be one DCC on a drive object.

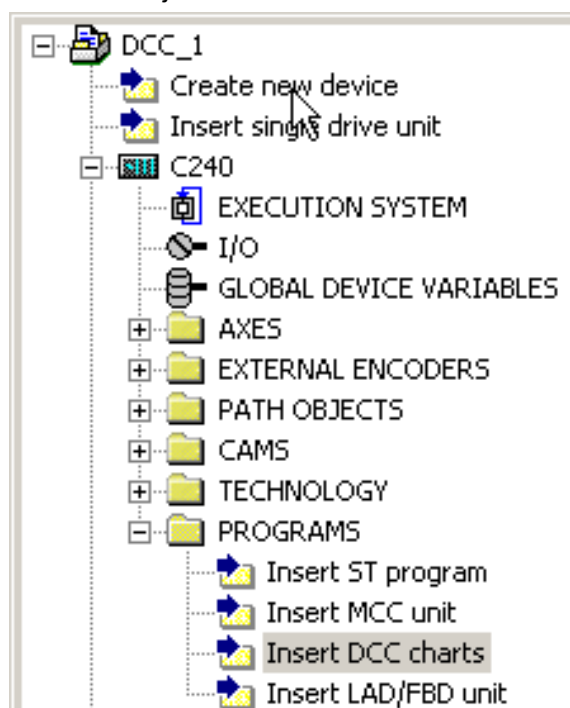


Figure 2-5 SIMOTION SCOUT: Inserting a DCC

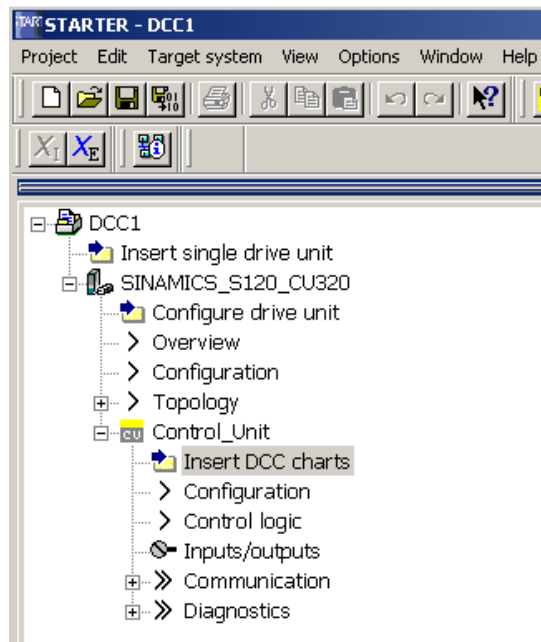


Figure 2-6 Inserting a DCC in the CU drive object of a SINAMICS CU3x0.x drive unit with STARTER

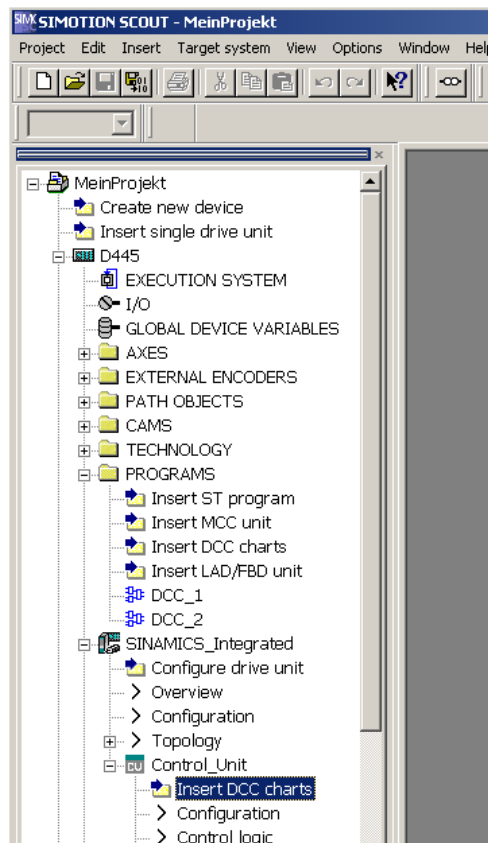


Figure 2-7 SIMOTION D4xx: Inserting a DCC SINAMICS chart on the drive object of the CU with SCOUT

3. The **Insert DCC** window appears.

The screenshot shows a dialog box titled "Insert DCC chart". At the top, there is a "Name:" label followed by a text input field containing "DCC_1". Below this, there are two tabs: "General" (which is selected) and "Compiler". In the "General" tab, there are three input fields on the right: "Author:", "Version:", and "Know-how protection:". Below these is a large empty rectangular box labeled "Existing Objects". At the bottom of the dialog, there is a "Comment:" label followed by a large text area. Below the comment area is a checkbox labeled "Open editor automatically" which is checked. At the very bottom, there are three buttons: "OK", "Cancel", and "Help".

Figure 2-8 Window - Insert DCC

4. Assign a name to the DCC.

The DCC has now been created.

If you have selected the **Open editor automatically** option in the **Insert DCC** window then the DCC editor opens automatically. When opening the DCC for the first time, the **Import DCB library** window is automatically displayed.

Note

Convention for assigning names to charts

The chart name may contain a maximum of 22 characters.

Explanation of the various types of chart

There are three different types of chart:

- Basic chart
- Chart partition
- Subchart

Charts that are visible within SIMOTION SCOUT/STARTER or the SIMATIC Manager are designated as basic charts. Every basic chart has up to 26 chart partitions and each of these partitions comprises six sheets. Embedded charts - the subcharts - can be used within each sheet. Each of these subcharts may also have its own chart partitions and subcharts. A maximum number of eight nesting levels with subcharts is possible.

Subcharts are not visible as charts in SIMOTION SCOUT/STARTER or in the SIMATIC Manager.

The following graphic clarifies the connection between the three types of chart.

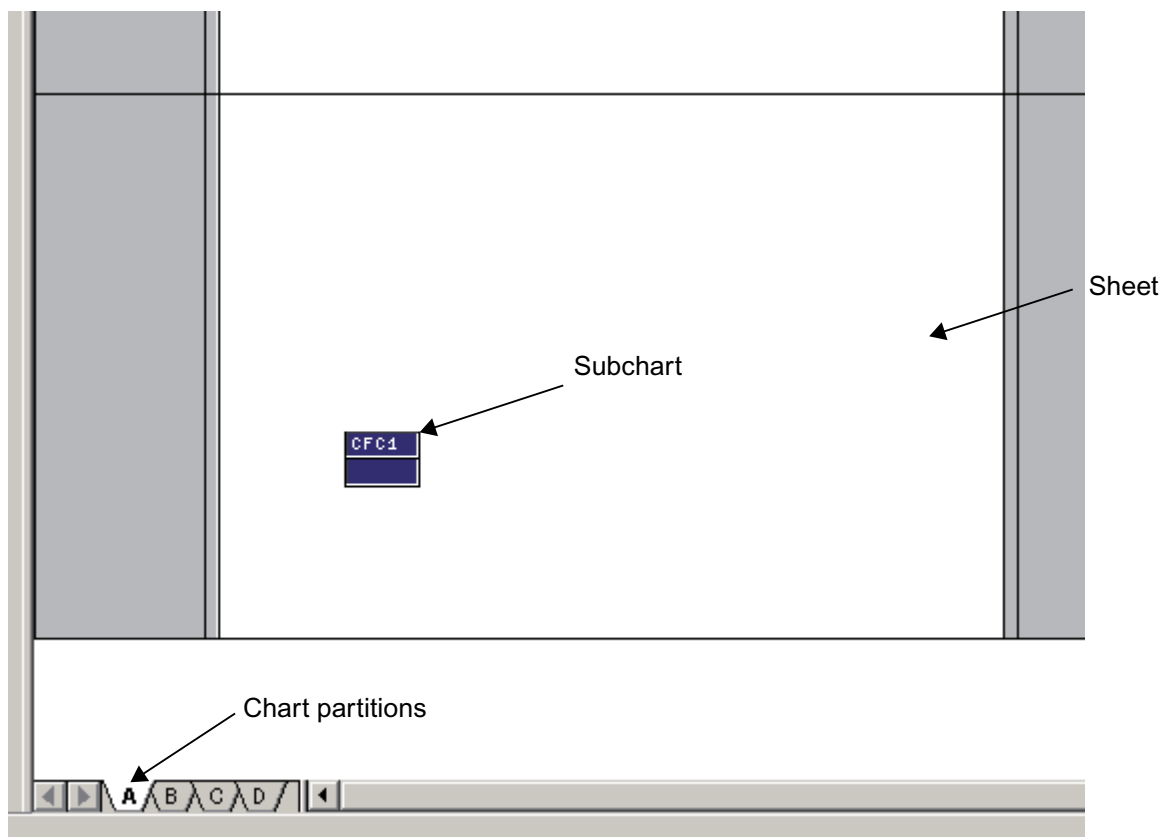


Figure 2-9 Connection between the types of chart in the DCC editor

2.3.4 Inserting a new chart (subchart)

A chart (subchart) can be inserted in another chart (chart-in-chart-technique). Hierarchical structures can be formed here. Each inserted chart can be opened and, like every other chart, edited further and therefore changed individually. A chart can be encapsulated for further use, i.e. chart connections added. Which block connections are provided at the chart connections can also be specified individually.

Requirement

You have already created a DCC in the SIMOTION SCOUT/STARTER engineering system which is opened in the DCC editor.

Procedure

1. Use **View > Overview** or the  button to switch from the page view to the chart overview. The six pages of the selected DCC are now shown.
2. You can now insert new subcharts using the menu item **Insert new chart** in the context menu and open these by right-clicking the subchart and selecting the **Open** command in the context menu.

2.3.5 Inserting new chart partitions

Requirement

You have already created a DCC in the SIMOTION SCOUT/STARTER engineering system which is opened in the DCC editor.

Procedure

1. Insert a new chart partition in the desired position using the menu items **Insert > Chart partition > In front of current chart partition** or **At the end**.
2. Alternatively, you can right-click an already existing chart partition on the tab and select **Insert chart partition in front of current chart partition** or **Insert chart partition at the end**.

2.3.6 View and representation

Going into the page view or overview representation

To change to the page view from the overview representation, right-click an empty space in the chart and select **Display this page** in the context menu that appears. The names of the block connections are displayed in this enlarged view.

To change to the overview representation from the page view, right-click an empty space in the chart and select **Overview** in the context menu that appears.

You can also switch to the page view and back to the overview again by double-clicking an empty area on a page.

It is also possible to switch between the page view and overview representation using the **View** menu.

You can switch to a page view with the block catalog displayed on the left using the **View -> Catalog** menu.

2.4 Library handling

2.4.1 Importing block libraries

When first creating a chart in SCOUT/STARTER, you may be prompted to import a block library for the selected device platform. The block types are imported in the language that has been set. You can select this in the dialog box. If the desired language is not available, English is used instead.

How to import block libraries:

- Open a chart from the relevant configuration and click **Options > Block types**. The **Import** window appears. Click **OK** to close the window. In the window **Import DCB Libraries**, the block libraries installed on your programming device are listed under **Libraries installed in SCOUT/STARTER**. Under **Libraries imported in the chart**, all libraries which have already been imported into this configuration are listed.
- Select the library to be imported under **Libraries installed in SCOUT/STARTER** and click **>>**. The import process is triggered when you click **Accept**.

Checks are performed when a block library is imported.

The names of the block libraries are defined on the basis of a naming convention. The individual parts of the name are explained in the following table. The part in question is shown in bold each time.

Table 2- 1 Naming convention for block libraries

Part of the block library name	Meaning
TPdcbli b _SIMOTION_4_1_2.2.0 [7.0]	Library identifier
TPdcbli b _SIMOTION_4_1_2.2.0 [7.0]	Target platform
TPdcbli b _SIMOTION_4_1_2.2.0 [7.0]	Target system version
TPdcbli b _SIMOTION_4_1_2.2. 0 [7.0]	Library version
TPdcbli b _SIMOTION_4_1_2.2.0 [7.0]	Build version

2.4.2 Updating the block library

You can update the block libraries via the menu **Options > Block types**.

How to update block libraries:

- Open a chart from the relevant configuration and click **Options > Block types**.
- The **Import** window is displayed. Click **OK** to close the window. In the window **Import DCB Libraries**, the block libraries installed on your programming device are listed under **Libraries installed in SCOUT/STARTER**. Under **Libraries imported in the chart**, all libraries which have already been imported into this configuration are listed.

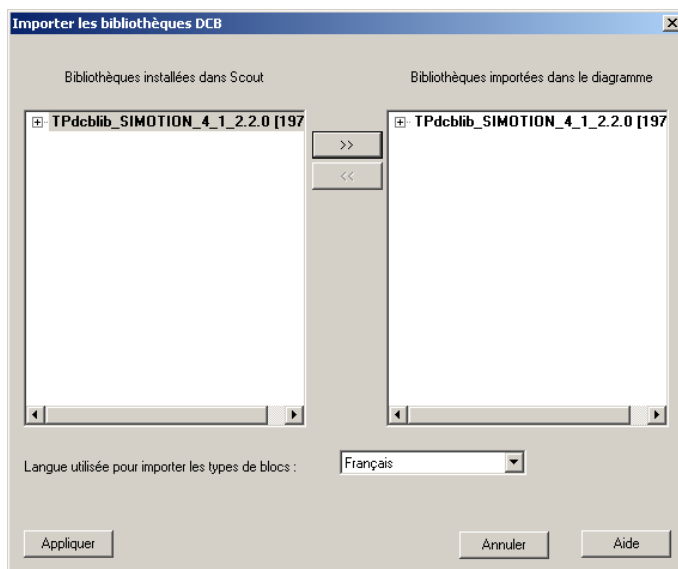


Figure 2-10 Updating the block library

Updating libraries after a device upgrade

After an upgrade of a device version, the library version must be updated manually - as opposed to copying DCC charts to other device types where compatible libraries are automatically updated.

Note

The standard library is automatically installed during setup, however if you accidentally uninstall the library it can be reinstalled via the DCC setup.

The libraries for SINAMICS can also be installed via the SSPs (SINAMICS Support Packages).

The libraries for SIMOTION are also available via the menu command **Extras -> Install libraries**. See also Creation of block libraries (Page 154)

- Check which previously imported libraries have a new version installed, by inspecting the version of the library under **Libraries installed in SCOUT/STARTER** and **Libraries imported in the chart**.
- Select the library to be updated under **Libraries installed in SCOUT/STARTER** and click >>.
- Click **Accept**.
- The progress of the update is displayed in a window.

Note

In SINAMICS, you can check a selected device in the engineering system using the **Select technology packages** function in the context menu as to whether all required block libraries and technology packages have been activated.

Note

In SIMOTION, the library is automatically loaded to the device during the project download. In SINAMICS, this action must be performed explicitly by the user before the project download. (Page 125)

The library to be imported is checked to see whether it contains a block type that has already been imported with another library.

Error log

If an error is detected when a library is being imported, an error log with details of the causes of the error is displayed.

2.4.3 Changing the block library language

You can customize the language of the block type comments via the menu **Options > Block types**.

How to change the language of the block types in a block library:

Checks are performed when a block library is updated.

- Open a chart from the relevant configuration and click **Options > Block types**.
- The **Import** window appears. Click **OK** to close the window.

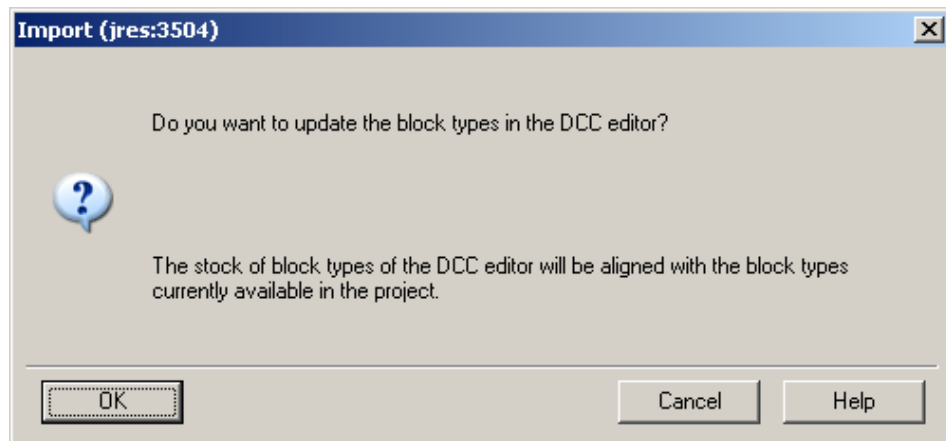


Figure 2-11 Import window

- The **Import DCB Libraries** window appears. The block libraries installed on your programming device are listed under **Libraries installed in SCOUT/STARTER**. Under **Libraries imported in the chart** all libraries which have already been imported into this configuration are listed.
- Select the desired language in the selection list of available languages. The selected language affects all imported libraries.

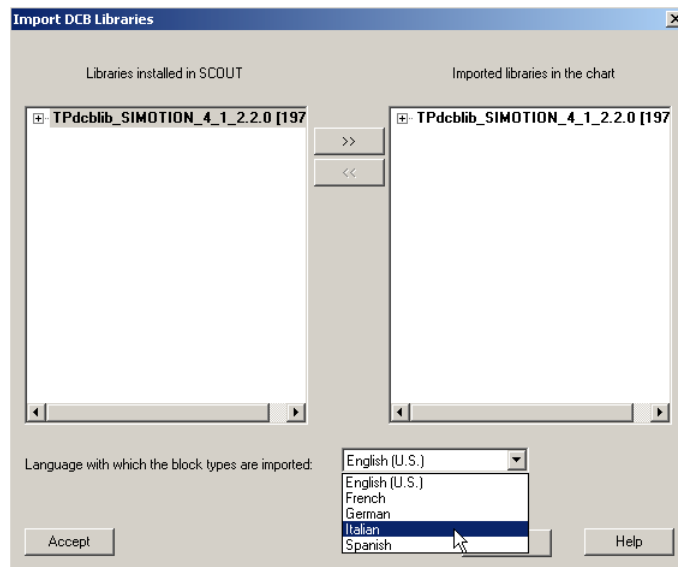


Figure 2-12 Changing the block library language

- Click **Accept**.
- The progress of the update is displayed in a window.

Note

Comments regarding the block types remain in the language in which they were created. The comments regarding block pins are translated if they have not already been altered manually.

2.4.4 Removing block libraries from the configuration

You can delete block libraries whose block types are no longer required in the configuration concerned via the menu **Options > Block types**.

How to delete block libraries:

- Ensure that the block types from the block library to be deleted are no longer to be used at all in the charts of the configuration.
- Open a chart from the relevant configuration and click **Options > Block types**.
- The **Import** window appears. Click **OK** to close the window.
- The **Import DCB libraries** window appears. The block libraries installed on your programming device are listed under **Libraries installed in SCOUT/STARTER**. Under **Libraries imported in the chart**, all libraries which have already been imported into this configuration are listed.
- Select the library to be deleted under **Libraries imported in the chart** and click **<<**. The deletion process is triggered when you click **Accept**.

Checks are performed when a block library is deleted. All unused block types of the library will be removed from the selection list of the block types in the DCC editor; used block types remain in the list. The chart cannot be compiled without an imported library if instances of the library are still present in the chart.

2.4.5 Exchanging the basic library version for installed libraries

The basic libraries are selected from the DCC charts used in the dialog **Options → Block types** of the DCC editor.

If there are no DCC sources available at the DCC charts, you can open a dialog via the context menu **Block types** of the DCC chart, where you can exchange the block libraries used.

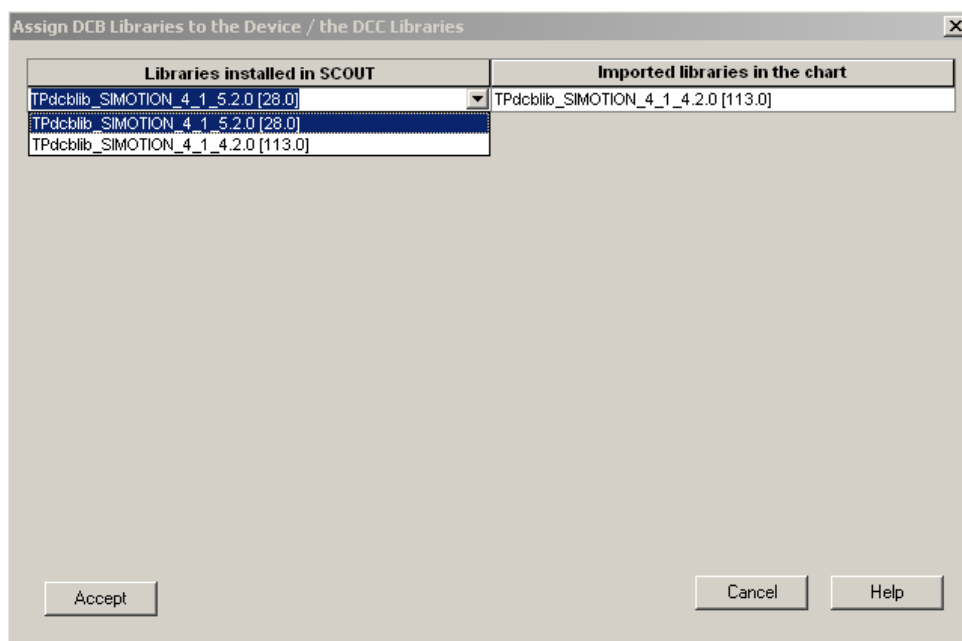


Figure 2-13 Block types

Mark the library in the left-hand column and press **Accept**.

2.5 Handling blocks

2.5.1 Introduction

In this section, you will learn which block types are available and how you can insert blocks in a DCC and delete them. You will also learn how to edit block connections.

Note

The online help provides detailed information (incl. timing diagram and plant view) for the individual blocks. To start the help, select the required block in the chart or in the block catalog and press the F1 key.

2.5.2 Inserting blocks in the DCC editor

Block types in the DCC editor

The block type inventory featured in the block catalog depends on both the device type and the version of the library. You will find the directories for the block families, as well as the directories **All blocks** (containing all blocks) and **Other blocks** (blocks that are not assigned to a family), in the block catalog. The names of the block families in the DCC editor are always in English.

Inserting a block

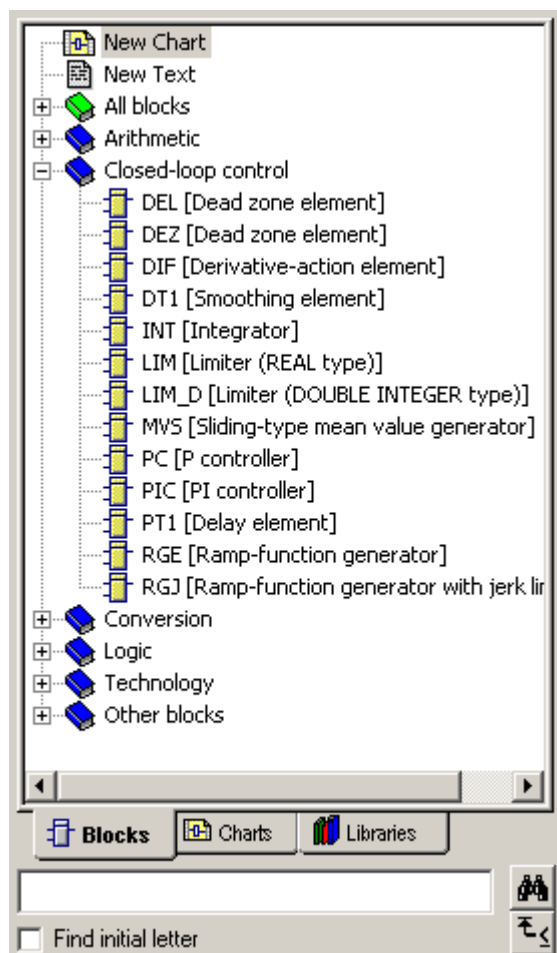


Figure 2-14 Inserting blocks

- Open a block family in the **Closed-loop control** family with the closed-loop control blocks.
- Select the required block and insert it in the chart using drag-and-drop. Only the outline of the block in dashed lines is displayed during the copying procedure. Release the mouse button at the required point.
- To search for a block, enter its name in the input field of the block catalog and click the **binoculars** button. The search process begins.

Note

If blocks are superimposed on the chart with other elements, such as other blocks or the sheet bar, the superimposed block will be displayed in gray and its connections will not be visible. You must reposition the blocks to ensure that all block information can be viewed.

2.5.3 Inserting text

You can add comments to your DCC that you enter in text fields. You can place these at any free position in the chart.

Procedure

Select the **New text** command via the directories of the block types and insert it in the chart using drag-and-drop. Release the mouse button at the required point.

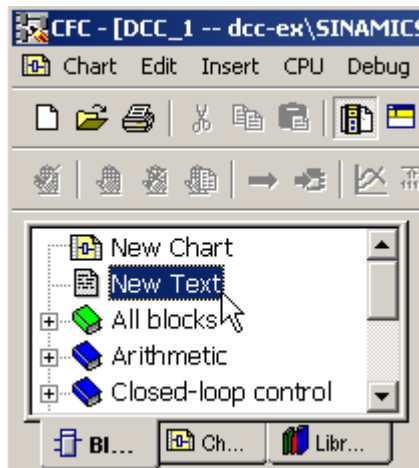


Figure 2-15 Inserting a text field

Alternatively, you can right-click at the desired position in the chart and select the **Insert new text** command in the context menu. Note that this option is only available offline.

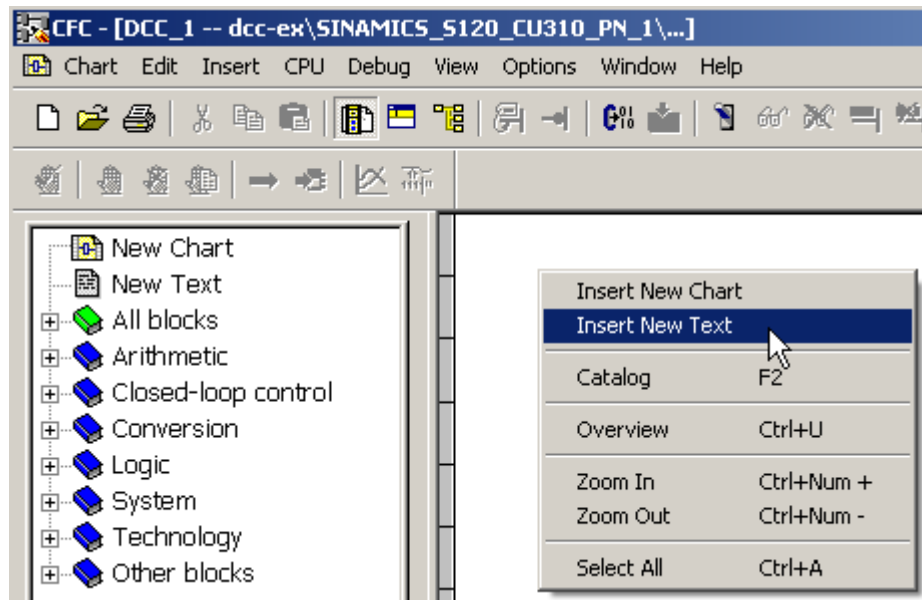


Figure 2-16 Inserting new text

You can adapt the size of the text field to your requirements by selecting the black points at the corners and the sides and dragging them to the desired size.

You can change your comments by clicking the text field and then entering or editing your text.

2.5.4 Specifying execution properties

You can display or change the execution properties of all the used blocks of the program. You can display the properties in the toolbar via **Edit > Execution sequence** or via the  button.

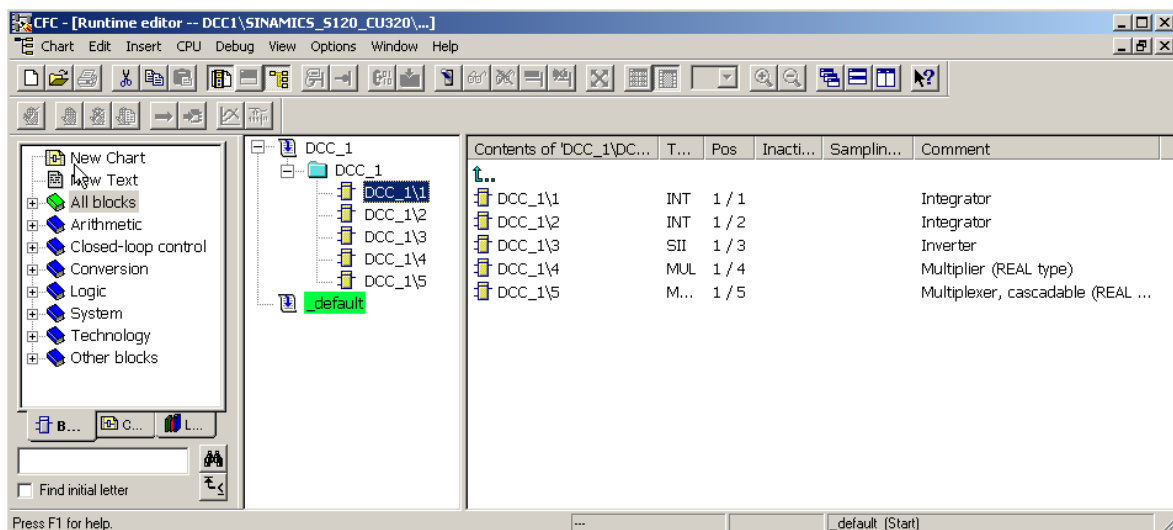


Figure 2-17 Execution editor of the dccReg1 chart with the execution groups Tsg_dccReg1 and Tsg2

In this window, you can also change the insert point in the execution sequence by dragging the block to the desired position. The assignment of a block to an execution group can be changed in the same way.

New inserted blocks are always placed in the execution system behind the block that is defined as predecessor. By default, this is always the block that has been inserted last. If a block is to be inserted in the execution sequence behind an already existing block, select the block with the desired offset in the overview, right-click and select the **Predecessor for insert point** function in the menu. The block now defined as predecessor block is displayed in light green in the DCC editor.

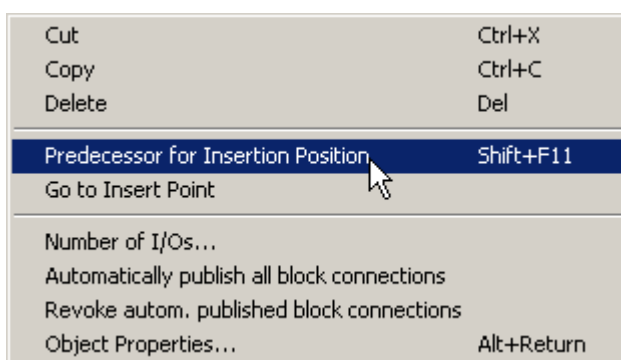


Figure 2-18 Block context menu: Defining predecessor for insert point

The active execution position is displayed in the status bar. This is at the bottom right in the execution editor.

2.5.5 Editing block connections

2.5.5.1 General

There are two types of block connection (inputs and outputs), each of which has a distinct function and is edited in a particular way.

The following sections contain further information about handling the block connections in the DCC.

2.5.5.2 Block connection properties

You can double-click each of the individual connections to parameterize it. Alternatively, the **Properties - Connection** window can also be displayed as follows:

1. Select the desired connection.
2. Select **Object properties** in the context menu.
3. The **Properties - Connection** window appears.

However, it is easier to parameterize the inputs as follows:

- Double-click a block header. The **Properties - Block** window appears. You can also open the **Properties - Block** window via the **Object properties** context menu command of the block or via the **Edit > Object properties** menu command.
- Click the **Connections** tab. The parameters in fields with a gray background cannot be changed.
- Enter the required values in the table and click **OK** to close the dialog box.

Input values

At the block inputs, a value can be entered in the **Value** field of the properties dialog box. If the input is not interconnected, it always has the specified value. With interconnected blocks, the output value of the upstream block always applies in the initialization phase and in the first cycle.

Output values

At the block outputs, a value can be entered in the **Value** field of the properties dialog box. In the first cycle however, the specified value is overwritten by the calculated value.

Note

Special feature with hidden block connections

In the DCC editor, you can hide block connections to improve the clarity of the configured charts. However, the hidden block connections remain active in the DCC, so their values are still evaluated. With generic blocks, you can also reduce the number of block connections in the DCC editor. The hidden connections are assigned default values. However, the hidden block connections remain active in the target system, so their values are still evaluated.

It should also be noted that the inputs of these blocks must be interconnected consecutively, starting from the first connection.

Number of block inputs

The number of inputs could be increased for the AND, ADD, MAS, MIS, MUL, NAND, NOR, OR and XOR blocks from the standard library. However, the DCC editor can only evaluate four input signals per block, and therefore this is not permitted. If the demand is greater, the block must be called several times. The note above also applies to data-type-specific variants of the blocks listed above.

Note

The specified limits for the possible number of connections 2...120 are not valid; the above-mentioned limit of four input signals applies.

Pseudo comments

Comments at block connections that start with @ are pseudo comments; they influence the function of the block connection and serve as interfaces to the basic system.

For further information about the pseudo comments, see:

- @ variables (SIMOTION) (Page 149)
- Creating customer-specific parameters ("declare") (Page 98)

Block connection units

The block connection units that can be set in the Properties dialog box serve only as comments in the DCC editor - the values are not used for automatic conversions.

Configuring the block display

You can change the display of the blocks. You can change the block width via **Options > Settings > Block / sheet bar width.....**

You can change the names of the block inputs/outputs via **Options > Settings > Display** in the submenu **Connections**.

The block type can be displayed in the form of both text and graphics. This can be configured via **Options > Settings > Display** in the submenu **Block headers**.

If you want to display more than the first eight characters of the comment, then select **Options > Settings > Block / sheet bar width**. Then in the **Blocks / Sheet Bars** window, set the block width to **Wide**. The first twelve characters of the comment are now displayed at the connection.

2.5.6 Interconnecting blocks

Blocks can be interconnected with one another. The outputs of a block then form the inputs for further blocks.

Requirement

The inputs and outputs of the blocks must possess compatible data types so that they can be interconnected. An overview of which data types can be interconnected is shown below:

Table 2- 2 Conversions

Input	Output	Description
WORD	INT	Interconnection of a word variable to an integer variable
INT	WORD	Interconnection of an integer variable to a word variable
DWORD	DINT	Interconnection of a double word variable to a double integer variable
DINT	DWORD	Interconnection of a double integer variable to a double word variable
BYTE	SINT	Interconnection of a byte variable to a short integer variable
SINT	BYTE	Interconnection of a short integer variable to a byte variable
USINT	BYTE	Interconnection of an unsigned short integer variable to a byte variable
BYTE	USINT	Interconnection of a byte variable to an unsigned short integer variable
USINT	SINT	Interconnection of an unsigned short integer variable to a short integer variable
SINT	USINT	Interconnection of a short integer variable to an unsigned short integer variable
UINT	WORD	Interconnection of an unsigned integer variable to a word variable
WORD	UINT	Interconnection of a word variable to an unsigned integer variable
UINT	INT	Interconnection of an unsigned integer variable to an integer variable
INT	UINT	Interconnection of an integer variable to an unsigned integer variable
UDINT	DWORD	Interconnection of an unsigned double integer variable to a double word variable
DWORD	UDINT	Interconnection of a double word variable to an unsigned double integer variable
UDINT	DINT	Interconnection of an unsigned double integer variable to a double integer variable
DINT	UDINT	Interconnection of a double integer variable to an unsigned double integer variable
SDTIME	REAL	Interconnection of an SDTime variable to a real variable

Procedure

Connect the output of the first block (source) with the input of the second block (sink). You can create this connection either by using drag-and-drop or by latching them together (clicking the relevant input connection and output connection once).

The connection line is automatically drawn from the output of the first block to the input of the second block.

2.5.7 Data type abbreviation in the DCC for connection and transformer blocks

Table 2- 3 Table of data types

Abbreviation	Keyword	Name	Bits
BO	BOOL	Logical number	8
BY	BYTE	Sequence of 8 bits	8
DI	DINT	Double integer	32
DW	DWORD	Sequence of 32 bits	32
I	INT	Integer	16
PC	LREAL	Double floating-point number Accuracy according to IEEE754	64
R	REAL	Floating-point number	32
SI	SINT	Signed short integer	8
TS	SDTIME	The SDTIME data type is derived from the REAL data type; 1.0 corresponds to 1.0 ms Negative values are not defined.	32
UD	UDINT	Unsigned double integer	32
UI	UINT	Unsigned integer	16
US	USINT	Unsigned short integer	8
W	WORD	Sequence of 16 bits	16

2.5.8 Interconnection to chart connections

A chart can be encapsulated for further use, i.e. chart connections added. Which block connections are provided at the chart connections can also be specified individually.

Procedure

1. Model your DCC in the DCC editor.
2. Select the Chart Connections window via **View -> Chart connections** or with the  button. The chart connections (IN, OUT, IN_OUT) are shown in this window.

Note

Please note that chart connections of type IN_OUT are not permitted in DCC!

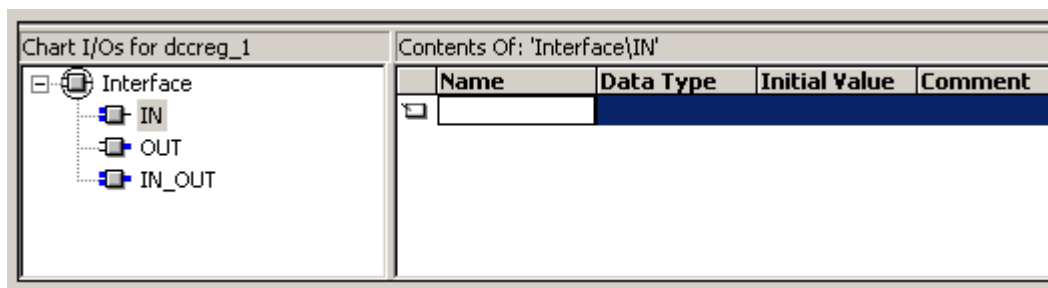


Figure 2-19 Screenshot showing chart connections area of the window

3. Define the required chart connections.
4. Interconnect the block connection to the chart connection. There are three ways of doing this:
 - Option 1:
Drag the block connection using drag-and-drop to IN (for inputs) or to OUT (for outputs) and keep the Ctrl key pressed. A chart connection is now created automatically.
 - Option 2:
Select the interface type (IN or OUT) from the left pane and then drag the desired block connection into the last (empty) line in the right pane using drag-and-drop.
 - Option 3:
Right-click the block connection to be interconnected and select **Interconnection to the chart interface** in the context menu. Then select the appropriate chart connection in the **Insert/Change Interconnection to the Chart Interface** window. Click OK to close the window.

Note

The interconnection to the chart connection by the context menu is available only when at least one chart connection exists already. The first interconnection to the chart connection functions only with drag&drop.

The block connection is interconnected to the chart connection. The assignment is displayed in the variables sheet bar of the DCC editor. The block interface is defined as part of assigning block connections to the interface.

Note

DCC SINAMICS: The use of chart connections is only permitted for subcharts.

2.5.9 Interconnection to global operands in DCC SIMOTION

Global operands are connection partners located outside of the DCCs.

Interconnections to global operands are entered in the sheet bar.

Where DCC is concerned, the interconnection of global operands serves as an interface to the basic system. You use this function to connect to ST/IO/system variables (in the case of DCC SIMOTION) or BICO parameters (in the case of DCC SINAMICS) in the drive.

Procedure

You can make an interconnection to a global operand as follows:

1. Open the DCC.
2. Right-click the block connection to be interconnected and select **Interconnection to operand** in the context menu.
3. Now select the global operand to be interconnected in the **DCC Signal Selection** window.
4. Click **OK** to close the window.

The block connection is interconnected to the selected global operand.

2.5.10 Deleting blocks

If you want to delete a block from the chart, select it and click **Edit > Delete**. You can also delete blocks using the context menu.

When blocks are deleted, the connections to the block connections are also removed. Output interconnections must first be deleted manually.

If you delete a block in online mode on which the outputs are interconnected with inputs of other blocks, the current signal values will become valid at the inputs of these blocks. These are taken over in the DCC and also saved to the card in the target system at the next RAM to ROM. In offline mode, the default values take effect again at the inputs after deletion of the upstream block.

If you delete a block on which the inputs or outputs are interconnected to chart connections, the connections are deleted, but not the chart connections. If these are not required, you must delete them separately in the Chart connections field.

2.6 Compiling

2.6.1 Consistency check without code generation

General

Some of the conditions that have to be met if a valid configuration is to be created from DCCs can only be checked once charts have been created.

This procedure is carried out automatically at certain points, e.g. when a project or charts are compiled.

Performing a consistency check

The contents of DCCs can be checked at any time.

To check consistency, click **Chart > Check consistency > Charts as program...**

The **Logs** dialog box is displayed automatically after the consistency check. Here, errors are indicated by an "E" and warnings by a "W".

Error log

You can also display the result of the consistency check via **Options > Logs** in the **Consistency check** tab.

Note

If charts are deleted from previously compiled and downloaded projects, this can lead to inconsistencies during the next compilation/download, as the overall chart execution system has to be adjusted.

Note

Cross-chart consistency checks are performed only after the compilation of the charts. The consistency check from the DCC chart always applies to the last compilation result.

2.6.2 Compiling the DCC in the DCC editor

Compiling

Please note that before the first compilation of a chart in a project, the project must first be saved in STARTER/SIMOTION SCOUT (via the menu command **Project** -> **Save** or with the  button).

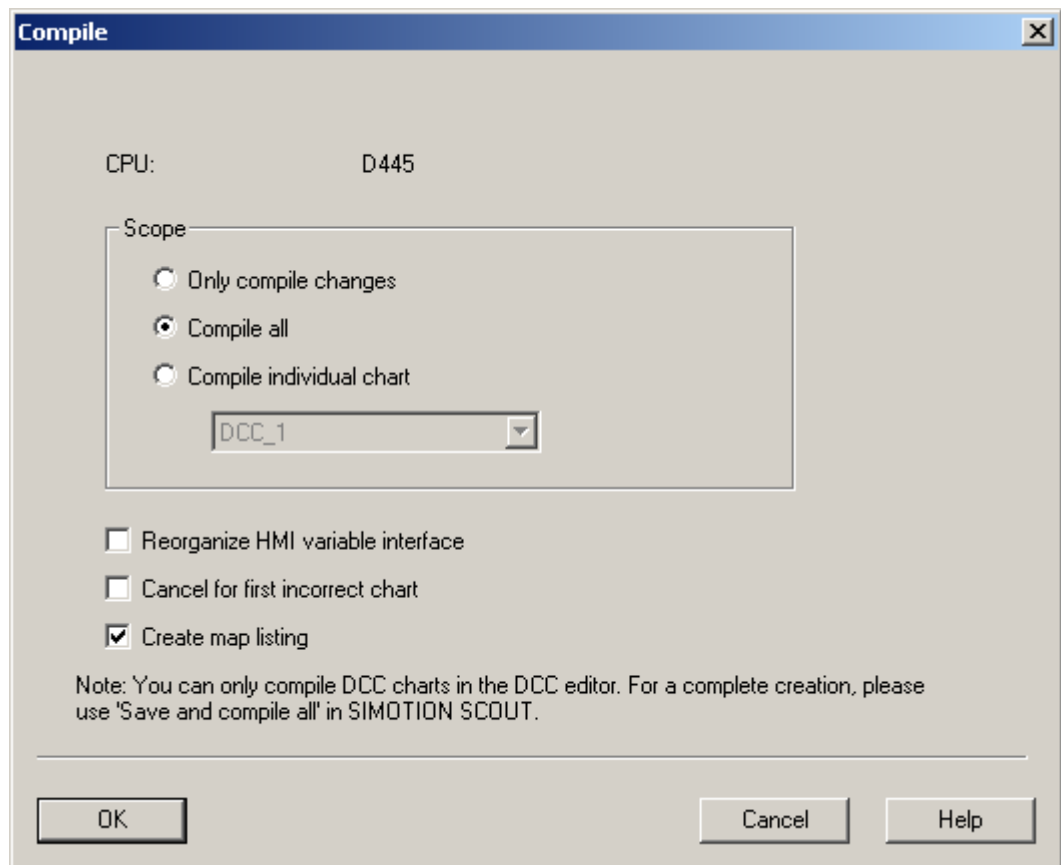


Figure 2-20 SIMOTION: Compiling a DCC

You can begin compiling with **Chart > Compile > Charts as program ...** or with the  button.

Compilation options

You can select the scope of the compilation:

- Scope: **Only compile changes**
Only those parts of the configuration that have been changed since the last compilation are recompiled. When recompiling, this option reduces compilation time.

Note

All the configuration charts will be checked for consistency, even if you select the **Only compile changes** option.

- Scope: **Compile all**
The entire contents of the configuration are compiled, regardless of whether the configuration has been changed since the last compilation.
- Scope: **Compile individual chart**
The selected chart is compiled, regardless of whether the configuration has been changed since the last compilation.

Additionally, you can select whether a map listing is to be generated and whether it should be canceled in the event of an error occurring in a chart. A map listing is a list of the global objects, the cross reference and the blocks used in the chart.

The map listing is saved in subfolder U7\debug in the project path.

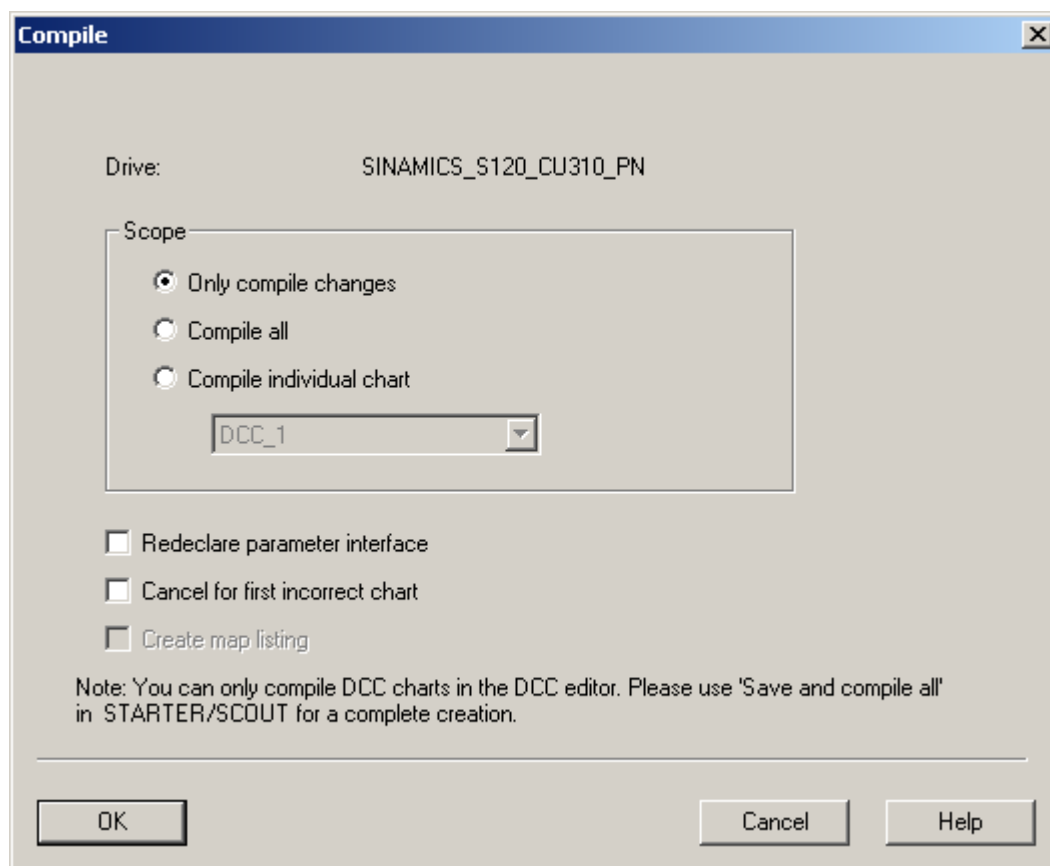


Figure 2-21 SINAMICS: Compiling a DCC

2.6.3 Error log

On completion of the compilation procedure, a detailed compilation log appears. Here, errors are indicated by an "E" and warnings by a "W".

- To navigate to the block that caused the error, select the error line in the log and click **Go to**, or double-click the error line.
- The log can be displayed again at a later point via the menu **Options > Logs** and, if necessary, can also be printed out.

2.7 Editing configurations further

2.7.1 Editing programs further: overview

This section describes more options for editing an existing program.

The following subjects are covered:

- Changing the block library
- Copy and paste charts or block groups between drive devices

2.7.2 Changing the block library

If you want to transfer an existing configuration to a new SCOUT/STARTER version, you can change the library version at a later point. The blocks will be adapted accordingly.

Procedure

1. Open a chart from the relevant configuration and select **Options > Block types** in the menu.
2. The **Import** window is displayed. Click **OK** to close the window. In the window **Import DCB Libraries**, the block libraries installed on your programming device are listed under **Libraries installed in SCOUT/STARTER**. Under **Libraries imported in the chart**, all libraries which have already been imported into this configuration are listed.
3. Check which previously imported libraries have a new version installed, by inspecting the version of the library under **Libraries installed in SCOUT/STARTER** and **Libraries imported in the chart**.
4. Select the library to be updated under **Libraries installed in SCOUT/STARTER** and click **>>**.
5. Click **Accept**.
6. The progress of the update is displayed in a window.

Response

The changes made compared with the old version are transferred to your existing configuration.

2.7.3 Copying of charts or chart sections

In the STARTER and SIMOTION SCOUT engineering systems, charts can be copied within a drive object (SINAMICS) or devices (SIMOTION), between various SIMOTION devices or SINAMICS devices as well as between charts. It is also possible to select block groups within a DCC and insert them in other charts of the same device family using copy and paste.

Note

The copying of charts or block groups between SINAMICS devices and SIMOTION devices and vice versa is not supported.

Copying a chart in SIMOTION SCOUT

To copy a chart in SIMOTION SCOUT, proceed as follows:

1. Open the SIMOTION SCOUT engineering system.
2. Select the source device from the project view.
3. Open the PROGRAMS subitem of the device.
4. Select an existing DCC and select the **Copy** command in the context menu of the chart.
5. Select the target device from the project view.
6. Open the PROGRAMS subitem of the device.
7. Select the **Paste** command in the context menu of the PROGRAMS subitem of the device.

The chart has been copied from the source device to the target device.

Copying a chart in STARTER

To copy a chart in STARTER, proceed as follows:

1. Open the STARTER engineering system.
2. Select the source drive unit in the project overview.
3. Open, for example, the Control_Unit subitem of the device.
4. Select an existing DCC and select the **Copy** command in the context menu of the chart.
5. Select the target drive unit in the project overview.
6. Open, for example, the Control_Unit subitem of the device.
7. Select the **Paste** command in the context menu of the Control_Unit subitem of the drive unit.

The chart has been copied from the source drive unit to the target drive unit.



WARNING

After copying, you must check whether the interconnections of published block connections in the chart copy have to be adapted to another drive object as a result of the copying. During copying, no automatic adaptation of the interconnections of the chart copy is made to the basic system or to other DCCs.

Overall chart interconnections are converted into text interconnections when copied.

Note

Only one DCC per drive object may be created in the STARTER engineering system.

Note

If charts are to be copied from one project to another, then STARTER or SCOUT must be started twice as only one project can be opened in each. Please note that, outside the engineering system, you can only copy and not cut.

Note

When copying a device, the associated charts are also copied.

An XML export or XML import of individual charts is not possible, but the associated DCCs are contained in the project export.

See Copying charts between projects (Page 104).

Copying block groups in the DCC editor

In the DCC editor, you can copy parts from one chart and insert them in another chart. To do this, you must open the source chart and the target chart in the DCC editor.

1. Use the lasso function to select the subsection of the source chart to be copied and select the **Edit > Copy** command in the menu bar.
2. Change to the target chart.
3. Select the **Edit > Paste** command in the menu bar.

The block group has been inserted in the target chart.

Block numbering

When copying blocks or block groups in the DCC editor, the name of the new block is formed as follows: if numeric characters are present at the end of the name, these are all deleted up to the first non-numeric character and replaced by the next free numeric character. It is therefore recommendable to allocate names consisting of letters (and numeric characters) such as block, block1, block2, etc.

Location of the insertion in the execution sequence

Blocks are always inserted into the execution sequence behind the selected block and the most recently added block is always selected automatically.

If a block is to be inserted in the execution sequence behind an existing block, select the block in the overview or in the chart and right-click the function **Predecessor for insert point** in the menu.

You can change the execution sequence at any time via the **Edit > Execution sequence** menu command or the  button.

See also Specify execution properties (Page 40)

2.7.4 Search in the project from STARTER/SCOUT

You can use the sheet bar to search for variables and SINAMICS parameters in DCC charts in the open project. The contents of alias definitions can also be found using the search function.

Open the dialog via the menu **Edit > Search in project** or using the shortcut **Ctrl + Shift + F**.

The results are displayed in the search results tab of the detail view.

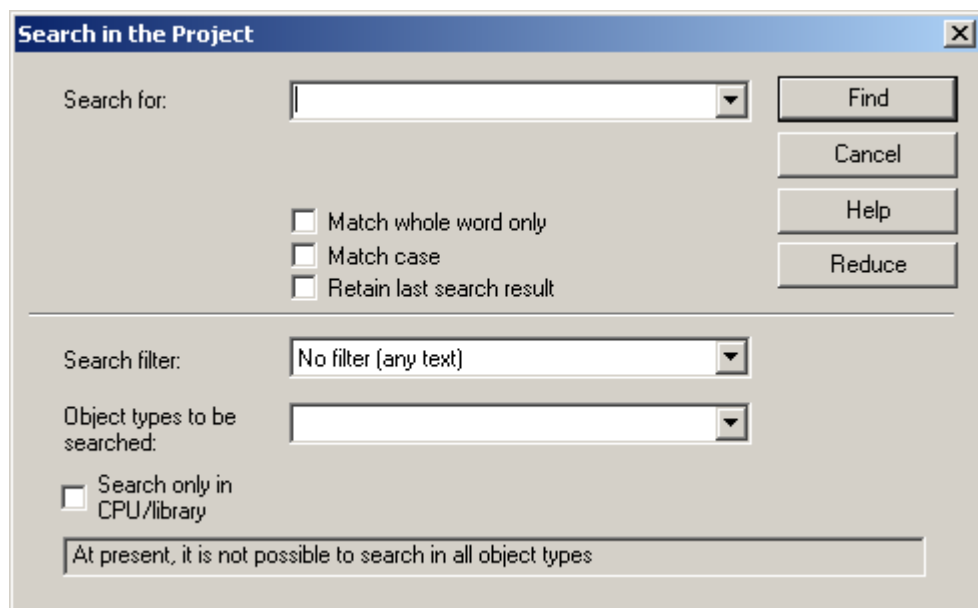


Figure 2-22 Searching in the project

2.7.5 Replace in the project from STARTER/SCOUT

The **Replace in project** function is based on the **Search in project** function.

You can use the **Replace in project** function to quickly adjust the interconnections to the system after copying and inserting DCCs.

Open the dialog via the menu **Edit > Replace in project** or using the shortcut **Ctrl + Shift + G**.

When you carry out a replacement, both the results found and the replacement term are displayed in the **Search result** tab of the **detail view**. The text can be edited again here.

Use the **Replace** button to replace all search results selected using the check box.

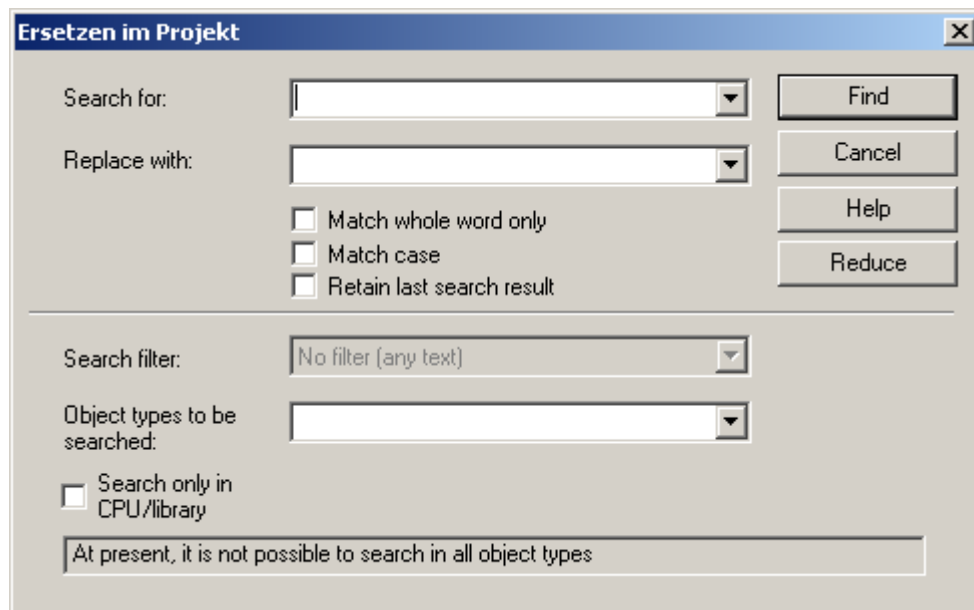


Figure 2-23 Replacing in the project

Note

Symbols that are defined from the DCC, so-called @ parameters, DCC parameters (SINAMICS), links to BICO parameters are not considered when searching/replacing.

Boundary conditions

- As of DCC 2.0.2, the symbols (variable in sheet bar) and alias definitions used in the DCC can also be searched for and replaced.
- The CFC editor must be closed during the search/replace operation.
- If the CFC chart sources have been deleted or the charts have know-how protection, searching/replacing of the sheet bar elements is not possible.
- It is not possible to undo these changes.

2.8 Test mode

2.8.1 Test modes

There are two types of test mode: .

- **Process mode**
Select this test mode if you want to monitor the behavior of individual instances, e.g. for the error analysis. When test mode is activated, all blocks are set to the status **Monitoring off**. In this test mode, you must select the relevant block connections and explicitly log them on for monitoring.
- **Laboratory mode**
Laboratory mode is used for convenient, efficient testing and commissioning. When test mode is activated, all blocks are set to the status **Monitoring on**.

You can select the desired test mode in edit mode using the menu commands in the **Test** menu. Once a test mode has been selected it is not possible to switch between them.

Test settings

You can specify the **Monitoring cycle** via the menu **Test > Test settings**. This displays the **Test Settings** window.

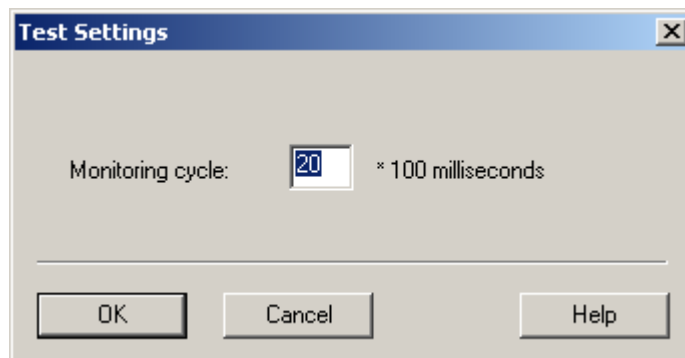


Figure 2-24 Test settings

In this window, you can set the **monitoring cycle** for the objects that have been logged on for testing, i.e. the cycle time for updating these objects. The monitoring cycles can be set in 1 to 100 steps each of 100 milliseconds (SIMOTION). Only whole seconds can be set as monitoring time for SINAMICS.

Note

If you have logged on many objects for the test, it is better to use a slower updating cycle.

2.8.2 Monitoring in laboratory mode

When test mode is activated, the **Monitoring on** function, or , will also be activated for DCC charts in **Laboratory mode**.

This means that, in test mode, you can also display the development of the values for those block connections that are logged on for display, i.e. the values will be read out and displayed cyclically. You can change the options relating to this dynamic value display and the connection parameters in test mode.

Saving settings

The logging on of blocks for monitoring is rejected when the online test is exited. The settings of the block interfaces with regard to the test are saved in the project.

Display of values during the test

Current values of block interfaces are displayed when they are logged on for the test and their block is logged on for monitoring.

2.8.3 Monitoring in process mode

In **process mode**, the **Monitoring off** function, or , is activated. This means that you must first select the blocks that you want to monitor (by highlighting them in the chart) before dynamic display can take place. You then need to execute the menu command **Monitoring on**.

Conversely, you can exclude individual blocks from being monitored if there are too many (highlight the block and execute the menu command **Monitoring off**).

NOTICE
If the monitoring function has been deactivated and you highlight a connection in order to log it on for testing via , monitoring is activated for this connection and for all previously logged-on connections of this block.

By activating test mode, connections are also established with the CPU for all connections listed within a value display window. Activate **monitoring** by selecting the individual connections in the **Monitoring** column.

2.8.4 Logging on/logging off connections for testing

In **edit mode** or **test mode** (process or laboratory mode), you can log on individual block or chart connections for testing:

- Highlight the connection and select **Test > Log on connections** or click the  button in the toolbar.

When this is performed in **test mode**, monitoring is also activated, i.e. the connection is displayed with a yellow background and with its current value. If monitoring has been deactivated for this block, it will also be activated for all connections that were logged on for testing prior to this.

In **edit mode** or **test mode** (process or laboratory mode), you can log off individual block or chart connections for testing:

- Select **Test > Log off connections** or click the  button in the toolbar.

Note

Alternatively, it is also possible to log on/log off all connections of a block for testing in edit mode by highlighting a block, selecting **Object properties** via the context menu, and selecting or deselecting the individual connections in the **For testing** column of the **Connections** tab.

2.8.5 Activating/deactivating connection monitoring

You can activate/deactivate monitoring (displaying the current values at the connection) of the block connections logged on for testing:

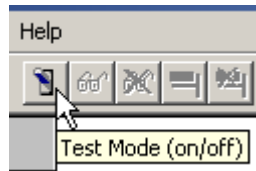
- Automatically, by activating test mode in laboratory mode.
- Via **Test > Monitoring on** or via the  button in the toolbar. In both **laboratory mode** and **process mode**, this function concerns only the blocks previously selected in the chart.
- You can deactivate monitoring (meaning that values at the connections will no longer be updated) via **Test > Monitoring off** or by using the  button. In both **laboratory mode** and **process mode**, this function concerns only the blocks previously selected in the chart.

All input and output values that have been activated for monitoring purposes are updated in accordance with the set monitoring cycle.

2.8.6 Activating test mode

To activate test mode, proceed as follows:

- Click the **Test mode** button in the toolbar



or click **Test > Test mode**.

Test mode is activated. The menu item is identified by a check mark. Depending on the selected test operating mode, the text **Test: RUN(laboratory mode)** or **Test: RUN(process mode)** is displayed in the status bar with a green background. Any menu functions that are not permitted in test mode are displayed as deactivated (grayed out).

Note

If the DCCs are different in the editor and in the target device, then they can behave differently, see Consistency of charts in test mode (Page 68).

Requirements for starting test mode

The use of test mode requires an online connection to the device.

As long as the DCCs are identical online and offline, test mode is activated immediately. If the DCCs do not have the same version in the RT system and in the engineering system, values can still be monitored in test mode, but in order to make further changes the consistency must first be restored by uploading the changes from the target system (in this case the changes are not visible in the editor) or by downloading the current version from the engineering system.

If you have made changes to connections or block types in the DCC, you will be informed that you can still monitor and trace values when test mode starts.

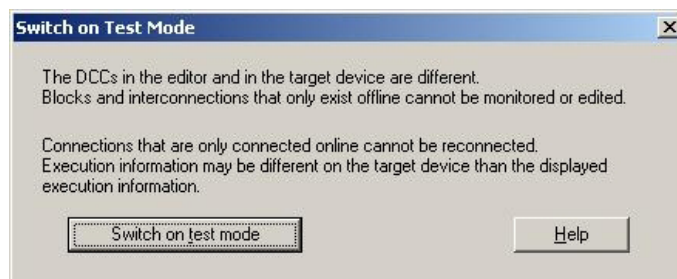


Figure 2-25 Test mode - changes to connections or block types

Online changes can only be performed after the DCC has been recompiled and downloaded to the target system. You are informed about this fact when starting test mode and can return to editing mode or activate test mode for monitoring purposes - as shown in the following dialog box.

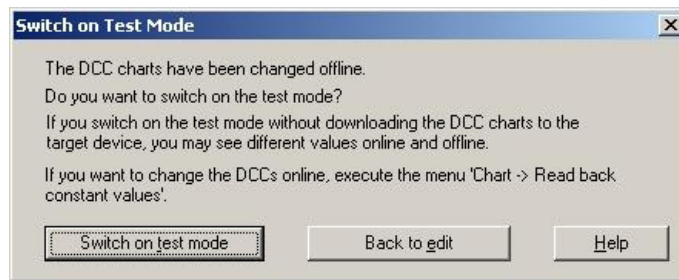


Figure 2-26 Test mode - loading changes to the target device

2.8.7 Monitoring test mode

In test mode, the values of the connections that have been logged on for monitoring are displayed with a yellow background.

Logging on connections for testing

You must explicitly log on to test the connections to be tested. For SIMOTION, the log on to the test can be performed either using the **Log on connection** button in the menu bar or using the context menu of the block connection. Select the **Log on connection** item in the context menu of the connection. For SINAMICS, go to the **Object properties** of the block connection (right-click) and activate the **For test** property. You can also log on the connections directly for the test by clicking with pressed Ctrl key.

Changing input values

You can also change all values of non-interconnected inputs in test mode. To show how changing a value affects execution, assign a new value to an input as follows:

- Double-click the input to be monitored.
- The **Properties - Connection** window appears. Enter the new value for the block input and confirm with **OK**.
- In the chart, you can now see how the value changes on the associated output.

Note

The changed value is only visible when the connection has been logged on for testing.

Logging off connections from testing

You can log off connections logged on for testing. The log off from the test can be performed either using the **Log off connection** button in the menu bar or using the context menu of the block connection. Select the **Log off connection** item in the context menu of the connection. You can also log off the connections directly from the test by clicking with pressed Ctrl key.

2.8.8 Enabling the value and trend display during a test

In test mode, you can use the value and trend display to analyze the input and output values of blocks.

Enabling the value and trend display

You can open the value and trend display using the **View > Value display** and **View > Trend display** commands in the menu bar of the DCC editor. Each of these is an autonomous program window. The windows can be arranged using the minimize and maximize functions.

The block interfaces can now be added to the value or trend display using the **Insert in value display** and **Insert in trend display** commands in the context menu. Current values are displayed when the test mode is switched on.

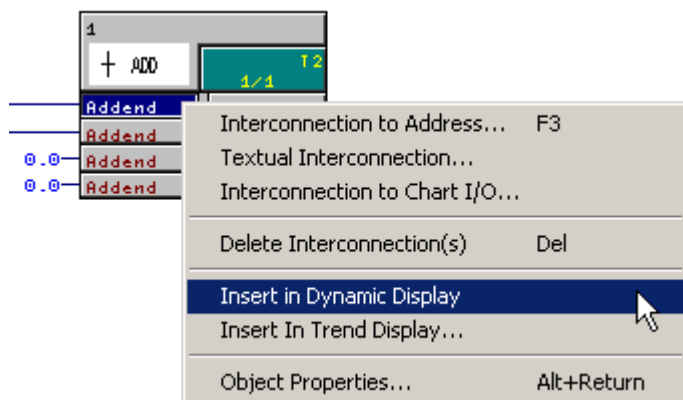


Figure 2-27 Inserting a block pin into the value or trend display

Settings in the Trend Display window

The **Trend Display** window displays the block connections added to the trend display in the form of curves. Each inserted block connection is called a **channel**, whereby a **lower limit** and an **upper limit** can be specified for each channel.

The number of sample values to be displayed on the time axis is specified in the **Display** area of the **Trend Display** window.

The desired trace parameters can be set in the **Trace Parameters** window, which can be accessed via the **Change** button.

2.8.9 Editing DCCs in test mode

To a large extent, you can continue to edit your configurations during test mode. An overview of the changes that can be made during test mode is shown below:

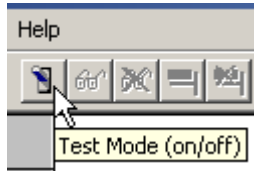
Table 2- 4 Further editing of configurations during test mode

Handling blocks (Page 36)	
Inserting blocks	Call the block catalog using the View > Catalog command. Open the block family and use drag-and-drop to move the selected block to the working area.
Deleting blocks	Select the block and use the Edit > Delete command to remove it. In DCC SINAMICS, blocks for which at least one connection has been published as a parameter cannot be deleted. The SAV, SAV_BY, SAV_D, SAV_I and STM blocks cannot be deleted or inserted online.
Moving blocks	Select a block and drag it to the desired position in the chart.
Renaming blocks	Select the block, right-click and select the Rename command. - The new block names apply only to the accesses to the input/outputs of the block using Java or Trace; the old names remain valid here. - For those blocks that use retain data, the retain data block on the device is also assigned the new name.
Interconnecting blocks (Page 44)	
Creating interconnections	In the DCC, select the block connections between which you want to establish a connection. Note that in DCC SINAMICS, block inputs published as BICO parameters can only be connected to block outputs published as BICO parameters or BICO outputs of the basic system. BICO inputs and BICO outputs of the basic system can still only be connected to block outputs and inputs published as BICO parameters in the DCC.
Deleting interconnections	In the DCC, select the block connection that you want to disconnect. Then select Edit > Delete to delete it.
Moving interconnections	Select the required interconnection and move it with drag&drop. When moving BICO interconnections, the signal can acquire the value 0 for a few cycles. The no longer interconnected input on the original connection is permanently assigned the value 0.
Changing the signal value of an input	Double-click the block input for which the value is to be changed. The "Properties - Connection" dialog box appears in which you can change the value.
Deleting global interconnections to the sheet bar	In the DCC, select the block connection that you want to disconnect. Then select Edit > Delete to delete it.
Comments (text)	
Inserting comments (text) in the chart	Select the New text command via the directories of the block types and insert it in the chart using drag-and-drop. Release the mouse button at the required point.
Changing comments (text) in the chart	You can move the text field in the DCC by selecting it and then dragging it to the desired position. You can change your comments by double-clicking the text field and then entering or editing your text.

2.8.10 Deactivating test mode

Deactivate test mode to return to edit mode.

- Click the **Test mode** button in the toolbar.



or click **Test > Test mode**.

Edit mode is reactivated.

2.8.11 Changing online during test mode

2.8.11.1 Preliminary remark

As a general rule, test mode is used for making online changes to values, interconnections and block instances without stopping the system.

It is not necessary to recompile after changes have been made online.

However, they must be saved with **Copy RAM to ROM** before Power Off , otherwise they are lost. This is especially important with regard to SAV blocks.

Note

Please note that online changes can only be made in test mode.

2.8.11.2 Changing values at block inputs online

Requirement

Only signal values at block inputs that are not interconnected can be changed online.

Procedure

The **Properties - Connection** window is opened by double-clicking the desired block connection. A new numerical value can now be entered for the block input in the **Value** line. The new value takes effect and is displayed in the chart when **Accept** is clicked. The window is closed by clicking **OK**.

Note

DCC-SINAMICS 2.5: Only signal values of block inputs can be changed online that have not been published as BICO parameters.

From SINAMICS 2.6. onward, all non-interconnected block inputs can be changed in test mode. They do not have to be published.

Note

DCC SIMOTION: The signal value at the block input can be changed online, but with active execution group is overwritten in the next cycle. Whether the signal value at the block input can be changed online does not depend on whether it has been declared as an HMI variable or not.

2.8.11.3 Deleting an interconnection online

Procedure

In the DCC, select the block connection that you want to disconnect. Then remove this with **Edit > Delete** or with the Del key.

Result

The connecting line between the connections disappears and the last value that was transferred on the connection appears as input value at the connection.

Note

DCC SIMOTION: Interconnections to chart connections cannot be deleted online!

2.8.11.4 Establishing an interconnection online

Procedure

In the DCC chart, select the block connection from which you want to establish an interconnection and drag an interconnection to the block connection to which the interconnection is to be established.

Result

The connecting line between the selected connections is established and the current value that has just been transferred appears at the output.

Note

Up to and including DCC-SINAMICS 2.5, a block input published as a BICO parameter may only be interconnected with outputs published as BICO parameters or with BICO outputs of the basic system.

Block connections cannot be published in test mode, i.e. new @ parameters inserted.

Note

DCC SIMOTION: Connections to published block inputs in the DCC or global operands cannot be recreated online.

2.8.11.5 Moving interconnections online

Procedure

Select the required interconnection and move it with drag&drop.

Note

When moving BICO interconnections, the signal can acquire the value 0 for a few cycles.

The no longer interconnected input on the original connection is permanently assigned the value 0.

2.8.11.6 Inserting a block online

Procedure

Call the block catalog using the **View > Catalog** command. Open the block family and use drag-and-drop to move the selected block to the working area.

The block instance is calculated with the next cycle.

The inserted block instance is assigned a standard name that can subsequently be changed online: Select the block, right-click and select **Rename**. Recompilation and loading is only required if the trend display, trace or monitoring of block connections has been activated.

Note

Whether a block can be added or deleted online is described under "Configuration data" in the online help for blocks.

Note

Renaming blocks

When blocks are renamed in online mode, a new block is created that replaces the existing block. This is then recalculated. The initialization values are newly set for the block.

Online renaming allows meaningful names to be assigned also for the online inserting of module instances.

2.8.11.7 Deleting a block online

Procedure

First delete all output interconnections by selecting the connection and then selecting **Edit > Delete** in the editor menu or with the Del key. Then select the block and remove it with the Del key or the **Edit > Delete** command.

Note

In SINAMICS, blocks for which at least one connection has been published as a parameter cannot be deleted.

Note

DCC SIMOTION: Blocks with interconnections to chart connections cannot be deleted online.

2.8.11.8 Inserting comments in the chart

Procedure

Select the **New text** command via the directories of the block types and insert it in the chart using drag-and-drop. Release the mouse button at the required point.

2.8.11.9 Changing comments in the chart

Procedure

You can change your comments by clicking the text field and then entering or editing your text.

You can adapt the size of the text field to your requirements by selecting the black points at the corners and the sides and dragging them to the desired size.

2.8.11.10 Moving blocks in the chart

Procedure

Select the block and drag it to a suitable free position in the chart.

2.8.11.11 Consistency of the charts in test mode

As long as the DCCs are identical in the DCC editor and in the target system, DCCs can be monitored and changed in test mode.

If the DCCs are different in the engineering system and in the target system, changes can still be made online as long as the DCCs are not compiled.

If the charts are different in the editor and the target device and the charts have not yet been compiled, the following dialog box appears:

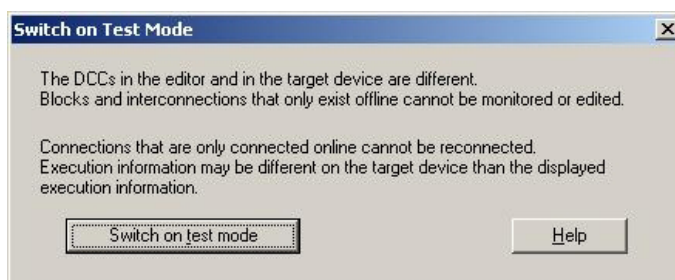


Figure 2-28 Activating test mode with inconsistencies

Click "Switch on test mode" to confirm the dialog.

If only the constant values of the chart are different in the CFC and in the target system, the user can recompile the DCC chart and transfer the changes from the target system to the CFC with **Chart > Read back constant values**. Monitoring, tracing and further online changes can then be performed in test mode.

If the DCCs differ online and offline and the DCCs have already been compiled, a prompt appears that the DCCs are different online and offline.

- If only the constant values of the charts are different, you can restore the consistency using the menu command **Chart > Read back constant values**.

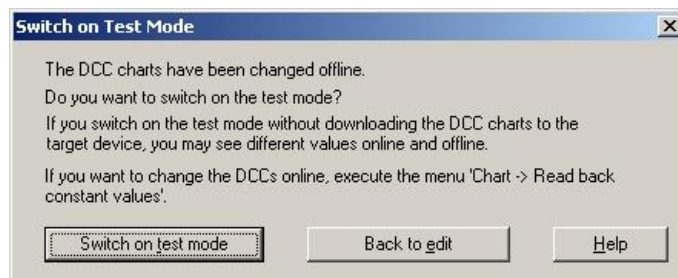


Figure 2-29 Activate test mode - Read-back constant values

- If the DCC charts differ online and offline and they have already been compiled, a prompt will inform you of this differentiation. Download the changes into the target system so that monitoring and changes can be performed in test mode.

The download is performed when the target device is stopped. With SINAMICS, the download only functions in the "Power-on inhibit" and "Ready for power on" operating modes.

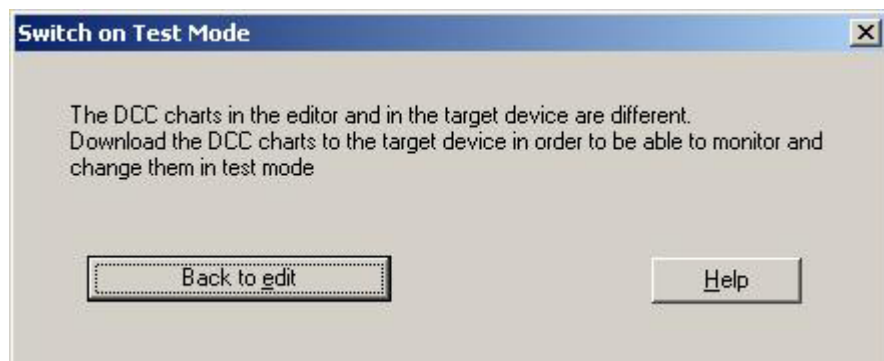


Figure 2-30 Activating test mode

2.9 Reference data

2.9.1 Chart reference data

As well as displaying the following information graphically in the DCC, you can use the "Chart reference data" (**Options > Chart reference data**) function to display it in the form of a list and print it out:

- Cross references of accesses to different objects
- Execution sequence
Graphic representation of the entire run sequence of a CPU.

You can use this to check your configuration structure.

You can display and print out the following lists of chart reference data:

- Operand cross references
This list displays all global operands used on the CPU along with the elements that access them.
- Execution group cross references
This list displays the existing accesses of any DCCs for all execution groups.
- Block type cross references
This list displays the block types used and the positions (on the DCC) where they are used.

Note

You can also generate the reference data of a DCC or a DCC library via the context menu with the **Reference data > Generate** command and then display it via **Reference data > Display**.

You can then always display the reference data again, but you only have to generate it once when first called or after changes to the @ variables.

2.9.2 List of block types

The list of block types shows where they are used. Unused block types can be deleted.

You can display the list by opening the **Chart Ref: Display Chart Reference Data** window via the **Options > Chart reference data** menu item. Now select the **View > Block types** menu item or use the  button in this window.

Column heading	Meaning
Block type	Block name (e.g. ADD)
Chart	Name of the DCC in which the block is used
Block	Name of the block instance (e.g. Integrator1)
Block comment	Block comment

Note

Double-clicking one of the displayed block types opens the associated chart and the selected block type is highlighted.

2.9.3 Cross References List Execution Groups (only for DCC-SIMOTION)

The list shows the existing accesses from arbitrary DCCs to the enable attribute of the execution group for all execution groups of the active CPU (the enable attribute can be used to switch individual execution groups on and off, see Enable attribute, execution groups (Page 174)).

You can display the list by opening the **Chart Ref: Display Chart Reference Data** window via the **Options > Chart reference data** menu item. Now select the **View > Chart element cross references > Execution group** menu item or use the  button in this window.

Column heading	Meaning
Execution group	Group name
(R/W)	Read (R) or write (W) access
Chart	DCC name
Chart element	Name of the interconnected block and connection
Element comment	Block comment
Type	Block type (e.g. ADD)

Note

The names selected in the execution system (i.e. the names of the execution groups and execution levels) for each device must be unique.

2.9.4 List of operand cross references

This is a list of the global operands used on the CPU or the drive unit, along with the block connections that they are connected to.

You can display this list when you display the **Plan Ref: Chart Reference Data** window from the **Options > Open chart reference data** menu item. Now select the **View > Chart element cross references > Operand** menu item or use the  button in this window.

Table 2- 5 Global operands

Column heading	Meaning
Symbol	Name of the global operand
Address	Blank
Data type	Keyword of the data type used
(R/W)	Read (R) or write (W) access
Chart	DCC name
Chart element	Name of the interconnected block and connection
Element comment	Block comment
Type	Block type (e.g. ADD)

2.10 Know-how Protection

2.10.1 Information on know-how Protection

The SIMOTION SCOUT/STARTER Engineering System provides know-how protection functions. This protects individual DCCs to prevent unauthorized access to chart information.

Setting the know-how protection function

A chart can be protected as follows:

1. Select the chart in the SIMOTION SCOUT/STARTER.
2. In the context menu of the DCC, execute the **Know-how protection > Set** function in offline mode.

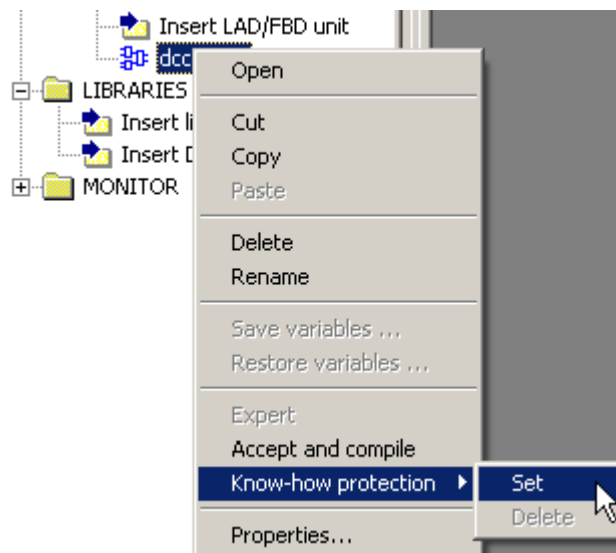


Figure 2-31 Setting know-how protection for a DCC

3. When using the know-how protection function for the first time, you will be prompted to enter the required access data (login & password) for protection of the chart in the **Know-How Manager** window. Close the **Know-How Manager** window via the **OK** button.

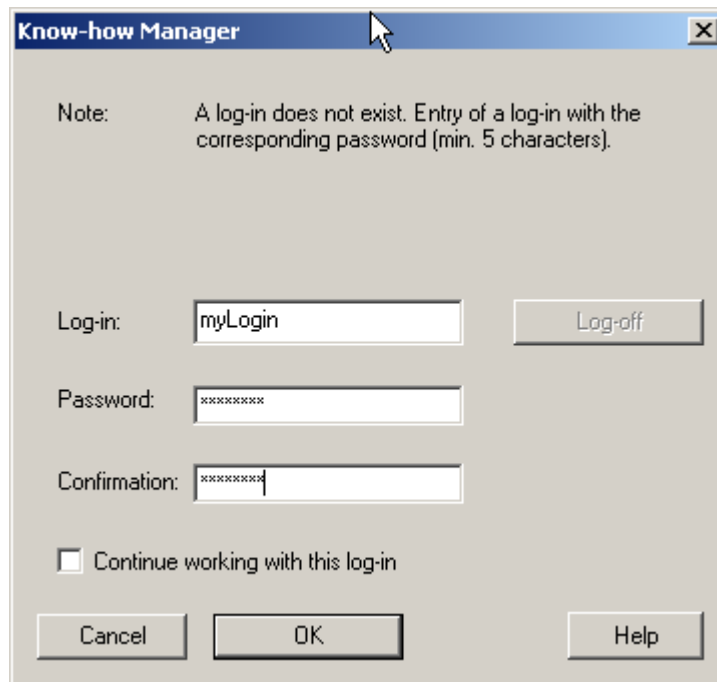


Figure 2-32 Know-how manager

The chart is now shown grayed out.

You can also define a general logon with password in the **Project -> Know-how protection** menu item, which then is used automatically for setting the know-how protection on the selected chart. Although you are then automatically logged on with this logon, you can at anytime log off from the **Project -> Know-how protection** menu item.

Activating the know-how protection function

To activate the know-how protection, you must log off the logged-on user in the **Know-how manager** window (access to the **Know-how manager** is via **Project > Know-how protection** in the menu). The protected DCCs of the opened project are then locked and cannot be opened. Only by entering the password can you remove the lock and open the DCCs.

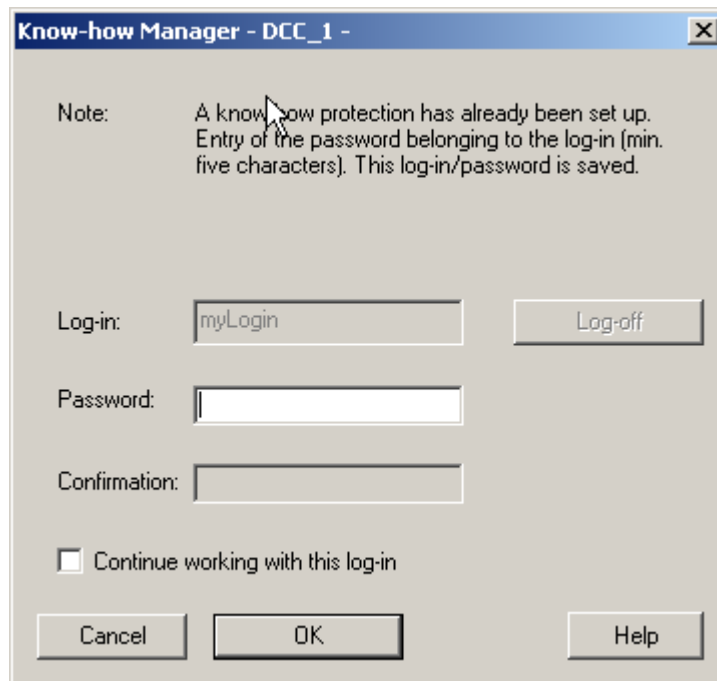


Figure 2-33 Know-how manager

Deleting the know-how protection function

To delete the know-how protection, you must first be logged on (if you are not already logged on, you are automatically prompted to do this during the procedure described below).

You can delete the know-how protection of a chart as follows:

1. Select the chart in the SIMOTION SCOUT/STARTER.
2. In the context menu of the DCC, execute **Know-how protection > Delete**.
3. Close the **Know-how manager** window via the **OK** button after entering the password.

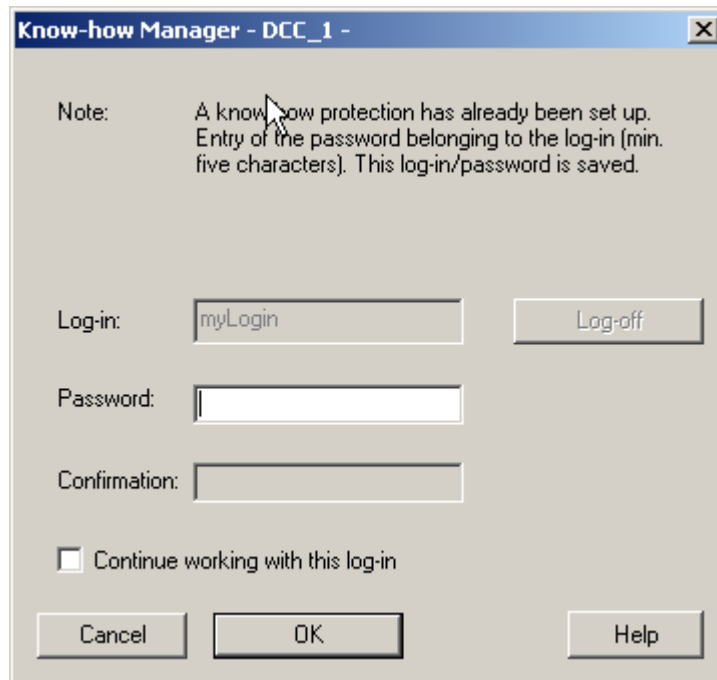


Figure 2-34 Know-how manager

Note

The know-how protection is not automatically reactivated in SIMOTION SCOUT/STARTER when you close the DCC.

Canceling the know-how protection

If you want to open a know-how-protected DCC, you can do this by double-clicking as usual. If you are already logged on, the know-how protection is automatically canceled, if you are not logged on, you are automatically prompted to do so when opening.

2.11 Startup behavior

With regard to the startup behavior, note that the respective initialization value is active at the block connections during startup and then the calculated value in each following cycle.

For more detailed information on the start-up behavior, please see Chapter Characteristics of block connections (Page 41).

2.12 Software upgrade and module exchange

Scout or Starter projects that contain DCC charts and are compiled with V4.1.1 or V4.1.2 can be opened and loaded with Scout / Starter V4.1.3, even without a DCC license.

This applies regardless of whether or not the DCC chart sources (i.e. original project) are available.

Note

This specification is applicable as long as the device version does not change. Following a replacement with a new device version, the DCB library versions present on the new device must be imported and the DCC recompiled. The corresponding DCC license (CFC for SIMOTION / SINAMICS) is required for this.

2.13 Version information

Proceed as follows to display the version information:

1. Left-click the *Help -> Info...* menu item in STARTER/SCOUT
2. Click the *System information...* button in the following *Information* dialog

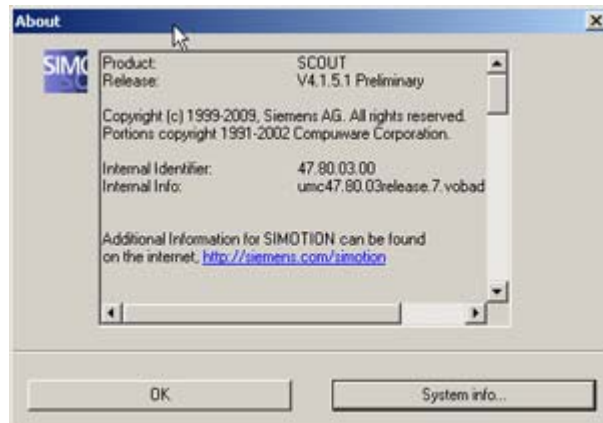


Figure 2-35 Version information - Information

3. The *System Information* dialog with the system information appears.

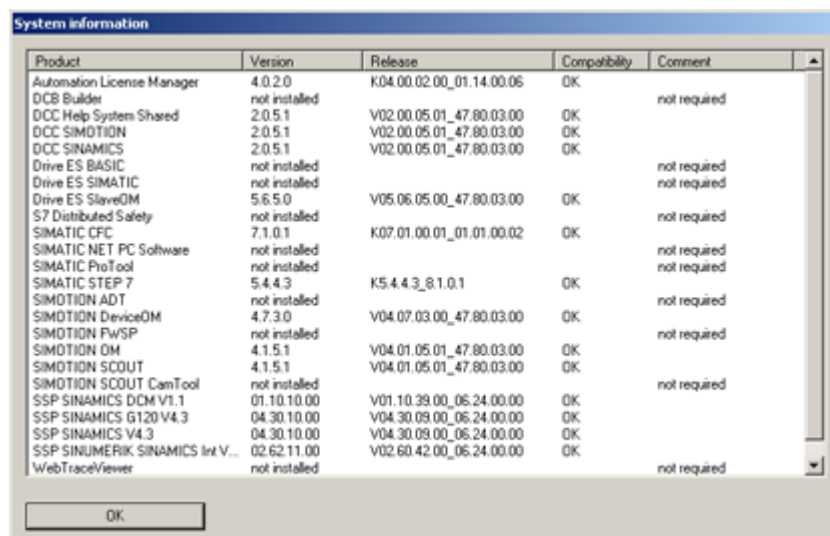


Figure 2-36 Version information - System information

DCC for SINAMICS

3.1 Overview

3.1.1 Introduction

This product brief is intended for first-time users who are not yet familiar with the DCC program package. Using a short example, you will find out how to create a new DCC, interconnect DCBs (drive control blocks; function blocks), compile the chart, download it to the target device and test it.

Note

In most cases, there are a number of options for working with the DCC editor (e.g. using the keyboard). In this example, one option is used. Apart from a few exceptions, no alternative methods of operation are explored here.

3.1.2 Software requirements

The software requirements for DCC are the same as those for SINAMICS STARTER.

You will additionally need:

- **DCC SINAMICS** software option package

Note

The DCC SINAMICS option package contains on a USB stick the Step7 CFC license required for the DCC editor. The installation is made using the Step7 Automation License Manager (refer to the help of the Step7 Automation License Manager). The Step7 Automation License Manager program is installed automatically together with STARTER.

When the product is supplied, the DCC technology option is not yet available on the drive unit. The DCC technology option must be loaded onto the drive unit's CF card in a separate operation using SIMOTION SCOUT or STARTER. The SINAMICS CU3x0, SINAMICS CUD or the SIMOTION D4xx then must be switched off and on once. Only then can the DCCs be downloaded and run on the drive objects.

There is only ever one DCC on a drive object. DCC can be activated simultaneously on several drive objects on a drive unit. DCC is not available on SINAMICS S110 drive units (CU305 module).

3.1.3 SINAMICS system integration

3.1.3.1 Applications and features

Application

A logic operation, which connects several states (e.g. access control, plant status) to a control signal (e.g. ON command), is required for controlling the drive system in a wide variety of applications.

As well as logic operations, a number of arithmetical operations / storing elements are increasingly becoming a requirement in drive systems.

Note

This additional functionality increases the computing time load. This means that the maximum possible configuration for a Control Unit is restricted.

Drive Control Chart (DCC) functionality is available for every drive object of the drive unit listed in the following table.

Table 3- 1 Drive object types for DCC SINAMICS

Drive object type	Object number (r0107)	Meaning
CU_S	1	Control Unit SINAMICS S (SINAMICS S120/S150)
CU_G	2	Control Unit SINAMICS G (SINAMICS G130/G150)
CU_I	3	Control Unit SINAMICS Integrated
CU_CX32	4	Controller Extension for boosting the computing performance
CU_GM	5	Control Unit SINAMICS GM
CU_DC	6	SINAMICS DC MASTER Control Unit
CU_GL	7	Control Unit SINAMICS GL
CU_SL	101	Control Unit SINAMICS SL
A_INF	10	Active infeed control
SERVO	11	Servo control
VECTOR	12	Vector control
VECTORMV	13	Vector control for SINAMICS GM
VECTORGL	14	Vector control for SINAMICS GL
VECTORSL	16	Vector control for SINAMICS SL
DC_CTRL	17	Closed-loop control for DC drives
S_INF	20	Smart infeed control
B_INF	30	Basic infeed control
A_INF MV	40	Active infeed control for SINAMICS SM150
B_INF MV	41	Basic infeed control for SINAMICS GM150
TB30	100	Terminal Board 30

Drive object type	Object number (r0107)	Meaning
TM31	200	Terminal Module 31
TM41	201	Terminal Module 41
TM15DI/DO	204	Terminal Module 15 (for SINAMICS)
TM120	207	Temperature evaluation with safe electrical isolation
DO encoder	300	Drive object encoder

3.1.3.2 Execution groups in the DCC editor

Description

Exactly one DCC can be created per drive object (DO = drive object). This DCC can contain up to ten execution groups.

Execution groups in the DCC editor

Execution groups are groups of blocks. The blocks of an execution group are started in a defined sequence at a specified time and are calculated cyclically one after the other within a specified sampling time.

In the DCC editor a maximum of ten execution groups (execution group 1 to 10) can be defined per drive object (DO) and thereby per basic chart (see also: SINAMICS system integration > Execution sequence, creating new execution groups). In the project navigator of STARTER / SCOUT (context menu of the chart, menu command **Set execution groups ...**) fixed or free execution groups of the drive object can be assigned to the previously defined execution groups.

WARNING

If the assignment of an execution group is changed in the **Set execution groups** window (or in parameter p21000[]), the affected execution group is initially logged off during time slice management and then logged on once again with the new assignment. The execution group is not calculated during the period between logging off and logging back on. The log-on and log-off take place in a background process of the drive unit; the duration is therefore not defined and is determined by the current CPU time load of the drive unit. (This affects the path of the output signal in the case of time-dependent blocks, e.g. the DIF differentiator.) Prior to the first calculation cycle after logging back on, internal status variables of the blocks are partially reset. For both of these reasons, this can result in jumps in the output signal of blocks, which for example can affect the torque/force setpoint and (in the case of operated axes) also the torque/actual force value. Logic signals can also acquire an unexpected state at this point of operation.

If the change to the execution group also results in a change to the sampling time, internal constants or factors are automatically adjusted for time-dependent blocks (BF, DCA, DIF, DT1, INT, MFP, PCL, PDE, PDF, PIC, PST, PT1, RGE, RGJ, WBG).

Note that the execution groups created in the chart are only visible in the **Set execution groups** window after the chart has been compiled.

In the offline mode of the STARTER / SCOUT V4.1.x, r21002 and r21003 are always displayed with the value 0 in the **Set Execution Groups** window.

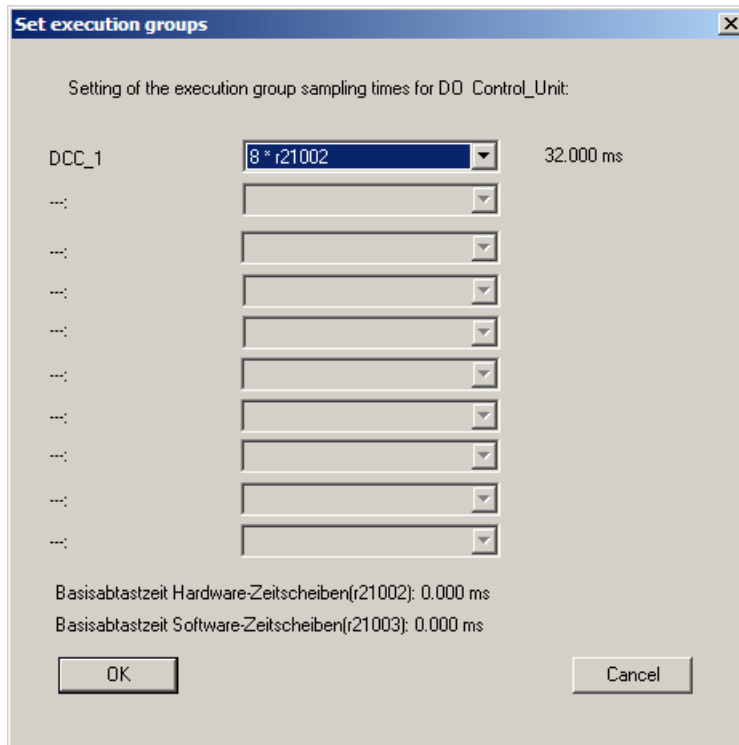


Figure 3-1 **Set Execution Groups** window

The execution groups created in the DCC editor must be assigned to the "Fixed execution groups" and the "Free execution groups" of the SINAMICS drive object.

Note

For the p21000 parameter (Properties execution group) and so also in the context menu of the "Set sampling times" chart, the write access for enabled controller is blocked in at least one drive axis or infeed. The old value automatically takes effect again in this case. If necessary, switch off all drive axes and infeeds, and make a new attempt.

3.1.3.3 Fixed execution groups

Fixed execution groups

A "fixed execution group" is called at a fixed position in the system execution using the sampling time of the corresponding system function.

The following fixed execution groups are available for SINAMICS as of FW Version 2.5:

- **Read in AFTER digital inputs**

This execution group is called after the current values of the digital inputs have been read in on this drive object type and the corresponding binector outputs have been written.

The sampling time of this execution group corresponds to the sampling time of the CU inputs/outputs (p0799) or the digital inputs/outputs of the TB30, TM31 and TM41 (p4099[0]).

- **Output BEFORE digital outputs**

This execution group is called before the digital outputs are output on this drive object type.

The sampling time of this execution group corresponds to the sampling time of the CU inputs/outputs (p0799) or the inputs/outputs of the TB30, TM31 and TM41 (p4099[0]).

- **Read in AFTER analog inputs**

This execution group is called after the current values of the analog inputs have been read in on this drive object type and the corresponding binector outputs have been written.

The sampling time of this execution group corresponds to the sampling time of the inputs/outputs of the TB30, TM31 and TM41 (p4099[1]).

- **Output BEFORE analog outputs**

This execution group is called before writing is performed on the analog outputs.

The sampling time of this execution group corresponds to the sampling time of the inputs/outputs of the TB30, TM31 and TM41 (p4099[1]).

- **BEFORE speed controller**

This execution group is called before the speed controller additional setpoint values "n_reg n_set1" (p1155) and "n_reg n_set2" (p1160) are read into FP3080. The sampling time of the speed controller (p0115[1]) produces the call, but a minimum sampling time of 1 ms is required.

- **BEFORE speed setpoint channel**

This execution group is called before function block diagrams 3010, 3020, 3030, 3040 and subsequent charts are calculated, if the setpoint channel has been activated (p0108.8 = 1). If no setpoint channel has been configured (p0108.8 = 0), calculation takes place before function block diagram 3095. The setpoint channel sampling time produces the call (p0115[3]).

- **BEFORE position controller**

This execution group is called after the actual position value preparation (function block diagram 4010) has been calculated and before the position controller (function block diagrams 4015, 4020 and 4025) is calculated. The sampling time of this execution group corresponds to the sampling time of the position controller (p0115[4]).

- **BEFORE actual position value**

This execution group is called before the actual position value preparation (function block diagram 4010) has been calculated and before the position controller (function block diagrams 4015, 4020 and 4025) is calculated. The sampling time of this execution group corresponds to the sampling time of the position controller (p0115[4]). (Available as of V4.3).

- **BEFORE basic positioner**

This execution group is called before the basic positioner function module (function block diagrams 3610 to 3650) is calculated. The sampling time of this execution group corresponds to the sampling time of the basic positioner function module (p0115[5]).

- **BEFORE standard technology controller**

This execution group is called before the technology controller function module is calculated (p0108.16 = 1) (function block diagrams 7950, 7954 and 7958). The sampling time of this execution group corresponds to the sampling time of the technology controller (p0115[6]).

- **Receive AFTER IF1 PROFIdrive PZD ¹⁾**

This execution group is called after cyclic process data has been received via communication interface IF1 (e.g. via the integrated PROFIBUS interface) and output to connector outputs r2050[...], r2060[...], binector outputs r2090 - r2093 and connector binector transformers r2094 and r2095.

The sampling time of this execution group corresponds to the PROFIdrive PZD sampling time. As of SINAMICS FW Version 2.5, communication interface IF1 (Interface 1) is always used by the PROFIBUS interface inside the CU or, if a CBE20 has been inserted in the option slot, PROFINET.

The parameter p0092 can be used to set whether the load calculation (r9976[]) evaluates the execution group for the isochronous operation (p0092 = 1) or the non-isochronous operation (p0092 = 0) during the SINAMICS startup.

Non-isochronous operation (r2043.1 = 0):

For SINAMICS **software versions V2.5 and V2.6**, the PROFIBUS data is sent and received immediately after each other at the start of the IF1 PROFIdrive PZD sampling time.

At the start of the IF1 PROFIdrive PZD sampling time set in p2048, the PROFIBUS data is sent first and then the process data received. In the case of non-isochronous cyclic communication at the IF1 communication interface, the execution group is cyclically calculated with the sampling time of the IF1 communication interface (p2048), i.e. after writing the connector outputs for the process data (PZD) r2050[.], r2060[.], r2090 - r2093 and after calculating the connector binector transformer r2094 and r2095.

As of SINAMICS **software version V4.3**, the PROFIBUS data is received and sent at two separate times. The process data is received at the start of the IF1 PROFIdrive PZD sampling time set in p2048. In the case of non-isochronous cyclic communication at the IF1 communication interface, the execution group is cyclically calculated with the sampling time of the IF1 communication interface (p2048) after receiving the data, i.e. after writing the connector outputs for the process data (PZD) r2050[.], r2060[.], r2090 - r2093 and after calculating the connector binector transformer r2094 and r2095.

For calculating the utilization of system r9976 correctly, p092 must be = 0 (== factory setting) in non-isochronous operation.

Isochronous operation (r2043.1 = 1):

In isochronous operation, the times at which the data is received from the master (T_o), the data is sent to the master (T_i) and the DP cycle time (T_{DP})³⁾ are configured in the master. Internally, the calls to T_o and T_i are realized using a state machine that is called cyclically with sampling time T_{Zu} , e.g. for standard servo drive objects with the current controller sampling time of 125 μ s. The sampling time $T_{Zu} = 125 \mu$ s, however, is at least as large as the largest current controller sampling time (e.g. for the vector, 250 μ s, 375 μ s or 500 μ s).

This means only integer multiples of $T_{Zu} \geq 125 \mu$ s can be set³⁾ as times T_i and T_o for the drive.

$$\rightarrow T_i = n_i \cdot T_{Zu}$$

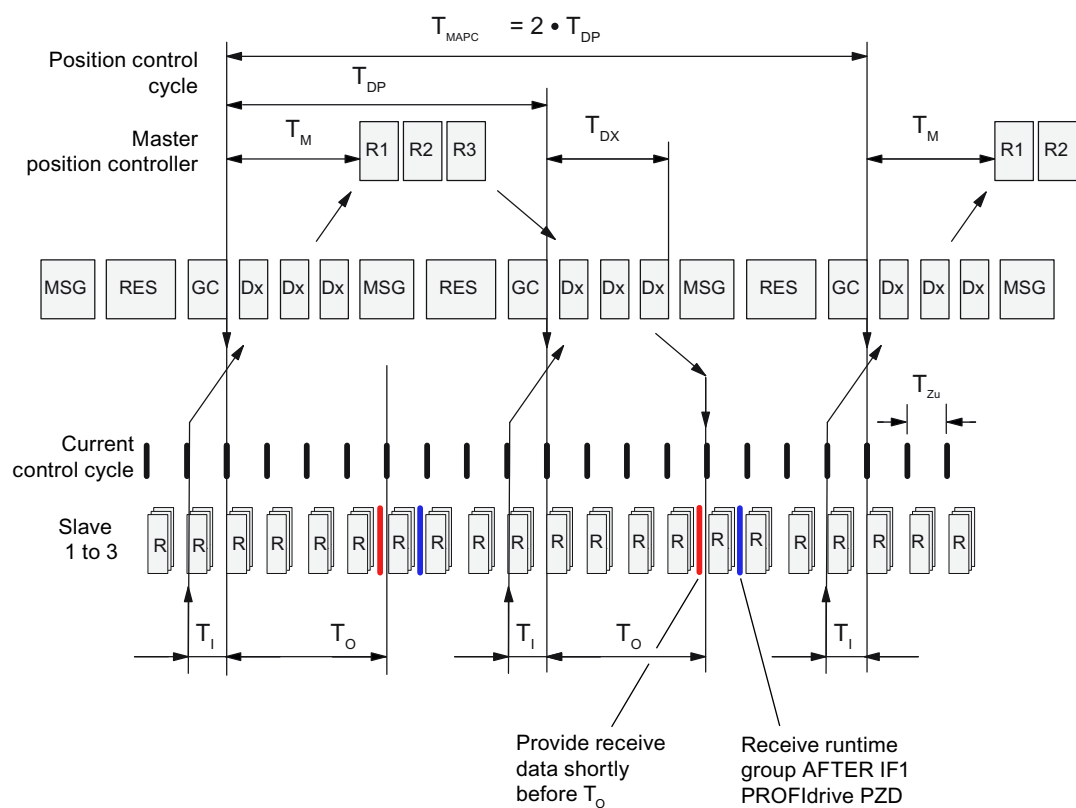


Figure 3-2 Calling the "Receive AFTER IF1 PROFIdrive PZD" execution group for isochronous PROFIBUS communication. See also ³⁾.

Received data (sent by the master) is always processed and made available at the end of the sampling time $(n_o - 1) T_{Zu}$, so that the received process data remains valid until the start of the next sampling period at the time $T_o = n_o \cdot T_{Zu}$ at r2050[.], r2060[.], r2090 - r2095. Thus, in the last sampling time T_{Zu} before T_o . The "Receive AFTER IF1 PROFIdrive PZD" execution group is then calculated in the following first sampling time T_{Zu} after T_o .

The calculation must be completed within the sampling time T_{zu} ; otherwise, warning A01053 "System overload detected" or a time-slice overflow (F01205) will occur. For this reason, only the absolute minimum number of DCBs that are necessary for the required function should be calculated in this execution group.

The CPU time available for this execution group falls the more drive axes are calculated on the CU since the current controllers (and also the speed controller for the servo) of the drive axes are also calculated in the sampling time $p0115[0] = T_{zu}$ ($\geq 125 \mu s$).

If the utilization calculation is to be executed during ramp-up of the CU for isochronous operation (unfavorable case regarding degree of utilization), $p092 = 1$ must be set. Otherwise ($p092 = 0$) the utilization in r9976 will be updated only after switchover to isochronous operation. The increased maximum overall utilization in isochronous operation will be displayed only in r9976.

If you want to calculate a larger number of DCBs, check whether you cannot configure this

better with the "Receive AFTER IF1 PROFIdr. flexible PZD" execution group.

Note

Note for this execution group that the higher computing time load in isochronous operation is considered by the load calculation only at the time of the transition (specified by the PROFIBUS master) into the isochronous operation. This means after the startup of the drive device, the load of the complete system initially still lies within the permitted range and only with the transition into the isochronous operation does the drive device shutdown with fault F1054 (system limits exceeded).

- **Send BEFORE IF1 PROFIdrive PZD ¹⁾**

This execution group is called before cyclic process data is sent via communication interface IF1 (e.g. the integrated PROFIBUS interface), i.e. before calculating the connector binector transformer p2080 – p2084 and reading in the connector inputs p2051[.], p2061[.].

The sampling time of this execution group corresponds to the PROFIdrive PZD sampling time. As of SINAMICS firmware Version 2.5, communication interface IF1 is always used by the PROFIBUS interface inside the CU or, if a CBE20 is inserted in the option slot, PROFINET.

The parameter p0092 can be used to set whether the load calculation (r9976[]) evaluates the execution group for the isochronous operation (p0092 = 1) or the non-isochronous operation (p0092 = 0) during the SINAMICS startup.

Non-isochronous operation (r2043.1 = 0):

For SINAMICS software versions **V2.5** and **V2.6**, the PROFIBUS data is sent and received (in this sequence) immediately after each other at the start of the IF1 PROFIdrive PZD sampling time.

The "Send BEFORE IF1 PROFIdrive PZD" execution group is processed at the end of the IF1 PROFIdrive PZD sampling time before the data is sent at the start of the next sampling time. After computing the execution group, at the end of the sampling time, the connector binector transformer p2080 – p2084 is still calculated and the connector inputs p2051[.], p2061[.] read in. This means all send data is present completely at the end of the sampling time and will be sent at the start of the next sampling time.

As of SINAMICS software version **V4.3**, the PROFIBUS data is received and sent at two separate times.

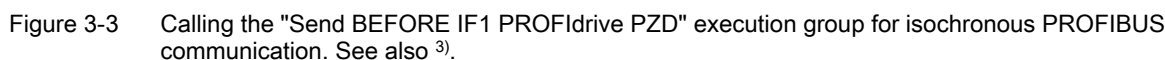
The "Send BEFORE IF1 PROFIdrive PZD" execution group is processed at the end of the IF1 PROFIdrive PZD sampling time before the complete data is still sent at the end of the sampling time. For the non-isochronous (cyclical) communication at the communications interface IF1, the execution group is calculated cyclically with the sampling time of the interface IF1 (p2048) (before sending the data), i.e. also before calculating the connector binector transformer p2080 – p2084 and reading in the connector inputs p2051[.], p2061[.].

For calculating the utilization of system r9976 correctly, p092 must be = 0 (== factory setting) in non-isochronous operation.

Isochronous operation (r2043.1 = 1):

In isochronous operation, the times at which the data is received from the master (T_o), the data is sent to the master (T_i) and the DP cycle time (T_{DP}) ³⁾ are configured in the master.

Internally, the calls to T_o and T_i are realized using a state machine that is called cyclically with sampling time T_{zu} , e.g. for standard servo drive objects with the current controller sampling time of 125 μ s. The sampling time $T_{zu} = 125 \mu$ s, however, is at least as large as the largest current controller sampling time (e.g. for the vector 250 μ s, 375 μ s or 500 μ s).

$$\rightarrow T_j = n_j \cdot T_{Zu} \text{ and } T_{DP} = n_{DP} \cdot T_{Zu}$$


The CPU time available for this execution group becomes smaller the more drive axes are calculated on the CU because the current controller (and the speed controller for the servo) of the drive axes are also calculated in sampling time $p0115[0] = T_{zu}$.

If the utilization calculation during the ramp-up of the CU has already executed the isochronous operation (the more unfavorable configuration regarding the utilization of the CU), then p092 must be set to = 1. Otherwise (p092 = 0) the utilization in r9976 will be updated only after switchover to isochronous operation. The increased maximum overall utilization in isochronous operation will be displayed only in r9976.

Note

Note for this execution group that the higher computing time load in isochronous operation is considered by the load calculation only at the time of the transition (specified by the PROFIBUS master) into the isochronous operation. This means after the startup of the drive device, the load of the complete system initially still lies within the permitted range and only with the transition into the isochronous operation does the drive device shutdown with fault F1054 (system limits exceeded).

- **Receive AFTER IF2 PZD ²⁾**

For the communications interface IF2, the CAN bus interconnection CBC10, the CBE20 or the PROFIBUS interface (see description p8839[.]) on the CU can be used. This execution group is available only on the CU_S, CU_G, CU_GM, CU_GL, SERVO, VECTOR, VECTORMV, VECTORSL and VECTOR_GL drive object types.

The execution group will be called once the connectors r8850[.], r8860[.] are written with the receive data and the calculation of the connector binector transformers r8894 and r8895 performed. The sampling time of this execution group corresponds to the IF2 PZD sampling time in p8848.

- **Send BEFORE IF2 PZD ²⁾**

For the communications interface IF2, the CAN bus interconnection CBC10, the CBE20 or the PROFIBUS interface (see description p8839[.]) on the CU can be used. This execution group is available only on the CU_S, CU_G, CU_GM, CU_GL, SERVO, VECTOR, VECTORMV, VECTORSL and VECTOR_GL drive object types.

The execution group is called before the binector connector transformers p8880 – p8884 are calculated and the connector inputs p8851[.], p8861[.] with the send data read. The sampling time of this execution group corresponds to the IF2 PZD sampling time in p8848.

- **Receive AFTER IF1 PROFIdr. flexible PZD ¹⁾**

This execution group is called after cyclic process data has been received via communication interface IF1 (e.g. via the integrated PROFIBUS interface) and output to connector outputs r2050[...], r2060[...], binector outputs r2090 - r2093 and connector binector transformers r2094 and r2095.

The sampling time of this execution group corresponds to the PROFIdrive PZD sampling time. The only difference from the Receive AFTER IF1 PROFIdrive PZD execution group is the way in which the execution group behaves in isochronous mode. Even in isochronous mode, this execution group is called using the PROFIdrive PZD sampling time configured in the Master as with any other time slot. This means that initially all time slots with a shorter sampling time are called at time T_0 depending on the validity of the receive data (current controller, possibly speed controller). This execution group is called as first only when the T_{DP} sampling time starts to be processed. You basically do not know how often the higher priority, shorter sampling times of the current and speed controller are calculated before this execution group starts to be processed. Processing of this execution group is also interrupted by shorter sampling times.

The benefit of this execution group is that considerably more blocks can be calculated than in the Receive AFTER IF1 PROFIdrive PZD execution group because the calculation does not have to be completed after current controller sampling time p0115[0].

- 1) IF1 is the abbreviation for communication interface 1. With SINAMICS V2.5 and SIMOTION V4.1, this refers to the integrated PROFIBUS interface or - if a CBE20 has been inserted in the option slot - the PROFINET interface. IF1 supports the PROFIdrive profile and isochronous operation.
- 2) IF2 is the abbreviation for communication interface 2. With SINAMICS V2.5, IF2 can only be used by the CAN bus when a CBC10 module has been inserted in the option slot.
- 3) For T_{DP} , T_i and T_o , the formulas and limit values listed in the Function Manual FH1 → Communication → Communication via PROFIBUS DP → Motion Control with PROFIBUS → "Time settings and Meanings" table apply.

Not every fixed execution group is available on every drive object type. This means, for instance, that SERVO, VECTOR, VECTORMV... drive object types do not contain any digital or analog inputs; therefore, the fixed execution groups for the digital inputs/outputs and analog inputs/outputs are not available here.

Please ensure that the sampling time for DCC SINAMICS is restricted to below 1 ms. As a property of an execution group, select a fixed execution group in which the sampling time for the assigned system function is < 1 ms. This execution group will thus only be called with the sampling time of 1 ms, and will thereby deviate from the assigned system function. Fault F51004 (see r0947) is set to indicate this deviation. The fault value of the fault (r0949) + 1 gives the number of the execution group where the deviation has occurred.

Example for the automatic limitation of the sampling time:

On the SERVO drive object type, select "BEFORE speed controller" as a fixed execution group.

The system function corresponding to this is the speed controller.

The sampling time of the speed controller is $p0115[1] = 125 \mu s$ in the factory setting of $p0112 = 3$.

In contrast to the sampling time of the speed controller, the sampling time of the DCC execution group is set automatically to 1 ms.

Please also refer to the description of parameter $p0112$ in the SINAMICS List Manual.



WARNING

In the case of an online change to the sampling time (execution group) the START method of the block is called. Initializations in the START method can result in jumps in the signal path.

3.1.3.4 Free execution groups

Free execution groups

The "free execution groups" are only defined by their sampling time ($p21000 = 1$ to 256 and $p21000 = 1001$ to 1096). The sampling times are provided centrally for all drive objects on a SINAMICS drive unit. A sampling time can be used simultaneously by several drive objects and several DCC execution groups. The sampling times are created in two different ways. The two methods have different limits for the maximum possible number of different sampling times.

- Free execution groups whose sampling times are created in the hardware:

With the hardware sampling times, in $p21000[0...9]$ each integer multiple of the basic sampling time (read-only in $r21002$) ranging from $1 * r21002$ to $256 * r21002$ can be generated with the following limits:

- Minimum sampling time = 1 ms
- Maximum sampling time = $(r21003 - r21002) < r21003$ (approx. 8 ms; on the D410 of the DP cycle).
- On the D410 drive unit, $r21003$ always matches the DP cycle (PROFIBUS T_DP cycle clock) or with the send cycle clock for PROFINET. If 1 ms is configured as DP cycle / send cycle for PROFINET, this means no hardware sampling times (but only software sampling times) from DCC are used on this device.
- The number of hardware sampling times on a drive device is limited. The number of still available hardware sampling times can be read from $r7903$ (as of SINAMICS V2.6). Further information is contained in Chapter DCC for SINAMICS → Overview → Computing time load, memory requirement and assignment of the HW sampling times → Number of possible different hardware sampling times (Page 112).

Note

As regards offline configuration using STARTER commissioning software, values 0 - 256 can be entered in $p21000[0..9]$, even if this violates the limits stated above for the hardware sampling times from 1 ms ... $< r21003$. This will only be detected once the Control Unit has been downloaded, and will result in the use of a replacement value and the F51004 fault. If the set value is too small, 1 ms will be set as substitute value; if it is too large, the next largest SW sampling time will be set.

- On the drive objects of the CU, TB30, TM15 DIDO, TM 31 and TM41, the sampling time for additional functions $p0115[0] = 4$ ms is preset. If you want to configure a DCC execution group with a smaller sampling time on these drive objects, you should first set the sampling time for the additional functions $p0115[0]$ on this drive object to the value of the smallest desired sampling time. To do this, $p0009 = 3$ must first be set. Only then is it possible to change $p0115[0]$. For the new value for $p0115[0]$ to take effect, $p0009 = 0$ must be reset.

- Free execution groups whose sampling times are created in the software:

The software sampling times are generated as an integer multiple of the basic value for software sampling times (read out in parameter r21003).

The possible set values for the software sampling times ($1 * r21003 - 96 * r21003$) can be taken from the parameter description for p21000 (see Function Manual, *Description of the standard DCC blocks*, in the appendix in the *Parameters* section).

WARNING

The START method of the block is called for an online change to the sampling time (execution group). Initializations in the START method can result in jumps in the signal path.

3.1.3.5 Execution sequence order, creating new execution groups

When a CFC is created, a new execution group with the same name as the chart will be created automatically. All blocks that are inserted into the chart will be automatically assigned in the insertion sequence of this execution group.

The computing sequence of the blocks within the execution group is specified by the configuration. If no other execution sequence is specified by the user, the execution sequence corresponds to the insertion sequence of the blocks within an execution group.

If the execution sequence with this execution group is to be changed, new execution groups added or the assignment of blocks to the execution groups changed, these properties can be changed in the execution editor. To do this, go offline in the STARTER and then open the chart. In the DCC editor, the **execution sequence**  button can be used to activate and deactivate the execution editor.

The center column of the execution editor contains the chart with its chart name. The execution group(s) are arranged hierarchically below. If you want to create additional execution groups, select the **Insert execution group...** item from the context menu of the chart (right-click the chart). The individual blocks can be moved between the execution groups or within an execution group using drag-and-drop. The details given in the Pos column (e.g. 1 / 2) correspond to the name in the bottom line in the right-hand part of the block header with a colored background. The 1st digit stands for the execution group (between 1... 10) and the 2nd digit for the sequence within the execution group. The execution sequence can be changed by dragging & dropping into the center column.

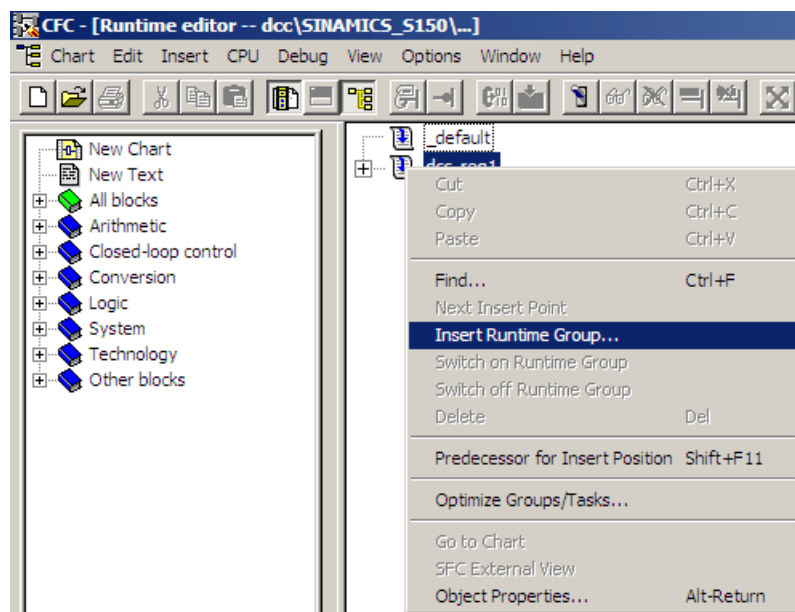


Figure 3-4 Context menu, **Insert execution group...**

Note

1. The assignment of identical names for DCCs in different drive objects is not permitted.
2. The assignment of identical names for execution groups in different drive objects is not permitted.

If the same fixed or free execution group of the drive object is assigned to several execution groups in the DCC editor, the execution groups of the DCC editor will be calculated in the sequence from top (calculated first) to bottom (calculated last) as shown in the "Set Execution Groups (Sampling Times)" window.

3.1.3.6 Creating customer-specific parameters ("declare")

The properties of the input and output connections of blocks can be edited in the DCC editor in the properties window of the block connection. Input connections may not be interconnected in the default setting of the DCC editor. The properties window of the connection is opened by double-clicking the connection or by selecting **Object properties** in the context menu of the connection.

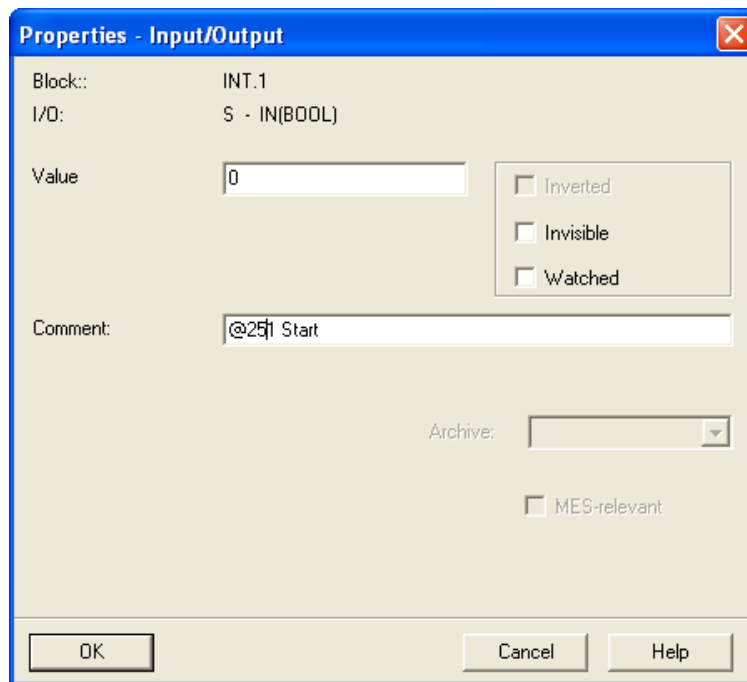


Figure 3-5 **Properties - Connection** window of the connection S (= Set) of the block INT (= INTEGRator)

In SINAMICS, input and output connections of blocks can be published as parameters. "Publish" means that users create the parameters themselves (parameter numbers and parameter text). This is the prerequisite in order to interconnect the block connections using BICO parameters to the basic system in order to assign the values to the input connections using adjustable parameters and to monitor the values of the output connections using parameters. The parameter number 251 in the above figure and in the following figure is arbitrary.

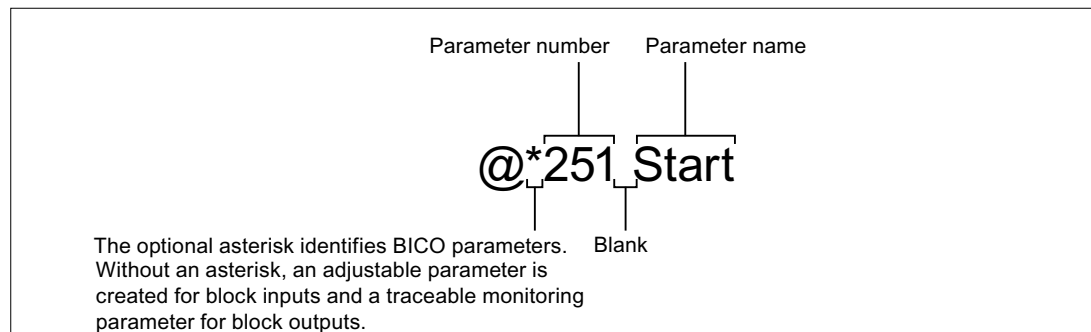


Figure 3-6 Structure of the block connection comment

The parameter name, which follows the parameter number separated by a space, is displayed in the STARTER expert list.

The data type of the published input/output is taken over by the block connection.

For the "publishing" of block inputs, a distinction must be made between adjustable parameters (without asterisk) and BICO parameters (with asterisk).

For the "publishing" of block outputs, the following distinction must be made:

- Without asterisk: pure monitoring parameter that cannot be interconnected and that can be recorded with the trace function.
- With asterisk: as for the case without asterisk, but additional interconnection possible.

Block inputs to be published may not be already interconnected in the default setting of the DCC editor. If necessary, an existing interconnection must first be deleted. To do this, click the interconnection in the chart to mark it and select **Delete interconnection(s)** in the context menu. In this DCC editor setting, the comment shown for a block input interconnected with a different block is always that of the connected block output.

If you want to remove this constraint, select **Options > Settings > Display ...** in the DCC editor and deactivate the **Interconnection comment** checkbox under **Parameters** and close the window with OK. Already interconnected block inputs can now be published without having to delete the interconnection first.

A parameter number may only be used just once in a chart. The DCC editor does not check when the parameter number is entered in the properties window of the connection whether this parameter number is already used in the chart. The multiple use of the parameter numbers will be indicated as error only when the chart is compiled.

The parameter numbers of the charts are represented on the number range from p21500 to p25999 of the expert list of the STARTER. With each DCC, it is possible to specify where the parameter range of the chart is to begin. This can be performed in the **Parameters** tab in the properties window of the chart in STARTER, using the parameter number base.

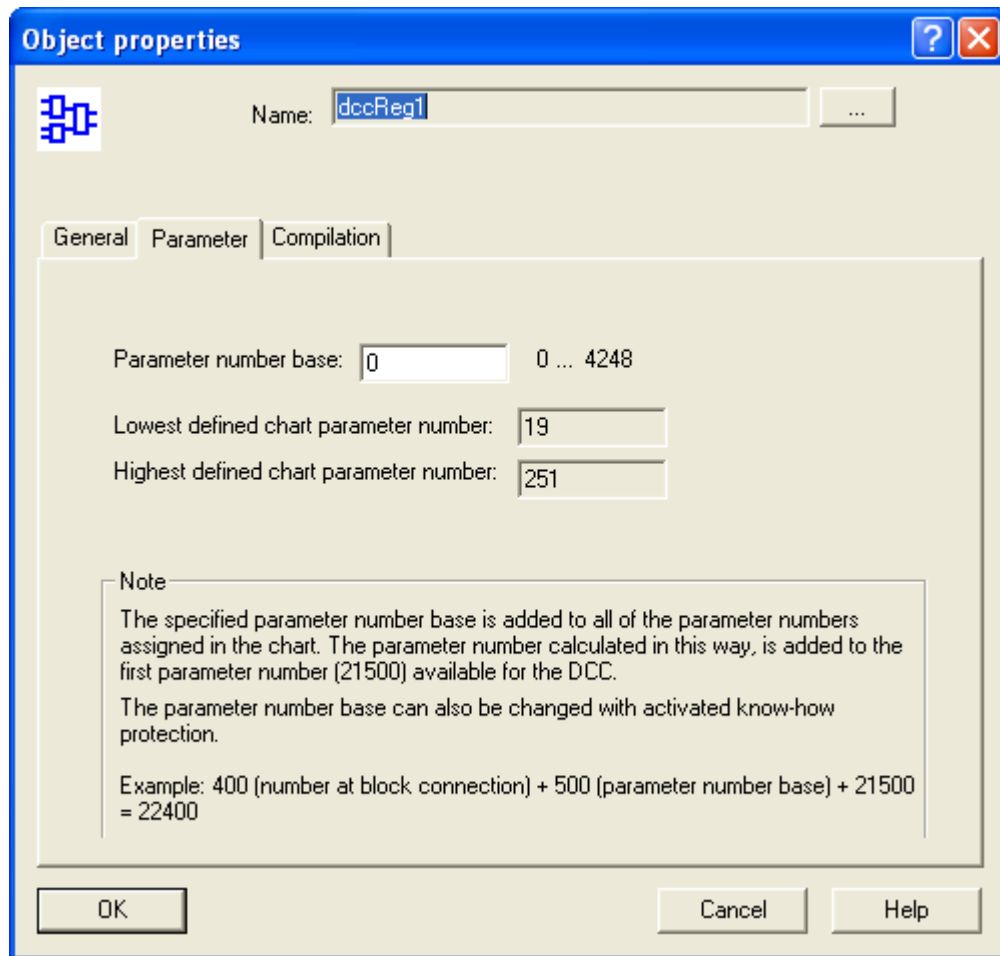


Figure 3-7 Property window of the dccReg1 chart for setting the parameter number base

The parameter number in STARTER has the following form:

- Parameter number in the chart + base parameter number + 21500;
where 21500 is the first parameter number available for DCC. The parameter number base must always be ≥ 0 . The parameter number base can be changed.

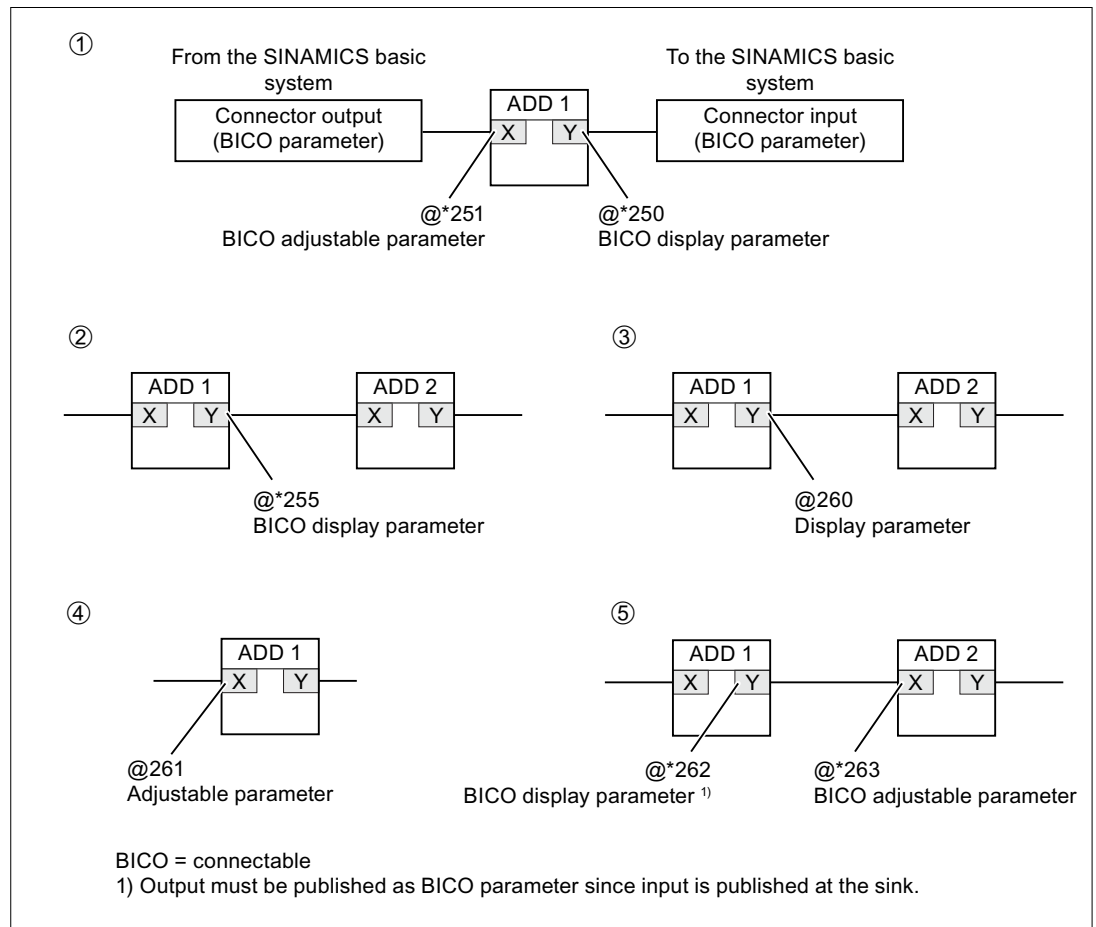


Figure 3-8 Examples of parameter definitions

Example 1: For an interconnection of the block connection with the SINAMICS basic system, the connection must be published as BICO parameter.

Example 2: Each block output connection can be published as BICO parameter. The signal on this output can be recorded with the trace.

Example 3: Each block output connection can be published as (not interconnectable with other BICO parameters) display parameter. The signal on this output can be recorded with the trace. The block input can be interconnected to every other non-published block input.

Example 4: The block input connection X of the ADD 1 block is published as adjustable parameter.

Example 5: Because the block input connection X of the ADD 2 is published as BICO parameter, the block output connection interconnected with it must also be published as BICO parameter. It is, however, possible in the DCC editor to interconnect ADD 1 Y with ADD 2 X when only the block input ADD 2 X is published. This results in an error message when the chart is compiled. The signal value of a block input published as BICO parameter cannot be recorded with the trace.

Note

No help function can be created for @parameters.

Display of the parameters

If the parameters configured on the block connections are to be displayed in the DCC editor, then select **Options > Settings > Display ...** and select **Comment** under **Connections**. The first eight characters of the comment are then displayed at the respective block connections in the block symbols.

3.1.3.7 Interconnection with SINAMICS parameters

You can interconnect with SINAMICS BICO parameters in the DCC editor.

Procedure

Interconnection can be performed as follows:

1. Creating customer-specific parameters ("Publishing", @ parameters); see Creating customer-specific parameters (Page 98).

Note

In the DCC editor, block connections can initially be connected with the drive's BICO parameters without the need to publish them as parameters. However, this will cause errors when compiling the chart, since only block connections that have been published as @* parameters can be connected to the drive's BICO parameters.

2. Select the block connection to be interconnected.

3. Right-click and select **Interconnection with operand...** from the context menu. The **DCC Signal Selection** window appears.

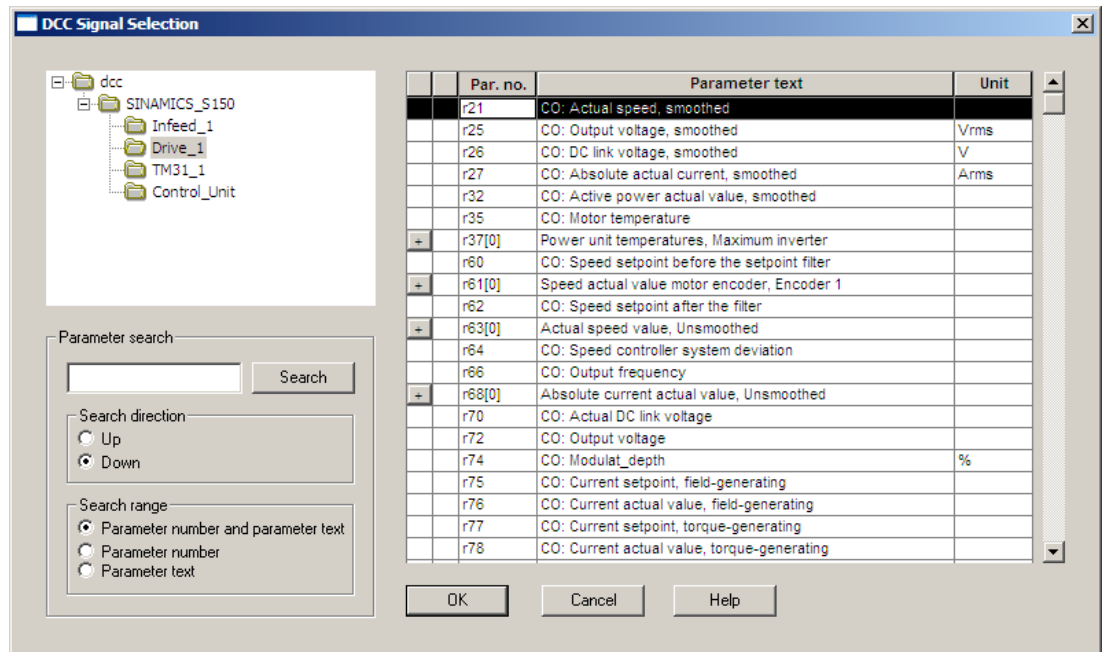


Figure 3-9 Signal selection

4. Select the parameter to be interconnected.
5. Click **OK** to close the window.

From this point on, the selected block connection is interconnected with the SINAMICS parameter and the parameter is displayed in the sheet bar of the DCC editor.

Note

BICO parameters defined in other DCC charts will not be displayed in the signal selection list.

Links to BICO parameters in other DCC charts can be drawn in the DCC editor directly between the charts.

3.1.3.8 Copy charts within a project

In STARTER, DCCs can be copied from one drive object to other drive objects.

1. Open your project in STARTER.
2. Open in the project navigator the drive object with the source chart.
3. Select the source chart and select the **Copy** command in the context menu of the chart.
4. Mark the target drive object.
5. Select the **Paste** command in the context menu of the target drive object.

The chart has been copied from the source drive object to the target drive object.



WARNING

After copying, you must check whether the interconnections from and to published block connections in the chart copy have to be adapted to another drive object as a result of the copying. During copying, no automatic adaptation of the interconnections of the chart copy is made to the basic system or to other DCCs.

3.1.3.9 Copy charts between projects

In STARTER, DCCs can be copied from one (source) project into another (target) project.

1. Open your project with the source chart in STARTER.
2. Open STARTER a second time and open there the target project in which the chart is to be copied.
3. Open in the project navigator the drive object with the source chart.
4. Select the source chart and select the **Copy** command in the context menu of the chart.
5. Mark the target drive object in the target project.
6. Select the **Paste** command in the context menu of the target drive object.

The chart has been copied from the source project to the marked drive object of the target project.



WARNING

After copying, you must check whether the interconnections from and to published block connections in the chart copy have to be adapted to another drive object as a result of the copying. During copying, no automatic adaptation of the interconnections of the chart copy is made to the basic system or to other DCCs.

3.1.4 Computing time load, memory requirement and assignment of the HW sampling times

The following statements govern the use of DCC-SINAMICS on the SINAMICS S120, S150, G130, G150 and (on the SINAMICS_Integrated in) SIMOTION D4xx devices. Details about other devices are contained in the associated device documentation.

3.1.4.1 Computing time load for the SINAMICS V2.5 and V2.6 software versions

The processing of blocks requires computing time. This means with the computing of DCC execution groups, it is no longer possible to calculate the maximum quantity structure for drive objects of a CU3x0, D4xx and CX32.

The resulting computing time load is dependent on the following:

- Number of calculated execution groups
- Sampling time of execution groups
- Number of calculated blocks
- Calculated block types
- For some blocks, it depends on the parameterization (e.g. the setting of enable connections)

Parameter r9976 (system load; can be found on the CU's drive object) is available in the system as an online tool. The average value for the computing time load in r9976[1] should **always** be below 85.0%. Likewise, the maximum value for the computing time load in r9976[5] should **always** be below 85.0%. Exceeding the computing time load limit of 90% triggers warning A50512. The warning is reset if the computing time load falls below 88%.

The computing time load should not fall below the limit of 85.0% so that the response time of the STARTER/SCOUT engineering tool is not too greatly restricted.

If the computing time load increases to the point that it is no longer possible for all blocks and system functions assigned to a sampling time to be completed within this sampling time, the drive unit switches off and outputs fault F01205: "CU: Time-slice overflow".

Offline mode enables the SIZER engineering tool to make an approximate statement regarding whether a configuration can be calculated on a CU3x0 or D4xx. The additional computing time load is **not** taken into account by activating the DCCs.

NOTICE

The computing time load reaches its maximum value when the following conditions are met:

- All necessary function modules are activated.
- All drives and infeeds are in the operational status.
- Closed-loop control in the final configuration is approved on all drives and infeeds.
- Configured isochronous data transfer is used.

The values displayed in r9976 are thoroughly smoothed internally. A change in the computing time load is therefore only displayed in full in r9976 after 2 - 3 minutes.

If, during OFFLINE configuration, it is not certain whether the limit value for the computing time load of 85% (r9976[1] and r9976[5]) will be adhered to, you can initially call the execution groups being used in a very large sampling time (e.g. $96 * r21003$). After checking the actual system load in r9976, you can then gradually select execution groups with shorter sampling times right through to the execution group that is intended for the application in question.

During OFFLINE configuration, the following should be carried out:

- Activate all necessary function modules on the drive objects.
- Assign all necessary blocks to their execution groups.
- Establish all connections, both between the blocks and to the drive objects (as long as this still makes sense even if an increased sampling time is being used).

The following applies as a rough configuration aid to SERVO drives:

If one A-INF infeed is configured on a SINAMICS CU320 with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu\text{s}$) and three SERVO drive axes are configured with the speed and current controller sampling times set in the factory ($p0112 = 3 == \text{STANDARD} \rightarrow p0115[1] = p0115[0] = 125 \mu\text{s}$) and the function modules and without isochronous PROFIBUS/PROFINET communication set in the factory, one DCC chart with 50 blocks can be calculated on each of the three drive objects in an execution group with a sampling time of 2 ms.

The following (functional single) blocks have been used in a chart in this case:

AND, OR, XOR, NOT, BF, CTR, MUX8, NAND, NCM, SH_DW, ADD, SUB, MUL, DIV, AVA, MAS, MIS, PLI20, DEL, DEZ, SUB_D, SUB_I, MUL_D, MUL_I, DIV_D, DIV_I, AVA_D, SII, STM, DLB, DX8, DX8D, MFP, PCL, PDE, PDF, PST, RSR, DFR, BSW, NSW, LIM, PT1, INT, DIF, LVM, LIM_D, PC, PIC, RGE

In a sampling time of 1 ms, only around half the blocks can be calculated. Around twice as many can be calculated in a 4 ms sampling time.

A SERVO axis with standard sampling times omitted from the maximum configuration thus permits the use of approximately 50 blocks (of the above-mentioned types) for a sampling period of 2 ms.

Approximately 50 blocks (of the above-mentioned types) in an execution group increase the computing time load of the CU by approximately 11%.

The following applies as a rough configuration aid to VECTOR drives:

If one A-INF infeed is configured on a SINAMICS CU320 with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu\text{s}$), two VECTOR drive axes are configured with the speed controller sampling times $p0115[1] = 2000 \mu\text{s}$ and the current controller sampling times $p0115[0] = 500 \mu\text{s}$ ($p0115[2] = p0115[3] = p0115[4] = 2000 \mu\text{s}$) and the function modules and without isochronous PROFIBUS/PROFINET communication set in the factory, one DCC chart with approx. 70 blocks (type selection, see above for SERVO) can be calculated on two drive objects in an execution group with a sampling time of 2 ms. This means approx. 70 blocks (type selection, see above for SERVO) increase the computing time load by approx. 15%.

In a sampling time of 1 ms, only around half the blocks can be calculated. Around twice as many can be calculated in a 4 ms sampling time.

The following generally applies:

You can of course use any blocks you want in your project. Using other block types may produce other results in terms of the number of blocks that can be calculated and the computing time load. Blocks with a very extensive functionality of course require a greater computing time. Note that the computing time needed for some blocks also depends on the values configured at the connections, e.g. whether an enable connection is set.

When measuring the computing time load, all connections should therefore be set to their final configuration values wherever possible.

When commissioning a configuration, the current computing time load must always be checked online on the drive unit for compliance with the limits stated at the start for r9976[1] and r9976[5].

3.1.4.2 Computing time load as of software version V4.3

With software version SINAMICS V4.3, DCC is available for the new higher performance CU320-2 module. As of software V4.3, after a download (DL) or a parameter change (e.g. of the sampling time of an execution group), the control unit (CU) uses the configuring data to determine the expected computing time load (including the loading by DCCs) and indicates this for the complete system in parameter r9976 (the system load that can be found on the drive object of the CU). If the calculated average computing time load for the complete system r9976[1] or the maximum load in a sampling time r9976[5] (including interruptions due to shorter sampling times) exceeds the maximum value of 100.00%, this causes the setting of fault F1054 (CU: System limit exceeded). The fault F1054 causes all infeeds and drive axes calculated on the CU to be switched off. The load is calculated on the CU, i.e. the load values are displayed in the STARTER/SCOUT only in online mode.

The resulting computing time load caused by DCC depends on the following:

- Number of calculated execution groups
- Sampling time of the execution groups
- Number of calculated blocks
- Calculated block types
- When using the execution groups "Receive PZD AFTER IF1 PROFIdrive" and "Send PZD BEFORE IF1 PROFIdrive" of the bus configuration (isochronous/ not isochronous; see chapter Fixed execution groups (Page 85))

The proportional computing time load caused by DCC is displayed on the drive objects on which DCCs are configured in the parameter r21005[0...9] for the execution groups 1 to 10 (as of SINAMICS V4.3). Note that the average computing time load for an execution group k can be calculated only when this is registered for cyclical processing ($p21000[k-1] \neq 0$, in the STARTER/SCOUT in the context menu of the chart → Set sampling times).

In contrast to software versions V2.5 and V2.6, a parameter change (in online mode of the STARTER) that affects the computing time load (e.g. change of the sampling time of a DCC execution group) causes an immediate recalculation of r9976 (and r21005) by the drive device. For parameters that may only be changed in the device states C1 (commissioning 1) or C2, i.e. may only be changed in STARTER/SCOUT offline mode (e.g. p0115), the new value of the computing time load in r9976 will be displayed only after the project download and the subsequent CU startup.

As of V4.3, the permitted maximum value of the computing time load r9976 increases practically to 99.99%. (For values > 100.00%, fault F1054 will be set; this simultaneously causes an OFF2 on all drive objects of the CU.)

As rough configuration aid, the following is true for **SERVO** drives with the current controller sampling time $p0115[0] = 125 \mu\text{s}$ and the speed controller sampling time $p0115[1] = 125 \mu\text{s}$:

1. If on a SINAMICS CU320-2 one A-INF infeed with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu\text{s}$), **six SERVO drive axes** with the speed and current controller sampling times set in the factory ($p0112 = 3 == \text{STANDARD} \rightarrow p0115[1] = p0115[0] = 125 \mu\text{s}$), the function modules set in the factory setting, one TB30 ²⁾ and with isochronous PROFIBUS/PROFINET communication with $T_{\text{DP}} \geq 2 \text{ ms}$ are configured, one DCC with approx. 50 blocks can be calculated in an execution group with a sampling time of 2 ms.
2. If on a SINAMICS CU320-2 one A-INF infeed with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu\text{s}$), **5 SERVO drive axes** with the speed and current controller sampling times set in the factory ($p0112 = 3 == \text{STANDARD} \rightarrow p0115[1] = p0115[0] = 125 \mu\text{s}$), the function modules set in the factory setting, 1 TB30 ²⁾, 3 TM31 ²⁾ and isochronous PROFIBUS/PROFINET communication with $T_{\text{DP}} \geq 2 \text{ ms}$ are configured, one (1) DCC chart with approx. 75 blocks can be calculated ¹⁾ in an execution group with a sampling time of 2 ms.
3. For each additional omitted SERVO axis, approximately 75 additional blocks ¹⁾ can be calculated in the sampling time of 2 ms.

As rough configuration aid, the following is true for **VECTOR** drives with the current controller sampling time $p0115[0] = 500 \mu\text{s}$ and the speed controller sampling time $p0115[1] = 2000 \mu\text{s}$ ($p0115[3] = 2000 \mu\text{s}$, $p0115[4] = 2000 \mu\text{s}$):

For more than three VECTOR axes on a CU320-2, the current controller sampling time will be increased automatically to the value $p0115[0] = 500 \mu\text{s}$. This automatically increases the speed controller sampling time to 2 ms.

1. If, on a SINAMICS CU320-2, one A-INF infeed with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu\text{s}$), **six VECTOR drive axes** with the current controller sampling times $p0115[0] = 500 \mu\text{s}$, the speed controller sampling time $p0115[1] = 2000 \mu\text{s}$, the functions and function modules set with the factory setting, one TB30 ²⁾ and with isochronous PROFIBUS/PROFINET communication with $T_{\text{DP}} \geq 2 \text{ ms}$ are configured, one DCC with approx. 50 blocks can also be calculated in an execution group with the sampling time 2 ms.
2. If, on a SINAMICS CU320-2, one A-INF infeed with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu\text{s}$), **five VECTOR drive axes** with the current controller sampling times $p0115[0] = 500 \mu\text{s}$, the speed controller sampling time $p0115[1] = 2000 \mu\text{s}$, the functions and function modules set with the factory setting, with isochronous PROFIBUS/PROFINET communication with $T_{\text{DP}} \geq 2 \text{ ms}$, one TB30 ²⁾ and three TM31 ²⁾ are configured, one DCC with approx. 75 blocks can also be calculated in an execution group with the sampling time 2 ms.
3. For each additional omitted VECTOR axis (with $p0115[0] = 500 \mu\text{s}$ and $p0115[1] = 2000 \mu\text{s}$), approximately 75 additional blocks ¹⁾ can be calculated in the sampling time of 2 ms.

As rough configuration aid, the following is true for **VECTOR** drives with the current controller sampling time $p0115[0] = 250 \mu s$ and the speed controller sampling time $p0115[1] = 1000 \mu s$ ($p0115[3] = 1000 \mu s$, $p0115[4] = 1000 \mu s$):

For up to three VECTOR axes on a CU320-2, the current controller sampling time and the speed controller sampling time have the values $p0115[0] = 250 \mu s$ and 1 ms, respectively, in the factory setting.

1. If, on a SINAMICS CU320-2, one A-INF infeed with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu s$), **three VECTOR drive axes** with the current controller sampling times $p0115[0] = 250 \mu s$, the speed controller sampling time $p0115[1] = 1000 \mu s$, the functions and function modules set with the factory setting, one TB30 ²⁾ and with isochronous PROFIBUS/PROFINET communication with $T_{DP} \geq 2 \text{ ms}$ are configured, one DCC with approx. 50 blocks can also be calculated in an execution group with the sampling time 2 ms.
2. If, on a SINAMICS CU320-2, one A-INF infeed with the sampling times set in the factory ($p0112 = 2 == \text{LOW} \rightarrow p115[0] = 250 \mu s$), **two VECTOR drive axes** with the current controller sampling times $p0115[0] = 250 \mu s$, the speed controller sampling time $p0115[1] = 1000 \mu s$, the functions and function modules set with the factory setting, with isochronous PROFIBUS/PROFINET communication with $T_{DP} \geq 2 \text{ ms}$, one TB30 ²⁾ and three TM31 ²⁾ are configured, one DCC with approx. 150 blocks can also be calculated in an execution group with the sampling time 2 ms.
3. For each additional omitted VECTOR axis (with $p0115[0] = 250 \mu s$ and $p0115[1] = 1000 \mu s$), approximately 150 additional blocks ¹⁾ can be calculated in the sampling time of 2 ms.

The following generally applies:

You can of course use any blocks you want in your project. The use of other block types may produce different results in terms of the number of blocks that can be calculated and the computing time load. Blocks with a very extensive functionality of course require a greater computing time.

¹⁾ The chart with 75 blocks consists of an execution group with the following blocks in the listed sequence:

AND, OR, XOR, NOT, BF, CTR, MUX8, NAND, NCM, SH_DW, ADD, SUB, MUL, DIV, AVA, MAS, MIS, PLI20, DEL, DEZ, SUB_D, SUB_I, MUL_D, MUL_I, DIV_D, DIV_I, AVA_D, SII, STM, DLB, DX8, DX8_D, MFP, PCL, PDE, PDF, PST, RSR, DFR, BSW, NSW, LIM, PT1, INT, DIF, LVM, LIM_D, PC, PIC, RGE, AND, OR, XOR, NOT, BF, CTR, MUX8, NAND, NCM, SH_DW, ADD, SUB, MUL, DIV, AVA, MAS, MIS, PLI20, DEL, DEZ, SUB_D, SUB_I, MUL_D, MUL_I, DIV_D.

The chart with 150 blocks consists of an execution group that contains the previously listed 75 blocks twice.

²⁾ All sampling times ($p4099[]$) of the TB30 and the TM31 have the value 4 ms (factory setting).

3.1.4.3 Memory requirement for software version SINAMICS V2.6

The blocks and @ parameters in the DCC charts require memory on the drive unit. To use DCC-SINAMICS, for SINAMICS G130, G150, S120, S120 Chassis, S120CM, S150 and SIMOTION D4x5, at least one drive axis must be excluded in comparison to the maximum possible configurations (1 ALM + 6 servo axes + TB30 or 1 ALM + 4 vector axes + TB30).

Maximum quantity structures with a topology of 5 servo or 3 vector axes with 1 ALM and 1 TB30			
Drive unit		CU320 ²	SINAMICS Integrated on SIMOTION D4x5
SINAMICS 2.6.1	Blocks ¹	350	120
	@Parameters ¹	350	120
SINAMICS 2.6.2	Blocks ¹	500	200
	@Parameters ¹	500	200

2) SINAMICS G130, G150, S120, S120 Chassis, S120CM, S150

Note

The specified maximum numbers for blocks and @parameters always apply for the entire drive unit and should be regarded as guide values. Only the factory-set function modules are active. The individual blocks and @parameters may be arbitrarily distributed across several charts. Conserving @parameters has little effect on the quantity structure of the blocks; the specified maximum numbers for the blocks should therefore not be exceeded.

Savings in further drive axes can increase the maximum limits by 50 blocks and 50 @parameters for each axis saved.

Maximum limits when just one axis is used (including 1 ALM + TB30)					
Drive unit		CU320 and CU310 ² with 1 servo axis	CU320 and CU310 ² with 1 vector axis	SINAMICS Integrated ³ with one servo axis	SINAMICS Integrated ³ with one vector axis
SINAMICS 2.6.1	Blocks ¹	550	450	320	220
	@Parameters ¹	550	450	320	220
SINAMICS 2.6.2	Blocks ¹	700	600	400	300
	@Parameters ¹	700	600	400	300

2) SINAMICS G130, G150, S120, S120 Chassis, S120CM, S150

3) S120i on D4x5

When expanding the topology on a drive unit with TMxx or DMC20 modules, one drive unit must be omitted for each additional TMxx module.

Please note when connecting CX32 modules to a D4x5 that memory and CPU time are also allocated for each CX32 on the D4x5. When using the DCC, another servo or vector axis must be omitted for every 2 CX32s (i.e. the axis must be omitted for the first CX32).

Example for SINAMICS 2.6.2:

D4x5 with 1 ALM + 4 servo axes + 2 CX32 + 1TB30 à
Permits DCC with 200 DCBs + 200 @parameters.

D4x5 with 1 ALM + 2 vector axes + 2 CX32 + 1TB30 à
Permits DCC with 200 DCBs + 200 @parameters.

When using DCC on the CX32 the same quantity structure applies as on the CU320.

Note

The final limits are determined by the total memory available on the drive unit and the computing time load. If the above-mentioned recommended maximum limits are exceeded, this can result in errors during upload or download (e.g. fault F1105 CU: insufficient memory) and the drive can no longer be switched on. If it is not possible to perform a new download with a suitably adjusted project, a power OFF/ON must be performed on the affected drive unit.

By activating additional function modules (e.g. basic positioner EPOS), additional memory is allocated and the specified maximum limits are decreased.

1) The charts for the above-specified quantity structures are formed from $n \times (\text{chart1} + \text{chart2})$:

Chart1 contains 50 blocks (of the following types: AND, OR, XOR, NOT, BF, CTR, MUX8, NAND, NCM, SH_DW, ADD, SUB, MUL, DIV, AVA, MAS, MIS, PLI20, DEL, DEZ, SUB_D, SUB_I, MUL_D, MUL_I, DIV_D, DIV_I, AVA_D, SII, STM, DLB, DX8, DX8D, MFP, PCL, PDE, PDF, PST, RSR, DFR, BSW, NSW, LIM, PT1, INT, DIF, LVM, LIM_D, PC, PIC, RGE) with 50 @parameters and approx. 90 connections (from block connector to block connector).

Chart2 contains 50 blocks (of block types AND, OR, ADD, MUL, DIV, B_DW, B_W, BY_W, D_I, D_R, D_UI, D_US, DW_B, DW_R, DW_W, I_D, I_R, I_UD, I_US, N2_R, N4_R, R_D, R_DW, R_I, R_N2, R_N4, R_UD, R_UI, R_US, UD_I, UD_R, UI_R, US_D, US_I, US_R, W_B, W_BY, W_DW, WBG, DCA, INCO, OCA, TTCU, ADD, ADD_D, ADD_I, ADD_M, AVA, AVA_D, RSS), 50 @parameters and approx. 90 connections (from block connector to block connector)

For example, 350 blocks and 350 @parameters are compiled from $4 * \text{chart1} + 3 * \text{chart2}$.

3.1.4.4 Memory requirement as of software version SINAMICS V4.3

The blocks and @ parameters in the DCCs require memory on the drive unit. A maximum of 1500 blocks and 1500 @-parameters may be configured in DCC SINAMICS with the CU320-2 module on SINAMICS S120, S150, G130 and G150.

3.1.4.5 Number of possible different hardware sampling times

The sampling times for the execution groups can be selected in p21000[x] as a multiple of r21002 (basic sampling time of hardware time slices), a multiple of r21003 (basic sampling time of software time slices), or on the basis of the sampling time of a basic SINAMICS system function (e.g. when p21000[x] = 9003 == "before setpoint channel" from the sampling time of the setpoint channel p0115[3]).

Only sampling times can be set as hardware sampling times, for which the following applies:

$1 \text{ ms} \leq T_{\text{sampling}} \leq r21003 - r21002$ in p21000[x]

Hardware sampling times, number, and assignment

During configuration, note that the number of different hardware sampling times ($1 \text{ ms} = \text{cycle duration } T_{\text{sampling}} < r21003 - r21002$) used by the basic SINAMICS system, active function modules (see r108) and Drive Control Chart is restricted as follows:

- CU310, CU320, D4xx --> no. of hardware sampling times = 13
- CU320-2 with SINAMICS V4.3 and higher --> no. of hardware sampling times = 25
- CUD (SINAMICS DC MASTER) --> number of hardware sampling times = 11

The assignment of the available hardware sampling times is displayed in r21008[0...24] as follows (STARTER/SCOUT: in online mode only):

- Value = 0.0 --> sampling time not assigned
- Value != 0.0 (not equal to 0.0) --> sampling time in ms
- Value = 9.9999e + 006 --> sampling time not supported

Hardware sampling times, usage

A sampling time can be used simultaneously by more than one execution group of DCC, function modules and the basic SINAMICS system.

For this reason, the execution groups should ideally be registered to existing sampling times or, if it makes more sense in relation to the function, the execution group "Before setpoint channel" should be used.

For internal reasons, the drive unit always requires at least one (or more, depending on how the basic sampling times p0115[0] of the drive objects are parameterized) free hardware sampling time, which is why the current number of free hardware sampling times can be read in r7903 (from SINAMICS V2.6 onward).

Note

Note that a long-term trace registers a sampling time of 2 ms and the trace registers sampling times in accordance with the selected trace cycle clock. If these sampling times have not already been registered by the basic SINAMICS system, "Free blocks" (FBLOCKS), or Drive Control Chart (DCC), these functions require additional free hardware sampling times.

The registered hardware sampling times can be read (if the DCC is activated) in r21008[0...12].

The current number of hardware sampling times to be assigned is displayed in r7903 (from SINAMICS V2.6 onward).

Project download, error message, and procedure

If too many different hardware sampling times are configured offline, an error message is not output until the project is downloaded.

In this case, proceed as follows:

1. 1. With the project in offline mode, set all the free execution groups to which hardware sampling times are assigned to software sampling times.
 - Hardware sampling times (DCC: p21000 < 256)
 - Software sampling times (DCC: p21000 ≥ 1001)
The assignment of fixed execution groups (DCC: p21000 ≥ 2000) does not need to be changed because the fixed execution groups use the same sampling time as the assigned basic SINAMICS system function.
2. Download the project again.
3. Once the project has been downloaded and the Control Unit has booted, check:
 - r7903: Number of hardware sampling times still available
 - r21008: Sampling times of the basic SINAMICS system already registered
 - Hardware sampling times.
4. Adjust the execution group parameters accordingly.

Note

The number of different hardware sampling times possible on a Control Unit is restricted. For this reason, software sampling times (multiple of r21003) or, if necessary, the fixed execution group (p21000[0...9] ≥ 2000) should ideally be used.

3.2 Working with DCC SINAMICS

3.2.1 Preliminary remarks on configuration

Preliminary remarks

The following is a brief explanation of what you will be configuring in this chart.

Configuration example

The requirement for this example is that there must be a STARTER or SCOUT version ($\geq$ V4.1.2) suitable for your SINAMICS SW version ($\geq$ V2.6) installed on your computer. The SINAMICS Support Package (SSP) V2.6 must also be installed. Your computer must also have a CFC license to use the DCC editor. You can install this license from the USB stick provided with the help of the Step7 Automation License Manager. A CU310 or CU320 is required with which STARTER/SCOUT can connect online, e.g. via PROFIBUS.

The configuration example deals with a straightforward oscillating circuit that creates sinusoidal oscillation at its output.

It will only take you a few minutes to create the chart; then you can execute it in test mode as a demonstration.

The following blocks are used:

- Two integrators (INT)
- One inverter (SII)

As indicated by the differential equation $f''(x) = -f(x)$, the oscillating circuit is comprised of two integrator blocks that are linked by negation.

The frequency of the oscillating circuit is determined by the integrating time constant at the integrators.

The oscillation amplitude is specified by the initial value at the integrator output.

Configuration example structure

The configuration example is divided into the following steps:

1. Creating a new project.
2. Inserting the DCC in the project.
3. Inserting blocks in a DCC.
4. Interconnecting blocks.
5. Parameterizing block connections in the chart.
6. Publishing block connections as parameters.
7. Setting execution sequence within an execution group
8. Compiling the DCC in the DCC editor.
9. Setting sampling time for an execution group
10. Loading the DCC technology option on to the drive unit's CF card.
11. Downloading the compiled DCC to the drive.
12. Displaying values at block connections online.
13. Recording signals from the DCC with the trace.
14. Archiving the project.
15. Creating documentation.

3.2.2 Creating a new project

- Create a new project in the SCOUT or STARTER engineering system, e.g. **dcc_ex**, see Creating a project (Page 20).
- Create a new device. To do this, insert a "SINAMICS S120 CUxxx" in the STARTER or SCOUT by double-clicking the "Insert single drive unit" command. Select "xxx" to suit your existing hardware (CU320, CU310DP, CU310PN).
- You can now insert a chart.

3.2.3 Inserting a DCC

- Open up the tree structure in the project navigator to the control unit.
- Call **Insert DCC** below the control unit (double-click).

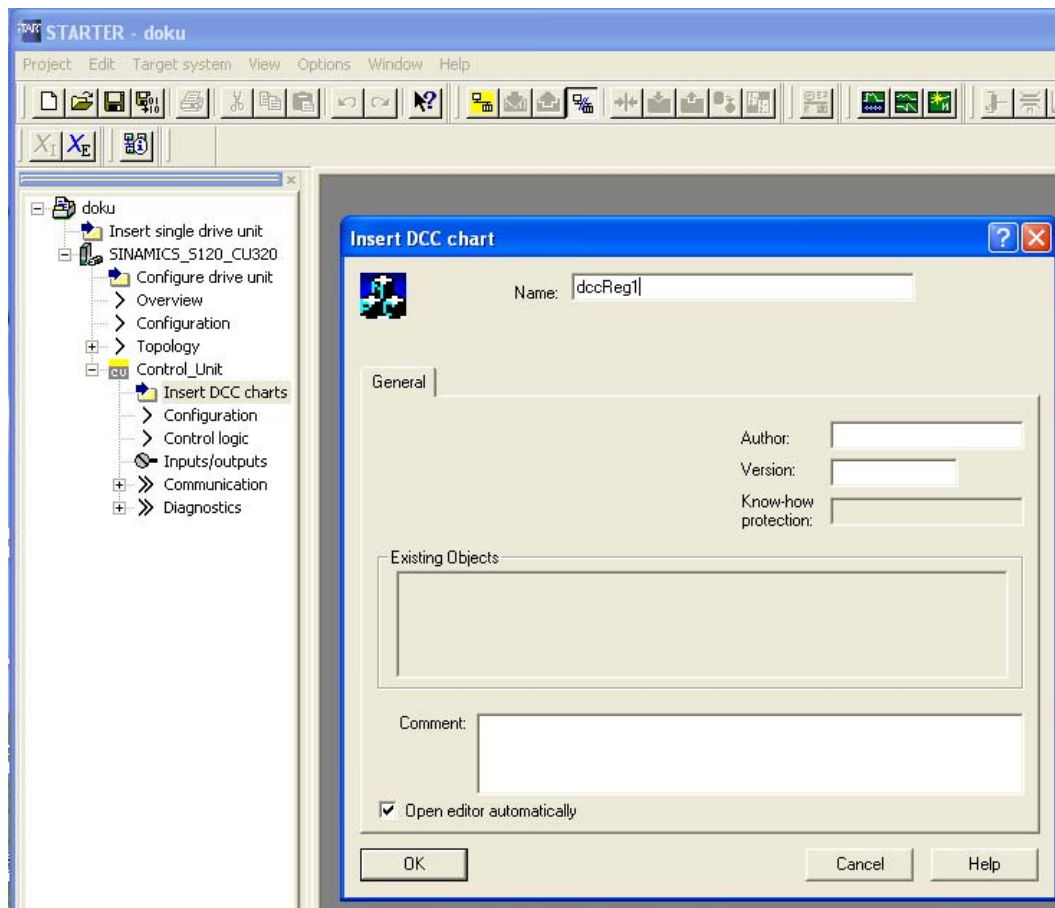


Figure 3-10 Inserting a DCC SINAMICS chart

- The **Insert DCC** window opens and you can enter a new name (of no more than 22 characters) for the chart (in our example *dccReg1*) and a comment. Please note that all characters after an underscore in the chart name can only be digits.
- Click OK to close the window. The DCC is opened when the **Open editor automatically** checkbox is activated

Alternatively, you can open the chart at any time by double-clicking on the chart symbol in the project navigator.

- When first creating a chart in a project, you will be prompted to import a block library.

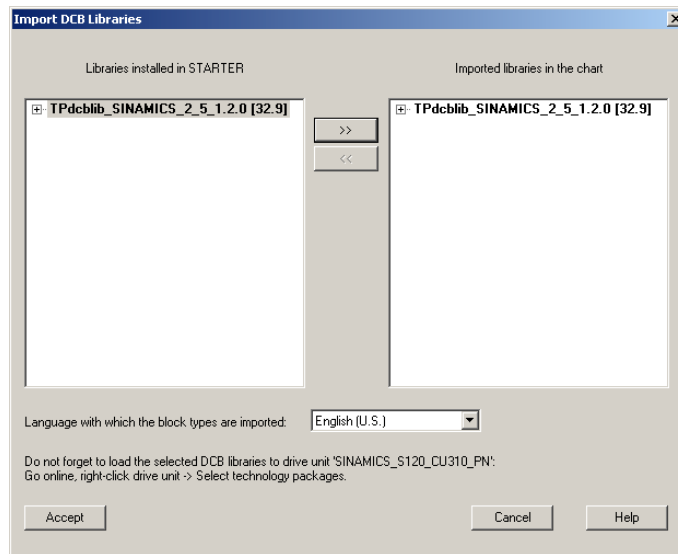


Figure 3-11 Importing a block library

- In the window **Import DCB Libraries**, select the block library in the left column **Libraries installed in STARTER**. Take the selected library into the right column by clicking the **>>** button.
- Close the window with the **Accept** command. The block library is loaded and the DCC editor opened with the chart.

The project structure has now been set up, a chart created and the block library loaded. All that remains is to create some activity within the chart, i.e. by inserting blocks and interconnecting them. Just one chart can be created for each drive object.

3.2.4 Inserting blocks

- Open a block family in the **Closed-loop control** family with the closed-loop control blocks.

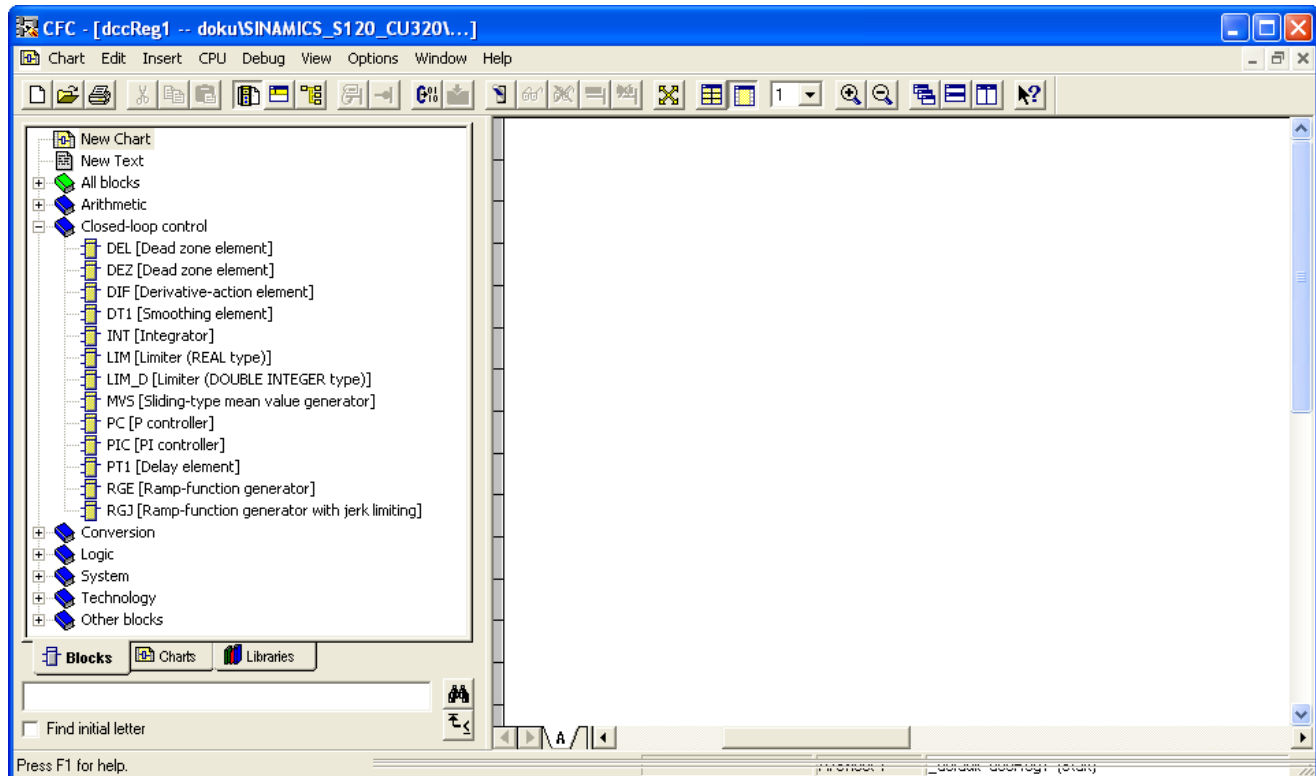


Figure 3-12 DCC editor with opened **Closed-Loop Control** DCB family

- Select the required block (e.g. INT) and insert it in the chart using drag-and-drop. Only the outline of the block in dashed lines is displayed during the copying procedure. Release the mouse button at the required point.

Note

If one block superimposes another block in the chart, the superimposed block will be displayed in gray and its connections will not be visible. You must reposition the blocks to ensure that all connections and block information are displayed.

3.2.5 Interconnecting blocks

Procedure

- Select the **Y** output of the first **INT**egrator, followed by the **X** input of the second **INT**egrator.
- Select the **Y** output of the second **INT**egrator, followed by the **X** input of the inverter (**SII**).
- Select the **Y** output of the inverter (**SII**), followed by the **X** input of the first **INT**egrator.

The autorouter creates the connecting lines from the outputs to the inputs and they are then interconnected.

3.2.6 Parameterizing block connections in the chart

For the first integrator INT 1/1, the following initial values should be assigned to the connections: LL = -10.0, LU = 10.0, SV = 2.0, Ti = 100 ms

For the second integrator INT 1/2, the following initial values should be assigned to the connections: LL = -10.0, LU = 10.0, Ti = 100 ms

To do this, open the **Properties - Connection** window of the respective block connection by double-clicking it. Enter the initial value for **Value** and click **OK** to close the window. Note that for the input of the values for Ti, the "ms" unit must follow the numeric value 100 without any spaces.

As the connections mentioned above are not interconnected, the entered values also remain valid after the initialization. The initialization of the block inputs and outputs is performed before the first cyclic call of the chart.

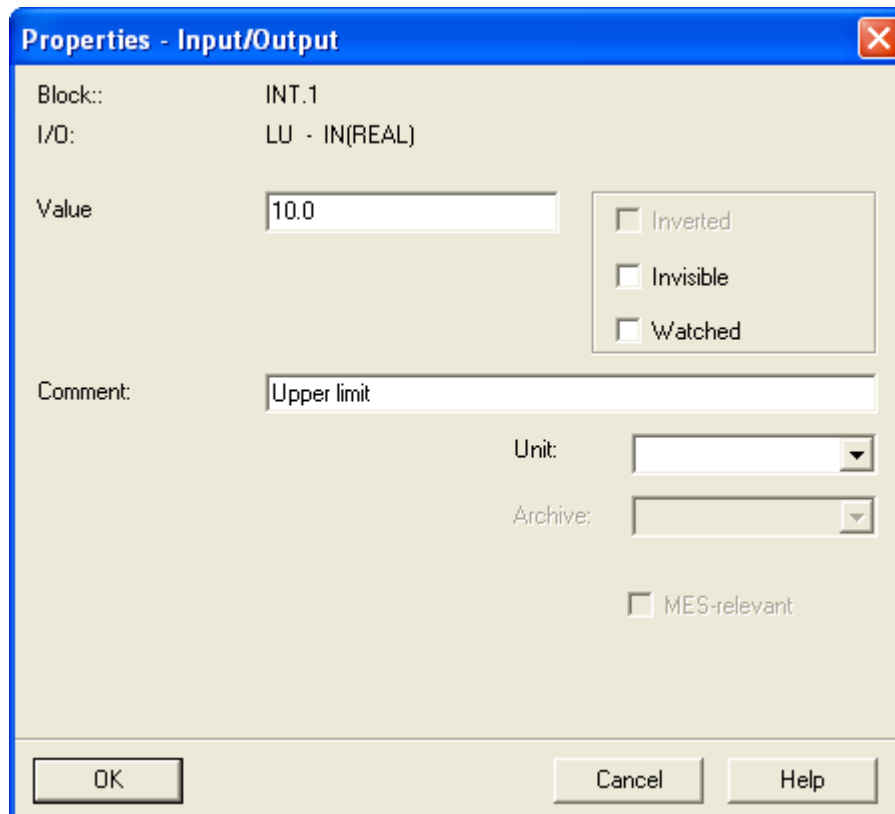


Figure 3-13 Properties window of the "LU" block connection

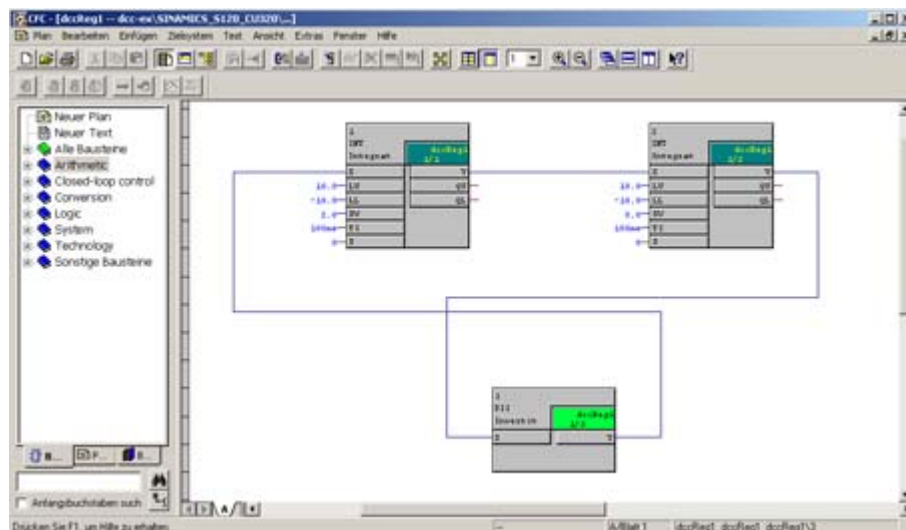


Figure 3-14 Chart "dccReg1" with interconnected blocks. Some connections have been assigned initial values that differ from the factory setting.

3.2.7 Publishing block connections as parameters

To be able to interconnect the output signal of the second (right-hand) integrator to the SINAMICS basic system, it must be published as an interconnectable parameters (i.e. as BICO parameter). To excite the oscillating circuit once, the integrator input of the first (left-hand) integrator must be published as adjustable parameter. The parameter numbers used in the following section have been chosen arbitrarily.

The Y connection of the second integrator is to be published as adjustable parameter. To do this, enter "@*20 output", for example, in the comment field in the Properties window (double-click connection Y). See Creating customer-specific parameters (Page 98).

Enter "@1 start" in the comment field of connection S (set) of the first integrator. This publishes the block connection as adjustable parameter.

As the default setting of the parameter number base of a chart is always 0, when the chart is compiled, the new parameters r21501 and r21520 are created and displayed in the expert list of the CU drive object in STARTER.

Note

To make this assignment of the @ parameters to the connections in the chart visible, the display form of the connections in the DCC editor must be changed. To do this, open the **Display Settings** window in **Options > Settings > Display**, change there under **Connections** the display from **Name** to **Comment** and click **OK** to close the window.

The chart will then be compiled.

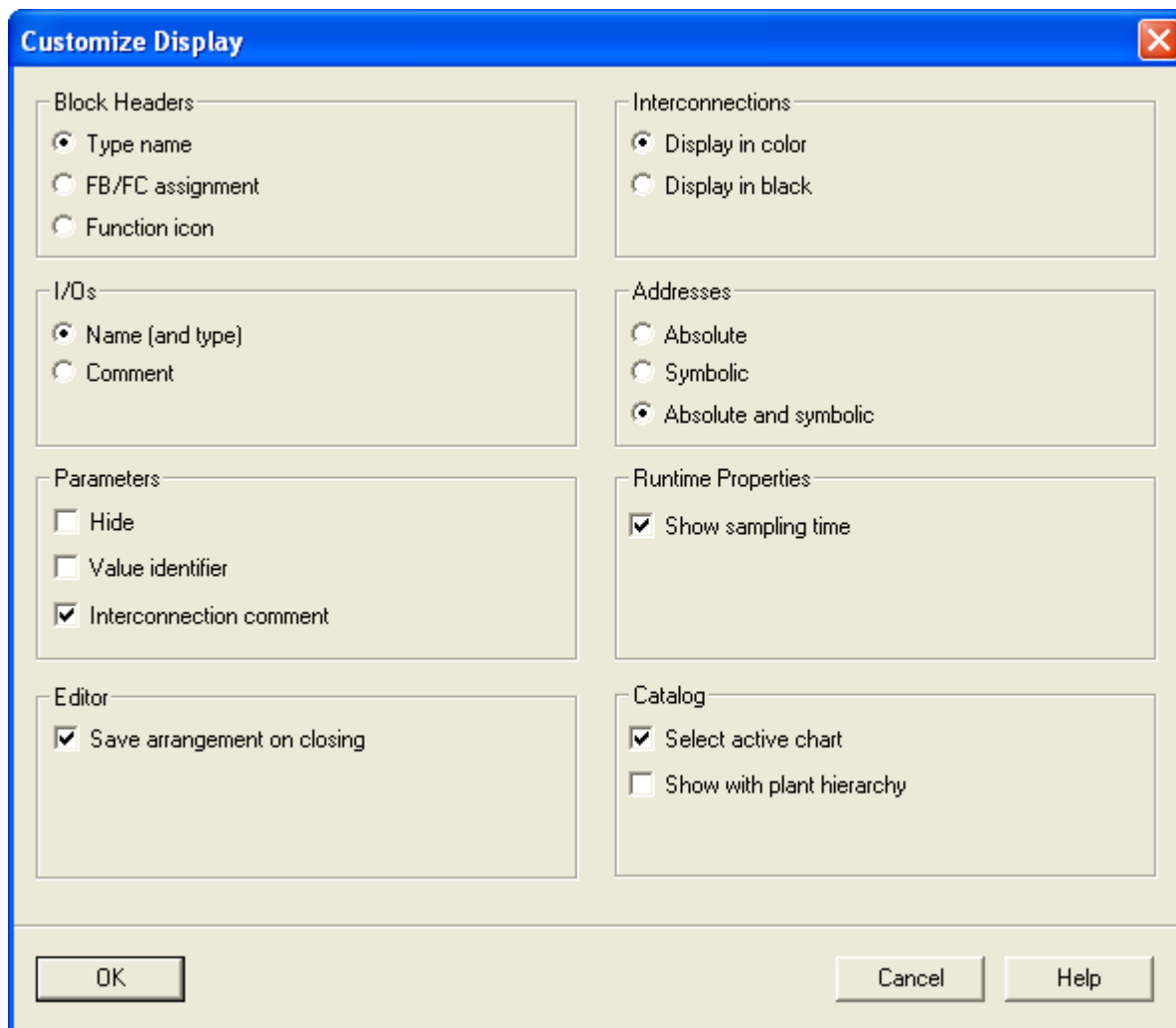


Figure 3-15 Display Settings window

p21000[0]	+	Run-time group properties, Run-time group 1	1 * r21002 (1)		Ready to run	1		
r21001[0]	+	Run-time group sampling time, Run-time group 1	4.000	System of units,		1		
r21002		Basis sampling time, hardware time slices	4.000	System of units,		1		
r21003		Basis sampling time, software time slices	8.000	System of units,		1		
r21008[0]	+	Hardware sampling times available	0.125	System of units,		3		
p21501	+	Start	0H	Operation	1	0H	1H	
r21520		OUT INT2	-9.403		1			

Figure 3-16 View from the expert list with the parameters defined in the DCC

3.2.8 Compiling the DCC chart in the DCC editor

Compiling

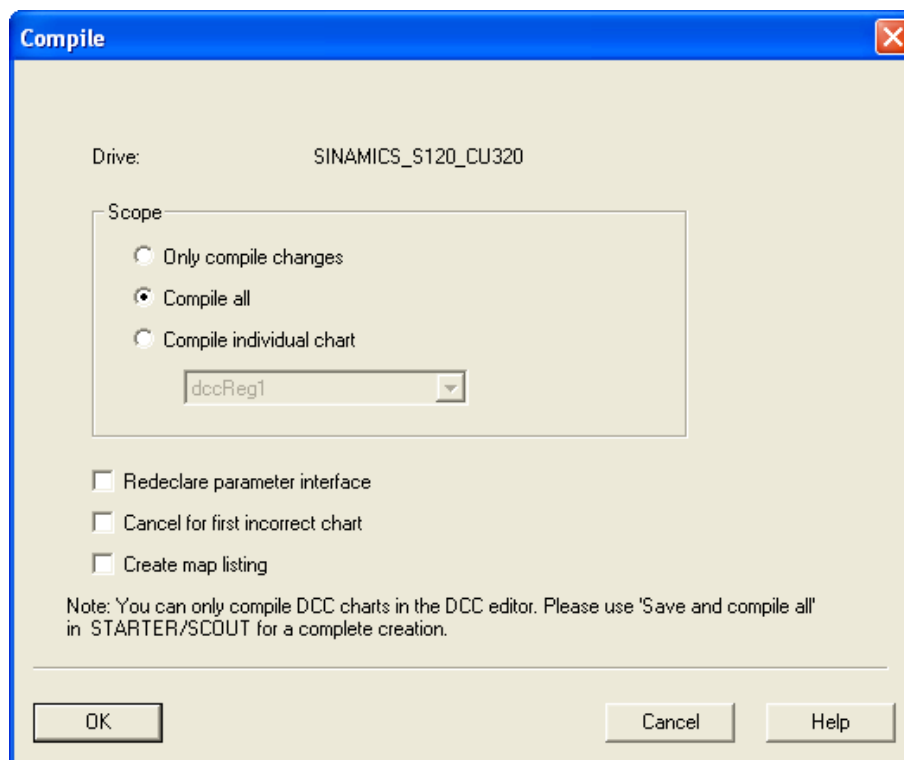


Figure 3-17 Compiling a dialog in the DCC editor

Before the first compilation from the DCC editor, the project must be saved once in STARTER ( button).

You can begin compiling from the DCC editor with **Chart > Compile > Charts as program...** or via the  button.

You can also start the compiling and saving of a project from the STARTER by clicking the  button.

If errors occur during compilation, the **Logs** dialog box will automatically be displayed at the end of the procedure (just as in the case of the consistency check).

Compilation options

For detailed information about the compilation options, refer to Compiling (Page 48).

After compilation

The compilation log is displayed after compilation. If error messages are displayed, the causes must be corrected before continuing.

3.2.9 Setting execution sequence within an execution group

These are automatically assigned to execution group 1 when the blocks are inserted in a new chart. Execution group 1 is automatically given the name of the chart. The sequence corresponds to the order in which they are inserted in the chart. The sequence can be displayed and edited in the execution editor. You can access the execution editor in the DCC editor via **Edit > Execution sequence** or with . If an execution group (light blue folder) is selected in the center column in the tree structure, the blocks contained in the execution group are displayed in the right-hand column.

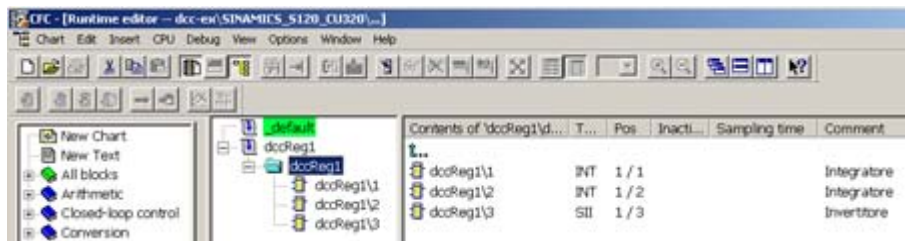


Figure 3-18 Execution editor with open execution group dccReg1

You can return to the DCC by selecting **Edit > Execution sequence** again or by clicking  again.

3.2.10 Setting sampling time for an execution group

The sampling time for an execution group is always set in the STARTER. The DCC you want is highlighted in the project navigator and the **Set execution groups** item called from the chart's context menu. The **Set Execution Groups** window opens.

Our chart was created on the CU drive object. $r21002 = 4$ ms in this case. A free execution group with a sampling time of $1 \cdot r21002$ is set.

Note

In the offline mode of the STARTER / SCOUT V4.1.x, $r21002$ and $r21003$ are always displayed with the value. The $r21002$ value is identical with the $p0115[0]$ value on the associated drive object.

When setting the sampling time, note that the minimum sampling time in the drive for DCC execution groups is 1 ms. Only times that are shorter than $r21003$ may be chosen as a multiple of $r21002$.

Please re-compile the chart once you have set the sampling time.

3.2.11 Loading the DCC technology option onto the CF card of the drive device

When the drive unit is supplied, the DCC technology option will not yet be present on the CF card of the drive unit. This is loaded to the CF card with the aid of SIMOTION SCOUT or STARTER.

To do this, call the STARTER, for example, and connect to the drive unit online. STARTER is in online mode.

Then in the drive unit's context menu, select the **Select technology packages...** command.

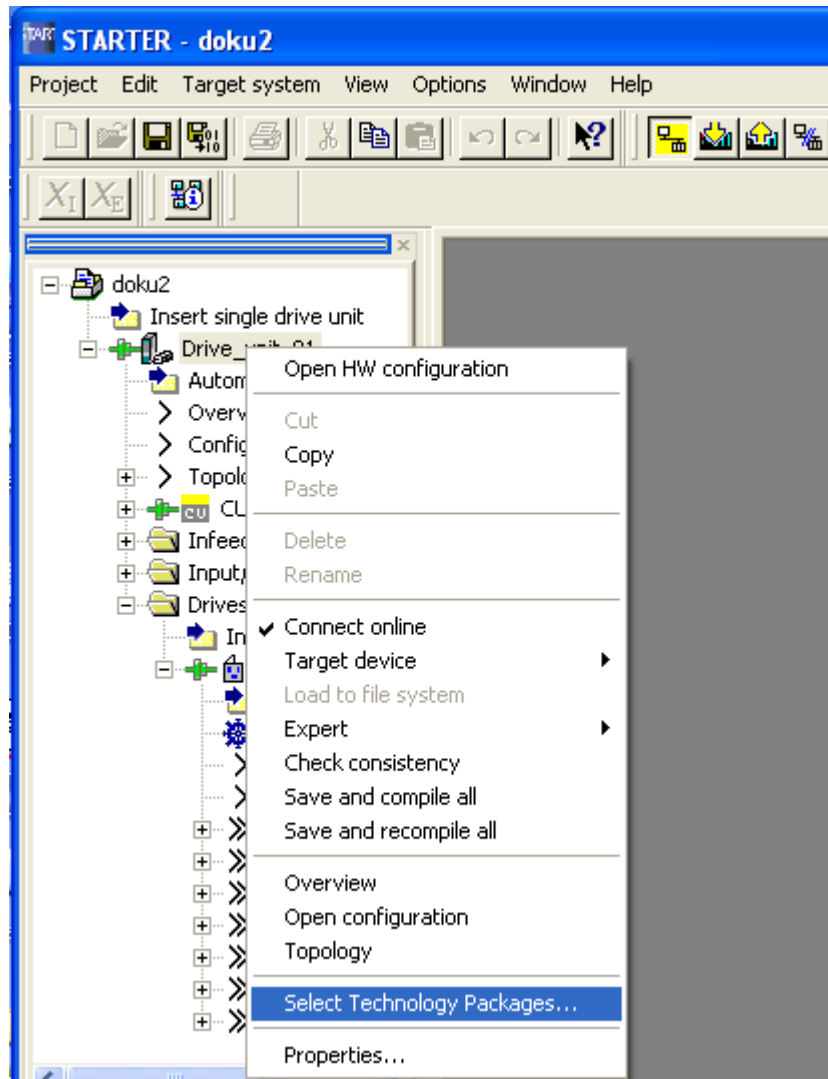


Figure 3-19 Context menu of the drive unit

In the **Select Technology Packages** window, select the desired block library and set **Download** for this library in the action column. Then click **Execute** in the bottom right of the window.

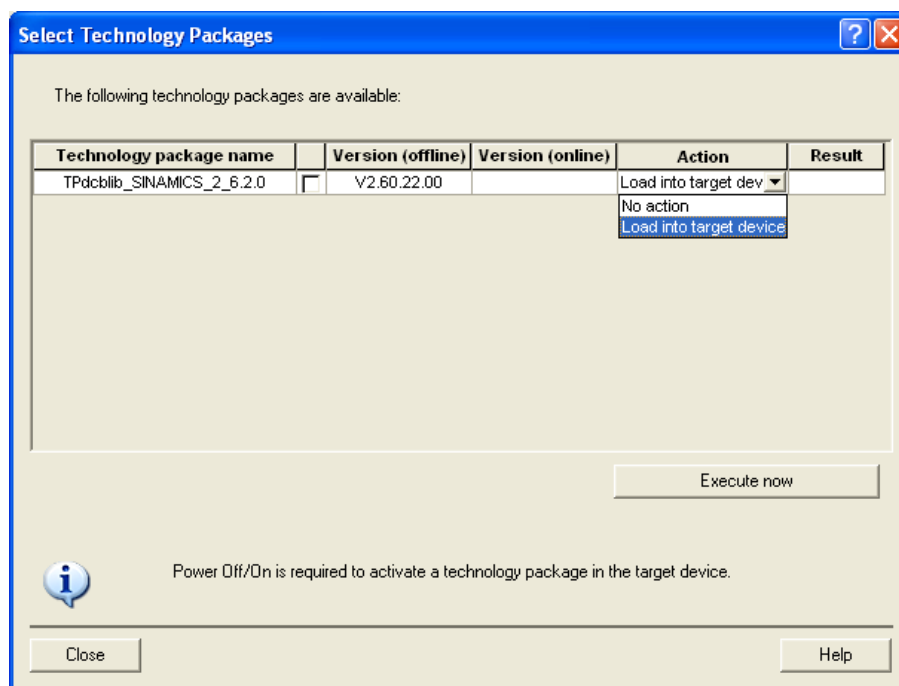


Figure 3-20 Select Technology Packages window

A progress bar shows how the downloading process is progressing. Once the downloading process is complete, the button in the bottom left changes from **Cancel** to **OK**.

At present, the **Load to PG** action is not available for DCC SINAMICS.

The window is closed by double-clicking **OK**.

NOTICE

Once the technology package has been downloaded, the drive unit has to be switched off and on again once. Only once the supply voltage to the CU and/or D4xx has been switched back on and it has been powered up can DCCs be downloaded to the drive unit and run.

Note

When downloading the technology package, as few people as possible should be connected to the bus, as the download time is significantly increased when multiple users are connected.

3.2.12 Downloading compiled DCC chart into the drive

To be able to execute the DCC program on a drive object, it must have first been downloaded to the drive unit. To do this, establish a connection to the drive unit and click the **Download** button in the Online/Offline Comparison window.

Downloading can take place in STARTER at any time via the **Download** function.

After the successful download, the DCC is calculated in cyclic operation on the CU drive object.

3.2.13 Displaying values of block connections online

As of SINAMICS V2.6 and SCOUT/STARTER V4.1.2, the DCC editor can display the values of block inputs and outputs online in test mode. The display is independent of whether a block input or output has been published as parameter. SCOUT/STARTER must be in online mode.

The signal value of block inputs (= connector inputs) published as BICO parameters cannot be displayed. Instead, the signal value of the supplying output can be displayed.

As soon as the test mode is exited again (e.g. by clicking ) , the yellow fields with the value display disappear.

If you want to display the values of the connections logged on once for display, by selecting the test mode again, you must set **Test > Laboratory mode** in the DCC editor. This setting is only possible when the DCC editor is not in test mode.

In our example, the output values Y of the two integrators are to be displayed. To do this, open successively in the DCC editor with a double-click on the connection of the **Properties – Connection** window and with a mouse click set a tick for **For test** on the right-hand side in the middle. This registers these outputs for display in test mode. Then use **Test > Test mode** or Ctrl-T or  in the DCC editor to switch to test mode. The values of the two block outputs are highlighted in yellow and refreshed with a monitoring cycle of 2 seconds.

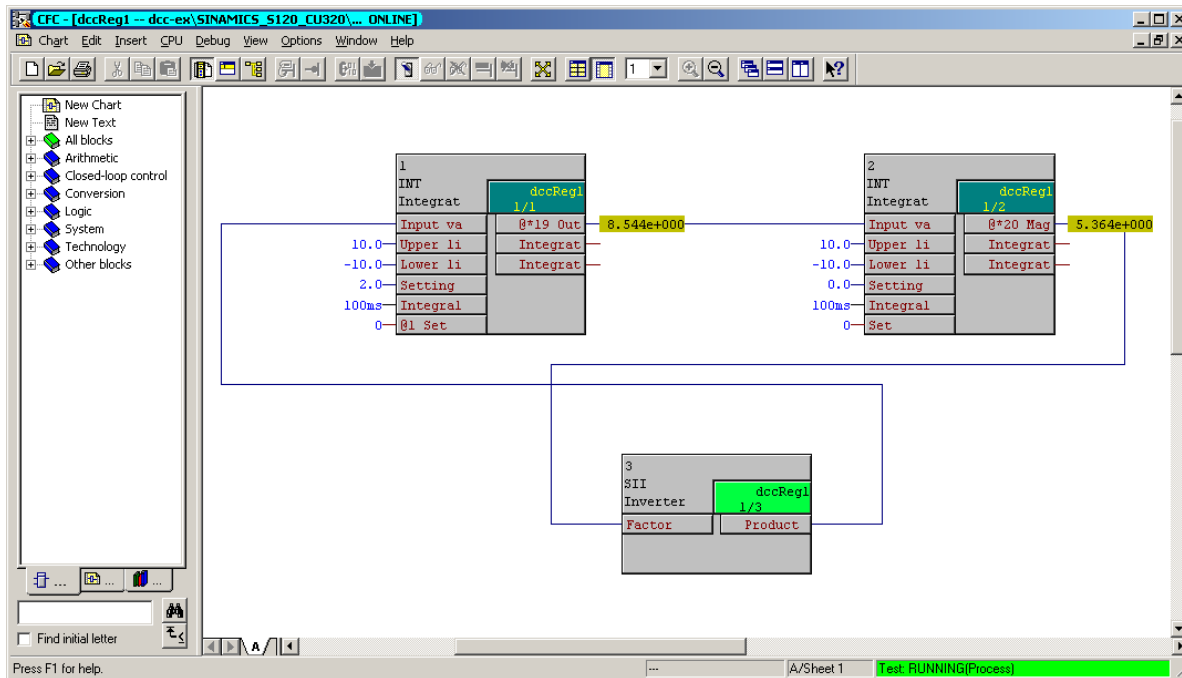


Figure 3-21 Display of the DCC dccReg1 in test mode

To make the oscillator vibrate, it must be excited once. To do this, set the p21501 parameter in the expert list of the STARTER once to "1" (the set value SV = 2.0 is applied at the output of the left-hand integrator) and then reset to "0". The oscillator vibrates and the output values of the integrators show changing values in the chart and in the expert list (r21520).

If you want to make further changes to the chart, you can exit the test mode with **Test > Test mode** or by clicking the  button. It can take several seconds before the test mode of the DCC editor is exited.

If you want to display the values of further block connections in test mode, right-click the desired block connection and select **Log on connection** in the context menu.

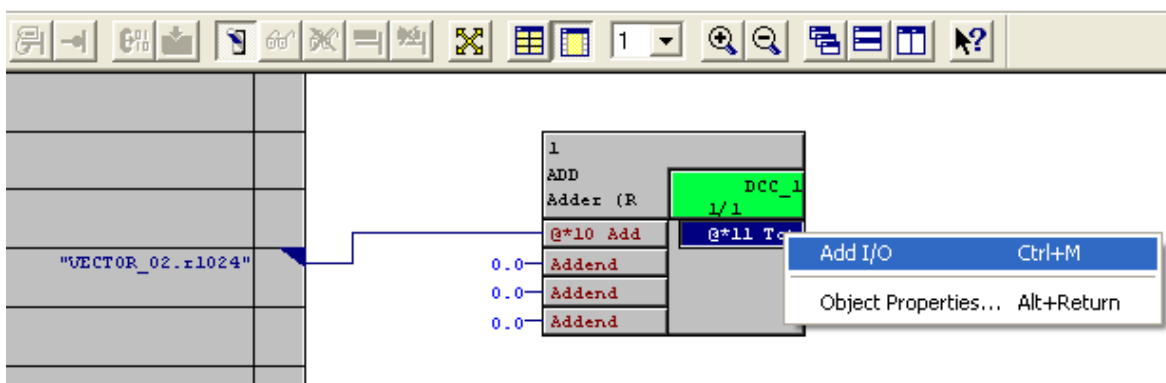


Figure 3-22 Log on connection context menu

If you want to display the signal value of a BICO output of the SINAMICS basic system, the signal value supplied to the DCC via the sheet bar cannot be displayed directly. In this case, an NOP_xx-block with the correct data type is inserted. The signal value of the output of the NOP_xx- can then be displayed online.

3.2.14 Interconnection to the BICO parameters of the basic system in DCC SINAMICS

Procedure

You can make an interconnection to the BICO parameters of the basic system as follows:

1. Open the DCC.
2. Select the DCB connection to be interconnected.
3. Publish the selected connection as a BICO parameter (the comment for the connection must start with "@*", see Creating customer-specific parameters ("publishing") (Page 98))
4. Right-click and select **Interconnection to operand...** from the context menu. The DCC Signal Selection window appears.
5. Select the parameter to be interconnected.
6. Click OK to close the window.

The block connection is interconnected to the selected BICO parameter.

Note

The interconnected block connections must be published as BICO parameters for connections to signal outputs or signal inputs of the basic system.

The connection can first be made in the DCC editor without publishing the block connections. However, the connection of an unpublished block connection to a BICO parameter of the basic system results in an error when compiling the chart.

Deleting and moving BICO interconnections

When moving BICO interconnections, the signal can acquire the value 0 for a few cycles. The no longer interconnected input on the original connection is permanently assigned the value 0.

3.2.15 BICO interconnections and reading back parameters

To display BICO interconnections (which have been changed online in the target device) in the DCC editor both online and up-to-date, the project must be read back from the target device. Proceed as follows:

Procedure

Subsequently changed BICO interconnections and parameter values can be read back from the target system via the menu command of the DCC editor **File -> BICO interconnections and read back parameters**.

Input and output BICO interconnections are read back. The read back process always applies to all DCC charts of a device.

Requirement

- The structure of the block instances and interconnections are the same both online and offline.
- The sources of the DCC chart must be available.
- It is not possible to read back BICO interconnections in the case of DCC charts that have been loaded from a target device into an empty project.

3.2.16 Record with the trace signals from the DCC chart

The integrator output published as interconnectable parameter acts like any other connector output of the SINAMICS basic unit. This means this signal can also be recorded with the trace. The signal characteristic at block connections (that are not published as parameters) can still be recorded with the trace. The trace is called in STARTER with **Target system > Trace** or . In this example, the outputs of the two integrators in the DCC are recorded with a cycle clock of 1 ms. The output of the first (left-hand) integrator is to be recorded as first signal in the trace. The signal selection is opened with  for this. Select the drive object on which the DCC is located (only CU_S_126). Open the directory tree and the dccReg1 chart is displayed. Open the directory tree of dccReg1 and the three blocks in the DCC are displayed. The desired connection is on block 1 (top left digit in the block symbol) of the dccReg1 chart. The designation of the connections in the signal selection of the trace is made up of the _ChartName_BlockNumber_, i.e. _dccReg1_1 in this case. The desired connection can now be selected in the list of block connections. The window is then closed by clicking **OK**. See the following figure.

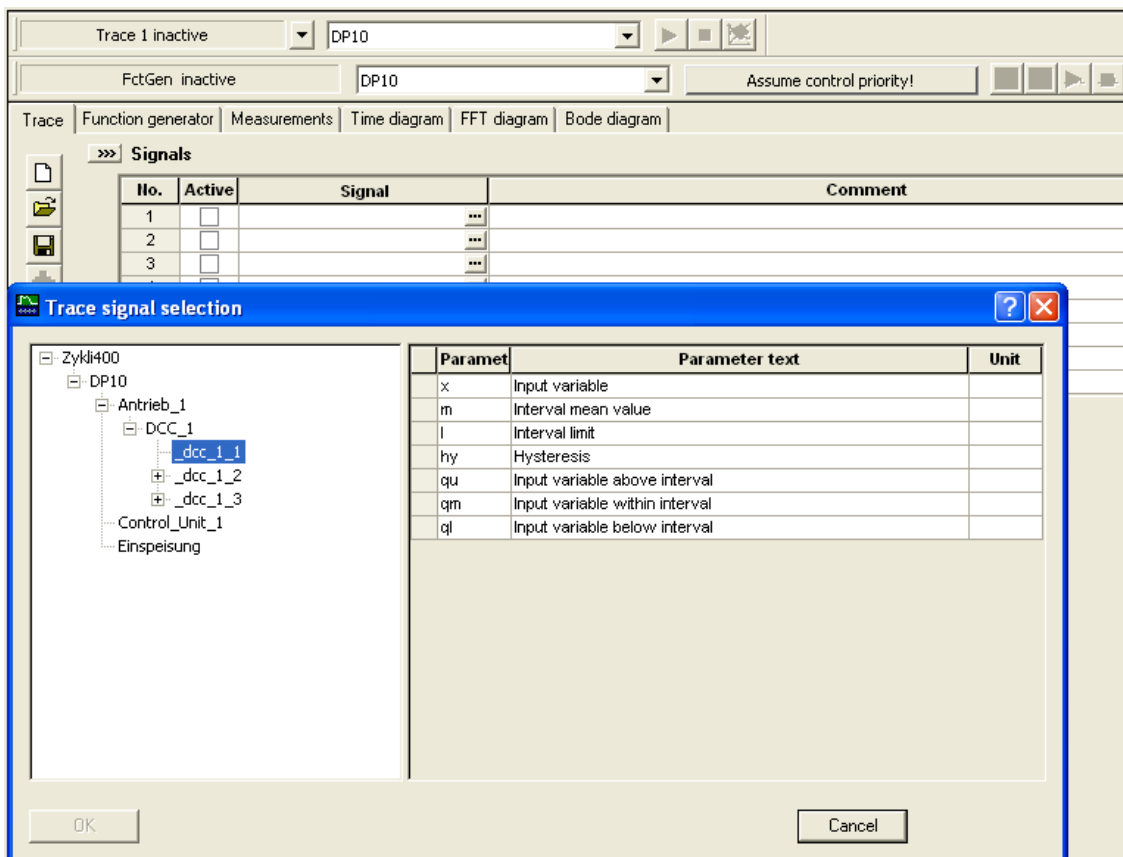


Figure 3-23 Trace signal selection for the block connections of the _dccreg1_1 block in the dccReg1 chart

The output of the second (right-hand) integrator is to be set as second signal. This block output is published as BICO parameter r21520. The recording can be started with the  button.

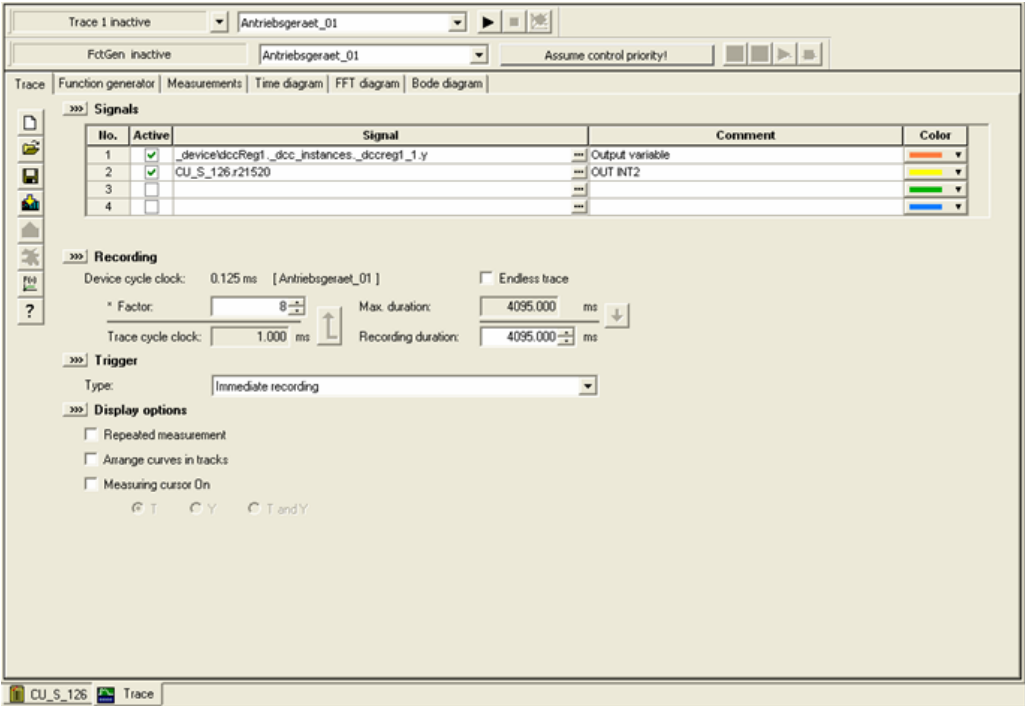


Figure 3-24 Parameterization of the trace

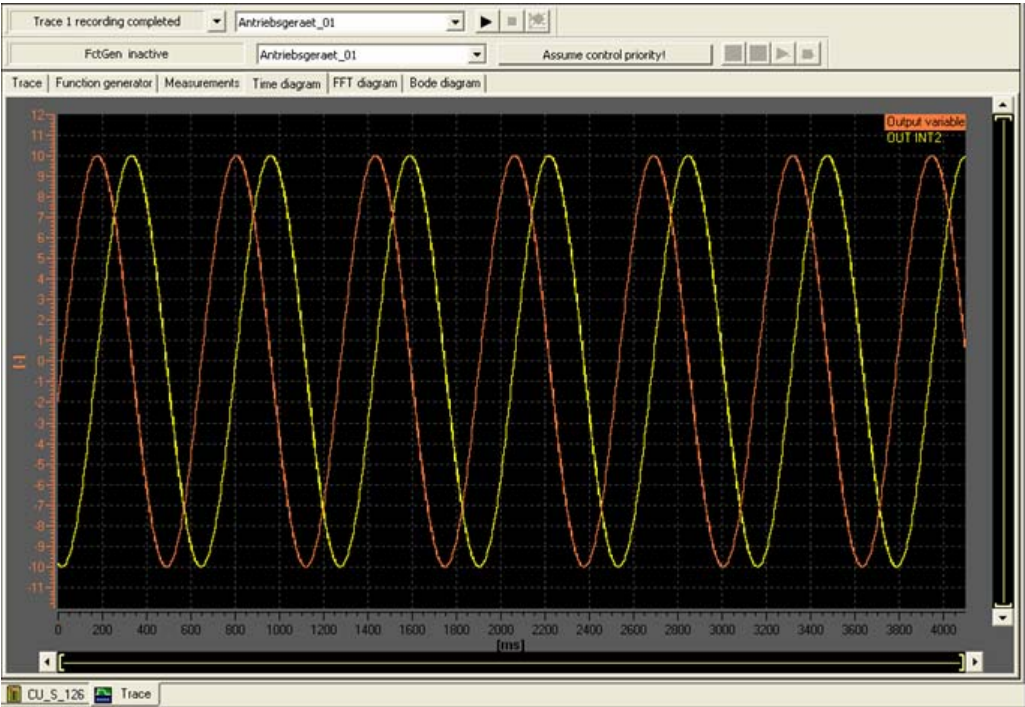


Figure 3-25 Display of the recorded signals as timing diagram

Note

When inserting block instances and connections online, the trace of the signal must be restarted.

3.2.17 Archiving a project

If during or after the commissioning, you want to archive the project, including the DCCs, on a data carrier, this is possible with **Project > Archive** in STARTER.

3.2.18 Creating documentation

Complete documentation

The example has been successfully configured and tested. You can now create the complete documentation for your example.

Chart reference data

Alternatively, the chart reference data also displays the execution groups with the block types and the execution sequence. Use **Options > Chart reference data** in the DCC editor to open the **Chart Ref** window.

In the **Chart Ref** window, click the  button to also display the blocks in the configured execution groups. To begin printing, click: 

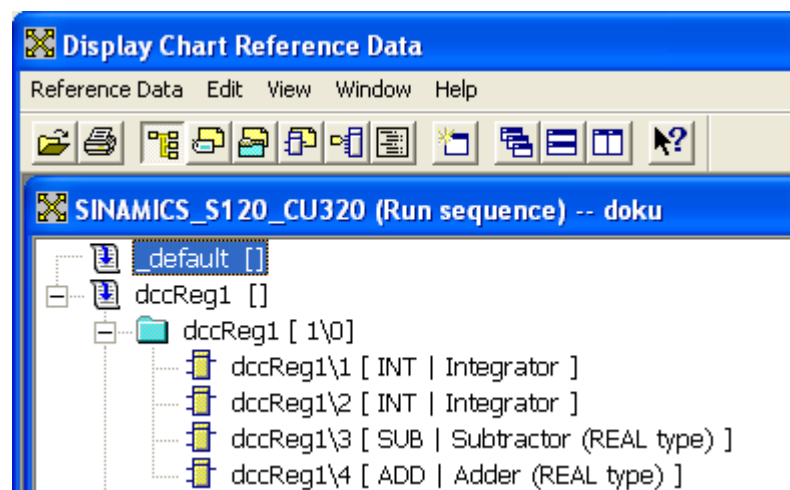


Figure 3-26 Chart Ref. window: Display chart reference data

Printing a chart

You should also print the chart to document the interconnection of the blocks. Because only one page was used on this chart, it suffices to print the current page. To begin printing, click:



If your chart consists of several pages, we recommend that you print them individually in the page view. Click the button to change to the page view.

Click or to customize the display so that all blocks are shown on a single page on the PC monitor.

Assignment of execution groups

To document the assignment of the execution groups, select the **Set execution groups** command in STARTER in the context menu of the chart. The **Set Execution Groups** window appears. Print a screenshot.

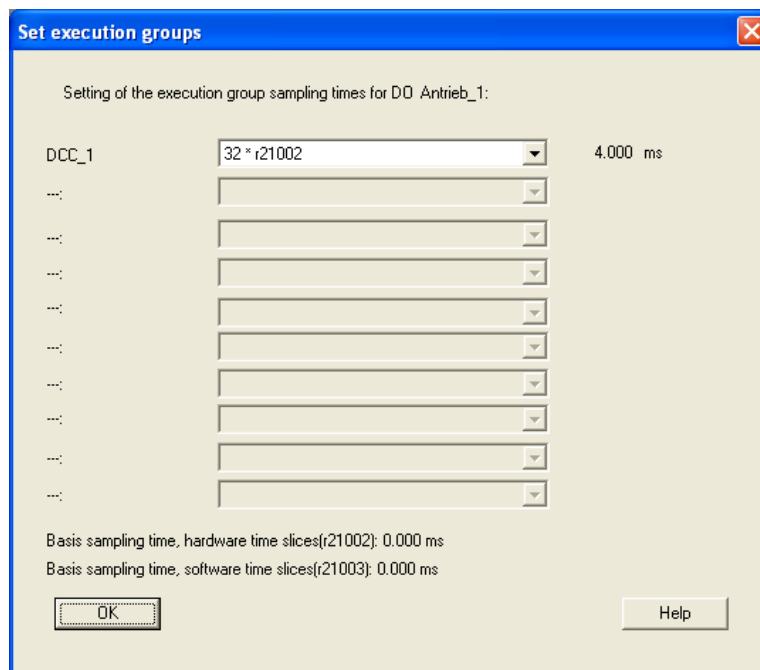


Figure 3-27 Set execution groups window

The change of the hardware r21002 basic sampling time displays the value for p0115[0] in the CU ramp-up of 1 ms. Here p0115[0] was changed from 4000 µs (factory setting) in offline mode to 1000 µs and then downloaded to the CU.

3.3 Connecting the DCC to the drive

3.3.1 Overview

Only DCB connections of a DCC chart declared as BICO parameters can be connected to the connector inputs (CI) and connector outputs (CO) of the drive.

All block interfaces declared as BICO parameters are per-unit variables. This means that calculations are performed with per-unit signal values within DCC. (1.0 corresponds to 100%). The conversion to the connector units of the drive is performed automatically.

3.3.2 Calculating a DCC chart with per-unit variables

Example 1.1 (interconnecting input value)

Disconnect any online connection to the drive unit in STARTER.

- p1020 = 1 and

- p1021 = p1022 = p1023 = 0

(function block diagram 2505) are set on a drive object of the SERVO (the "Extended setpoint channel" function module must be activated) or VECTOR type. Fixed speed setpoint 1 (p1001 in function block diagram 3010) is then output on r1024 (Speed setpoint active).

The following are also set:

- p1001 = 1500 rpm and

- p2000 = 3000 rpm (reference speed).

Insert a DCC on the drive object.

Insert the ADD block in this chart. Connections X1 and Y of the ADD block are published as (interconnectable) BICO parameters.

Right-click twice on input X1. The context menu opens. Select **Interconnection to operand....** In the signal selection window DCC now select r1024 (CO: actual speed value effective) and confirm with **OK**.

Compile the chart by selecting **Chart > Compile > Charts as program....**

Select Set execution groups... in the context menu of the DCC, set the only execution group that is present, e.g. to 20 * r21003 and confirm with **OK**.

Go online with STARTER and download the project to the target system.

Then online in the DCC editor, switch the test mode on with **Test > Set test mode**. Now right-click the output of the adder and select **Log on connection** in the context menu.

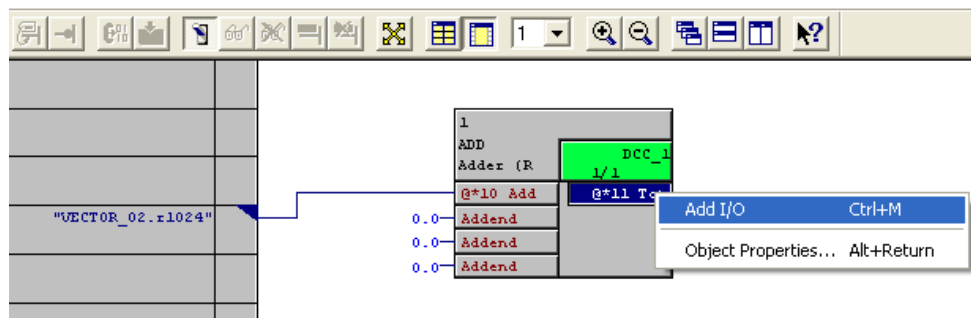


Figure 3-28 Log on connection context menu

The current output value (= per-unit speed setpoint = $1500 \text{ rpm} / 3000 \text{ rpm} = 0.5$) is then displayed.

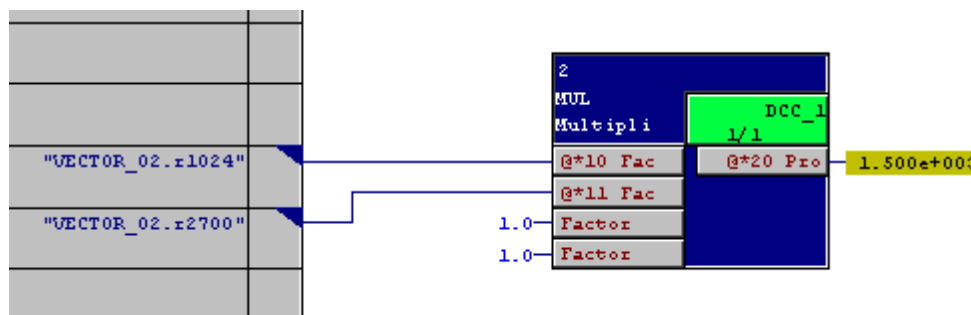


Figure 3-29 Connection value on example 1.1 is displayed

Example 1.2 (interconnecting the output value)

In the DCC chart, a value of e.g. 1.2 should be interconnected with the basic device. The value for this example will be configured as an initializing value to the X input of the NOP module. The module output (Y) has therefore the constant value 1.2, and will be configured as an interconnectable BICO parameter (@*20).

If a connector output of DCC is interconnected with the basic device, this will always be interpreted as a referred value of the basic system. The unit-related absolute value in the basic device now depends on which unit and therefore which reference variable the selected connector input has in the basic system:

When interconnecting the connector input for speed setpoint p1070 (function plan 3030), the referred output value from the DCC chart will automatically be multiplied with the reference variable "Reference speed" p2000 (in the example $p2000 = 1500 \text{ min}^{-1}$). In the SINAMICS basic system, a speed setpoint of 1800 min^{-1} ($= p2000 * 1.2 = 1500 \text{ min}^{-1} * 1.2$) will thus become effective.

For this reason, always take note in the case of BICO interconnections with the basic system if you work with absolute variables or with referred variables in DCC. Since the connector output of the DCC chart in the case of interconnection with a unit-related connector input of the basic system will always (internally) be multiplied with the reference variable (belonging to the unit).

3.3.3 Calculating a DCC chart with absolute variables

3.3.3.1 Example 2.1 (interconnecting input value)

If you want to work with absolute variables in the DCC (in our example using speed values), you have to convert the referenced value to which the variables are automatically converted by the drive into an absolute variable (in our example, a speed) using the reference variable. For DCC, the reference variables are provided as monitoring parameters r2700... r2707 which have no units and can be interconnected. Parameters r2700... r2707 have the special feature that the value is transferred 1:1 to DCC without being divided by the reference variable.

On SERVO type (the "Extended setpoint channel" function module must be activated) or VECTOR type drive objects,

– p1020 = 1 and

– p1021 = p1022 = p1023 = 0

(function block diagram 2505) are set. Fixed speed setpoint 1 (p1001 in function block diagram 3010) is then output on r1024 (speed setpoint active, function block diagram 3010).

The following are also set:

– p1001 = 1500 rpm and

– p2000 = 3000 rpm

r1024 (speed setpoint active, function block diagram 3010) then assumes the value 1500 rpm.

The interconnectable monitoring parameter for the r2700 reference speed indicates a value of 3000 in the expert system.

The multiplier MUL is dragged into the chart.

The 1st factor (X1) is published as a BICO parameter and interconnected with r1024. → Factor 1 = 0.5 (referenced speed setpoint active)

The 2nd factor (X2) is published as a BICO parameter and interconnected with r2700. → Factor 2 = 3000.0 (value of reference variable p2000)

At the multiplier output this results in:

$$0.5 * 3000.0 = 1500.0$$

The speed setpoint r1024 is available again in DCC in rpm.

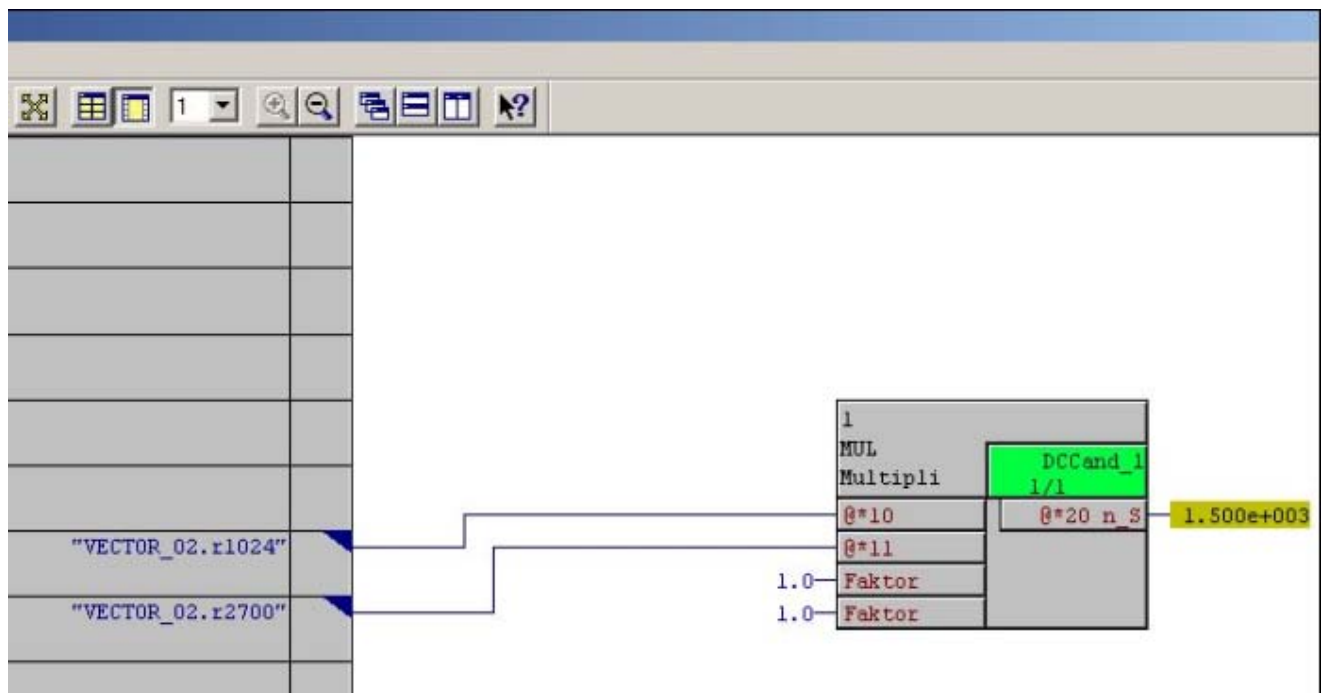


Figure 3-30 DCC for example 2.1 in online test mode

The block was automatically assigned to an execution group when inserted into the chart. The free execution group 20 * r21003 (selected at random) has been assigned to the execution group in the **Set execution group** menu.

3.3.3.2 Example 2.2 (interconnecting output value)

Absolute values are used in the DCC to calculate the torque in Nm units. The torque setpoint calculated is to be passed on to the "Extra torque M_extra 2" (p1513, function block diagram 6060) connector input. For this to happen, the absolute value must be converted into a referenced torque while still in the DCC. When connecting a real type (== floating point) connector output (CO) from the DCC with connector input (CI) p1513 M_extra 2 [torque in the Nm unit] on the drive, the automatic conversion in the BICO connection always assumes that DCC is providing a referenced signal.

Note

The LIM type block supplies the absolute torque of 0.204625 Nm calculated in the chart at its Y output. This absolute value is now divided by the reference variable for torque r2703 = 0.8185 Nm and the referenced torque value for interconnecting with the drive is thereby calculated. The referenced torque value is available at the block DIV's output and is published as BICO parameter r21530 (value = 0.2499 = 0.204625/0.8185). Block NOP1 has only been inserted so that the value of reference torque r2703 can be shown in the chart at its r21527 output published as a BICO parameter.

Monitoring parameter r1515 (total extra torque, function block diagram 6060) is only calculated if the speed controller is enabled and the drive is magnetized (r0056.4 = 1 = yes). (A machine must be connected to the motor modules and be running for this purpose.) Otherwise the value is set to zero.

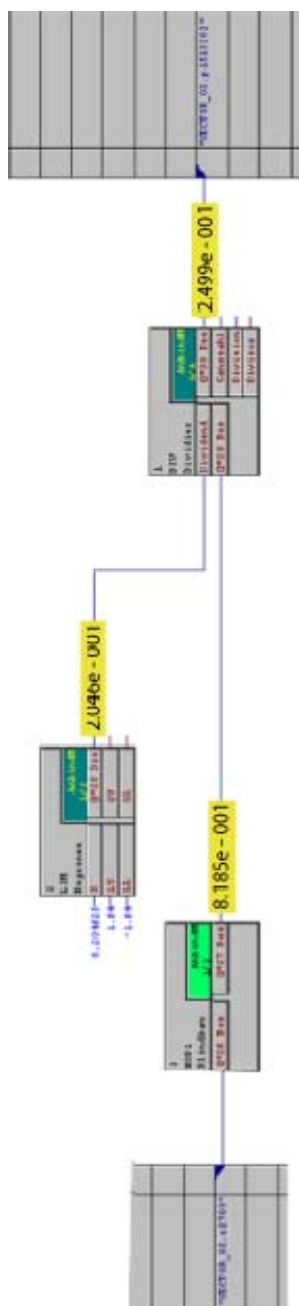


Figure 3-31 Interconnecting output value

3.3.4 Interconnecting DCC signals with communication interfaces IF1 and IF2

3.3.4.1 Preliminary remark

The basic system can be connected to process data interfaces IF1 and IF2 via free telegram configuration using BICO (p0922 = 999) or via (standard) telegrams depending on p0922. To interconnect with the basic system, the DCC connections needed must be published as BICO parameters. These DCC parameters should always be interconnected with the basic system in the DCC Editor by calling the context menu **Interconnection to operand**.

3.3.4.2 Interconnecting received process data with DCC

When interconnecting the received PZD data (see SINAMICS S Parameter Manual, function block diagrams 9206 and 9204), the particular way in which the connector outputs (COs) of PZD processing behave (for IF1 r2050, r2060 and for IF2 r8850 and r8860) should be noted. These COs (connector outputs) can either provide their signals in whole numbers (integers) or as floating point values (REAL). The data type provided is determined by the first signal linked to this CO. This can also be determined by a standard message frame having been selected (p0922 != 999) on a drive object. The PZD COs are thereby automatically interconnected according to the definition of the message frame selected. This interconnection is not canceled when resetting p0922 = 999 == free message frame configuration with BICO.

p1070 = 2060[2] is set for example. r2060[2] (PZD received word 3 and 4) is thereby connected to the main setpoint (in function block diagram 3030) of a REAL variable of the basic system. This means that only connector inputs from DCC with the REAL signal data type can be interconnected on r2060[2].

Note

Connections to integer inputs can be made both online and offline in the DCC editor; a corresponding error message is only issued when the chart is downloaded.

3.3.4.3 Interconnecting sent process data with DCC

The send data is interconnected as with any other BICO connection.

Detailed descriptions about this topic are contained in the *SINAMICS S120 function manual*.

3.4 DCC SINAMICS specifications

3.4.1 Rules for assigning names in the DCC editor

Names are used for data exchange between SCOUT/STARTER and the DCC editor.

The names in the DCC editor must therefore abide by the following rules:

- Basic chart:
 - No keyword or previously defined name permitted
 - Must start with a letter or underscore
 - May contain numbers, letters and "_"
 - "_" must be followed by a number
- Subchart:
 - Must start with a letter or underscore
 - May contain numbers and letters
 - "_" is not permitted
 - Keyword or previously defined name permitted
 - In the DCC editor, a check is performed to determine whether the name has previously been defined.
- DCB instance:
 - Can start with a number
 - May contain numbers and letters
 - "_" is not permitted at the beginning
 - Keyword or previously defined name permitted
 - In the DCC editor, a check is performed to determine whether the name has previously been defined.
- Execution group:
 - Keyword or previously defined name permitted
 - In the DCC editor, a check is performed to determine whether the name has previously been defined.
- BICO parameter:
 - No previously defined parameter numbers permitted
 - In the DCC editor, a check is performed to determine whether the parameter number has previously been defined.

3.4.2 Field/name lengths and conventions

Field/name lengths and conventions

Object	Length	Remark
Chart name	22 *)	May not contain the following characters: \\ . : / * ? " < > # % () Use of the "_" character is subject to particular specifications.
Chart comment	255	All ANSI characters are permitted.
Execution group	22	Character set as for Chart , including "_".
Block instance comment	80	All ANSI characters are permitted.
Parameter comment	80	All ANSI characters are permitted.
Block name (instance name)	16 *)	May not contain the following characters: \\ . : / * ? " < > # % Use of the "_" character is subject to particular specifications.
Global operand	Max. 49	Interconnection with BICO parameter

*) The chart name and block name must jointly consist of a maximum of 24 characters, including separating characters.

3.4.3 Representation of the dynamic value display

The values are output next to the connections, according to their data type. They are displayed on the screen with a colored background.

Table 3- 2 Representation of the dynamic value display

Representation	Meaning
Blue on white	Representation of the values in edit mode (offline)
Black asterisks on yellow	Values during transfer to the dynamic display
Black value on yellow	Representation of the values read from the drive object in test mode
#### on a red background	Representation of values while the dynamized values required from the drive object are missing (fault, overload)

DCC for SIMOTION

4.1 Overview

4.1.1 Introduction

This product brief is intended for experienced SIMOTION users who are not yet familiar with the DCC program package. Using a short example, you will find out how to create a project after starting up SIMOTION SCOUT, create a DCC chart, interconnect blocks, compile the chart, download it to the target device and test it online.

Note

In most cases, there are a number of options for working with the DCC editor (e.g. using the keyboard). In this example, the quickest or most suitable option is used. Apart from a few exceptions, no alternative methods of operation or procedures are explored here.

4.1.2 Software requirements

The software requirements for DCC are the same as those for SIMOTION SCOUT.

You require an appropriate license to use the DCC editor.

4.1.3 SIMOTION system integration

4.1.3.1 Execution level, execution group and execution sequence

DCC tasks and execution groups

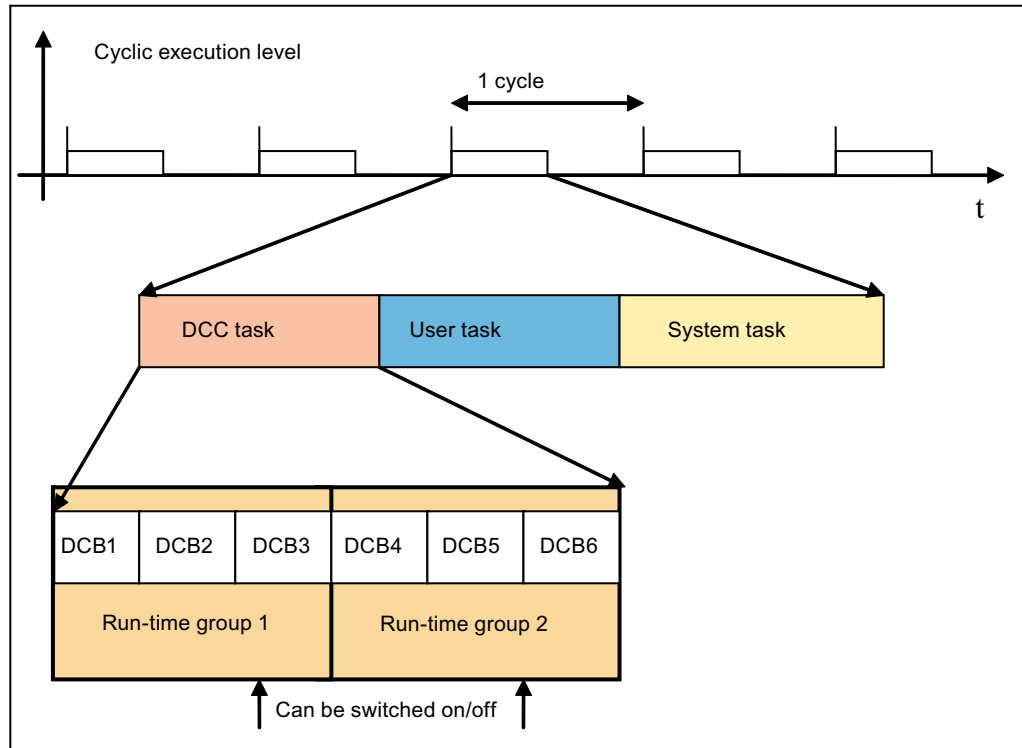


Figure 4-1 Sequence model for blocks

Task

Five DCC tasks are available in each execution level for user programming.

The main features of the tasks are:

- Start behavior: When and under what conditions is a task started?
- Priority: Which task is interrupted by which other task?

Execution groups

The available tasks (sampling time and execution sequence in the system: T1 to T5) are assigned execution groups. The blocks are embedded into these execution groups. The execution groups therefore allow the task to be structured or divided as required by the user; e.g. dancer control, setpoint processing, etc. The execution groups are embedded sequentially into the blocks. **Only blocks from one** basic chart may be contained in an execution group.

You can use the "Enable" attribute to activate and deactivate an execution group. Normally, execution groups are processed in cycles; however, the enable attribute can be used for switching individual execution groups on and off.

A BOOL-type block output can be connected to enable an execution group or a block group. To do this, highlight the connection to be interconnected and select the **Interconnect with execution group** command in the context menu.

Execution sequence

For tasks: The execution sequence corresponds to the sequence in which execution groups and blocks are inserted within a task.

The same applies to groups: The sequence in which the individual blocks are inserted is the execution sequence within the execution group.

You can change the execution sequence.

Setting the system cycle clocks

The properties that determine the execution behavior of a task are configured in SIMOTION. You can carry out parameterization in the properties dialog box of the CPU under **Expert execution system > Set system cycle clocks**.

No multiple insertion of blocks in different tasks

Inserting a block several times in different tasks is not permissible.

Changing the execution sequence

- Click **Edit > Execution sequence** or the  button.

The **Execution Sequence** window appears: In the left-hand side of the window, you will see the structure of the task. In the right-hand window, the content is displayed. The default integration position of the blocks is in cyclic task **T2**, in the default execution group that has the same name as the chart.

To move blocks that have already been inserted from task **T2** to higher-priority task **T1**:

- Double-click the **T2** symbol.
- The execution groups are displayed in the right-hand window:
 - When you highlight the execution group, the blocks it contains are displayed (including the chart/block name, comment and position).
 - Highlight the execution group that you want to move in task **T2** and use drag-and-drop to drag it to the task **T1** symbol in the left-hand window. The right-hand window is now empty.
 - When you double-click task **T1** to open it, the execution group that you have moved will be displayed there.
- Close the window via **Chart > Close** and proceed to the next step.

4.1.3.2 HMI variables (publishing of variables and @ variables)

HMI variables

You can declare block inputs and outputs as **HMI variables** and therefore generate a static interface for these for use in your system visualization. This interface is largely static from an HMI point of view, i.e. not every change made to the DCC configuration demands that the address information of the HMI system be reimported.

The memory image with the HMI variables is **not** automatically deleted during compilation. It is only deleted on request.

HMI variables that are removed from the DCC when the block that defined them is deleted remain in the memory image until **reorganization** is requested during compilation in the DCC.

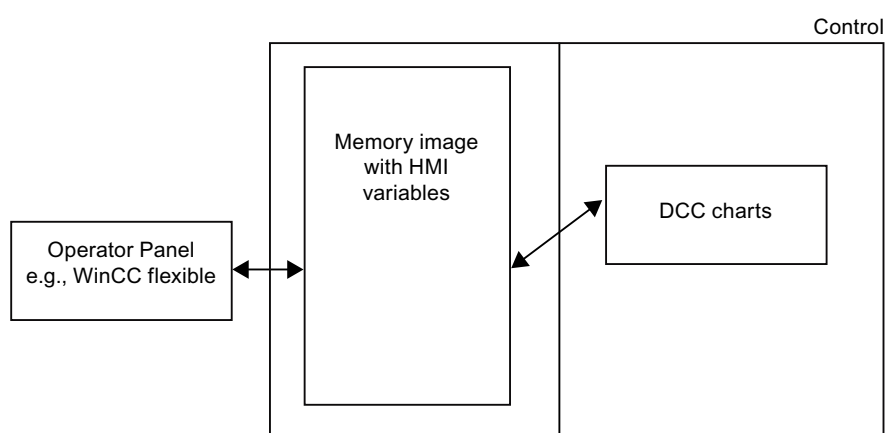


Figure 4-2 HMI variables

@ variables (SIMOTION)

Structure of the @ variables

In the comment, block inputs and outputs are published by the variable name preceded by @.

Table 4- 1 Structure of comments in SIMOTION

Comment	Meaning
SIMOTION: @name <<variable comment>>	The connection is entered as an HMI variable in SIMOTION. Name is the part before the first blank (whereby you must adhere to the ST conventions (Page 176)).

The text which follows the variable name (separated by a blank), is transferred as a variable comment. It is then displayed in the symbol browser. The variable identifier results from the variable name.

The data type of the published input/output is taken over by the block connection. Where necessary, it is mapped onto the appropriate data type of the engineering system.

No help function can be created for @ variables.

Exporting to WinCC

Procedure

A created DCC chart can be exported to WinCC, whereby the export specifications of the SIMOTION SCOUT also apply for DCC charts. However, for this to happen the OPC-XML export must be activated in the Properties dialog box of the chart in SIMOTION SCOUT.

In the engineering system, right-click the chart and select **Properties** in the context menu. The export is activated via **Enable OPC-XML** on the **Compiler** tab.

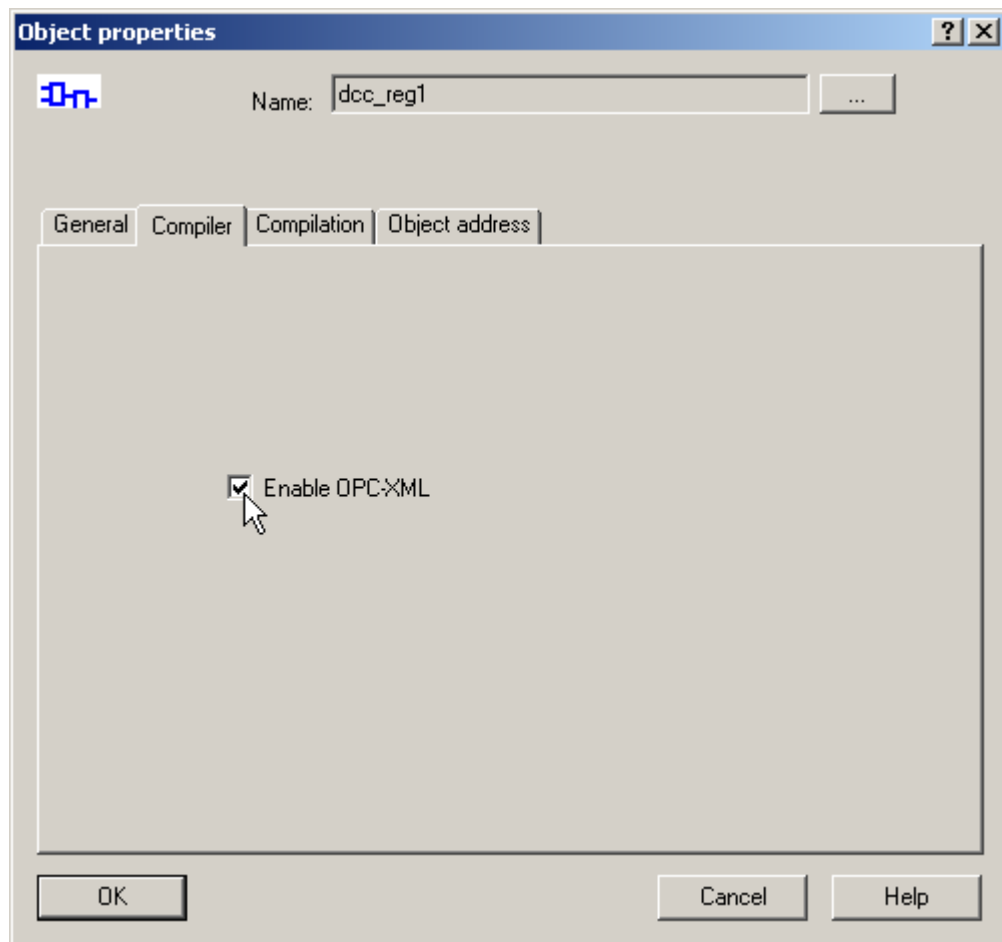


Figure 4-3 DCC chart properties - OPC-XML export

The export can be performed in SIMOTION SCOUT via **Options > Export OPC data**.

4.1.3.3 Interconnecting with SIMOTION variables

Interconnecting with SIMOTION variables

You can interconnect with global operands (SIMOTION variables) in the DCC editor.

Note

Please observe the Specifications for assigning names in DCC SIMOTION (Page 176).

Procedure

Interconnection can be performed as follows:

1. Select the block connection to be interconnected.
2. Execute the **Insert > Interconnection to operand** command in the menu. The interconnection to a global operand can also be performed via the context menu of the block. The **Symbol Input Help** window appears.

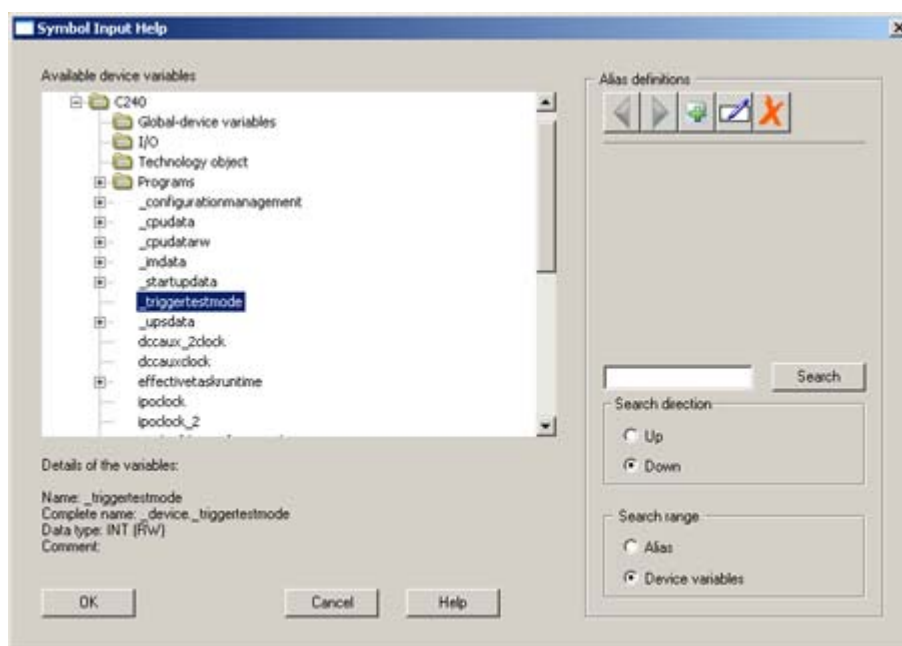


Figure 4-4 **Symbol Input Help** window

3. You can now navigate in the CPU or technology object assigned to the chart and select the device variable to be interconnected – the application highlights compatible device variables in bold-face type.
4. Click **OK** to close the window.

From this point on, the selected block connection is interconnected with the global operand and the global operand is represented by a sheet bar variable in the DCC editor.

Note

The global variables and the IO variables must first be created and then the interconnection performed.

Note

You can search for a variable name or a parameter text in the search field of the **Symbol Input Help** dialog box.

Interconnection to array elements

You can interconnect array elements of ST programs in the DCC editor.

Procedure

Interconnection can be performed as follows:

1. Create a new ST program with an array variable in the SIMOTION SCOUT engineering system using the **Insert ST program** command.
2. Right-click and select **Interconnection with operand** from the context menu. The **Symbol Input Help** window appears.

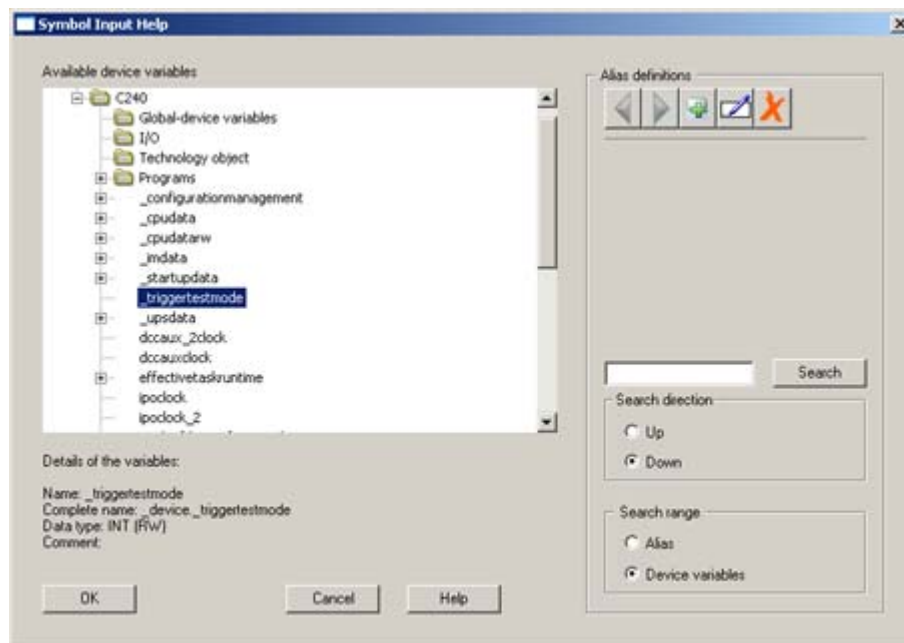


Figure 4-5 Symbol input help window

3. Select the array variable to be interconnected.
4. Enter the number of the array element to be interconnected.
5. Click **OK** to close the window.

From this point on, the selected block connection is interconnected with the array element and the array element is represented by a sheet bar variable in the DCC editor.

Alias parameter

You can assign alias identifiers in the DCC editor for variables. The reason for assigning an alias may be that a name is too long - the DCC editor only accepts a maximum of 49 characters for identifiers.

Procedure

An alias can be created as follows:

1. Right-click the block connection and select **Interconnection with operand** from the context menu. The **Symbol Input Help** window appears.
2. Select the connection that is to be defined by the alias.

3. Click the  button from the alias definition area.



4. Enter the alias name in the entry field.
5. Close the window.

The alias has been successfully created and can now be used - the computer now resolves the alias as required, e.g. within the framework of the compilation.

4.1.4 Creating block libraries

4.1.4.1 Fundamentals

The DCC editor offers a function for saving a chart as a **block library (DCB library)**. This kind of chart relates to a connection created by the user, which features an interface so that it can be reused and is saved in a library. The charts as block library (DCB library) function therefore include the option of know-how protection, since the configured DCCs are only available as a transparent block within a block library. The DCC configuration that has been created is hidden. The DCC chart created for a block can be permanently deleted via the function **Delete source of the block types...** in the context menu of the library. This command is only available if all blocks in the library have been compiled without errors. If it is executed, all the corresponding chart sources are deleted leaving only what has been compiled in the project. You cannot undo the action. By right-clicking on the DCC chart and selecting the **Block types** command, however, you can open a dialog in which the block libraries used can be exchanged.

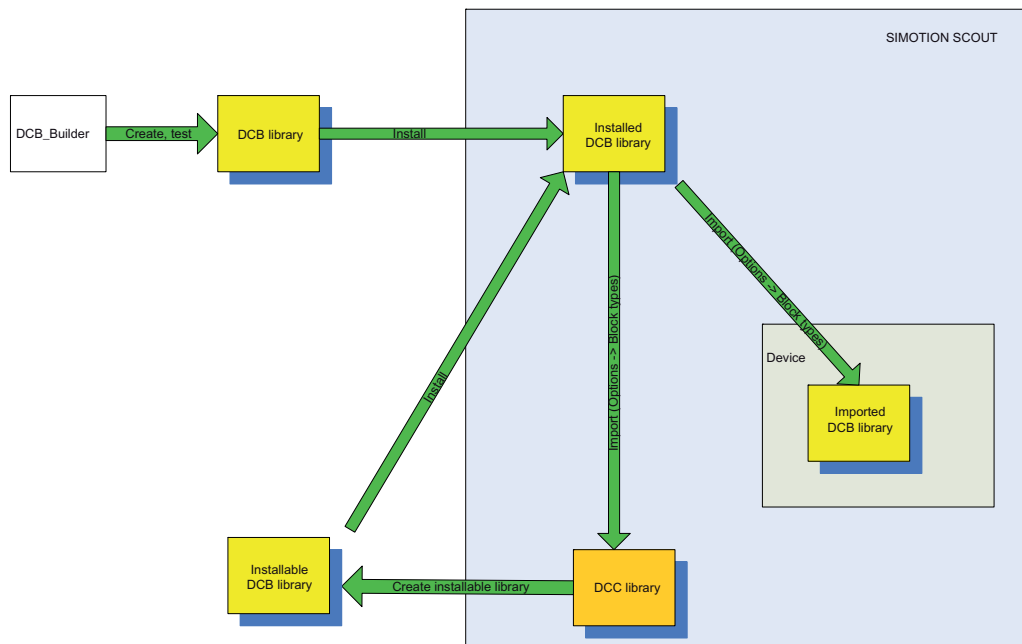


Figure 4-6 Installation of DCB libraries

Requirements

You have already created a new DCC library in the SIMOTION SCOUT engineering system.

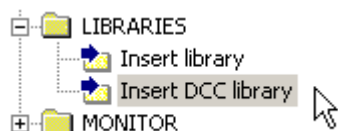


Figure 4-7 Inserting a DCC library

There is already a DCC available for saving in the DCB library.

The chart connections are displayed.

Note

You can display the chart connections via the  button in the toolbar.

Rules for creating DCB libraries

- It is only possible to compile charts in DCB libraries.
- The chart may be hierarchical (chart-in-chart).
- Interconnections between charts in the library are not permitted.
- No HMI variables or @ variables may be configured at the chart connections of the basic chart or of the blocks that it contains.
- If a chart connection input is interconnected with several (internal) block inputs, a suitable NOP block needs to be connected between them.
- All blocks may only be integrated into the same execution level / execution group. Here, the chart blocks must directly follow one another.
- Inside the chart folder that needs to be compiled, all of the blocks from a basic chart must be placed in the execution group with the same name as the chart.
- Global operands are not permitted, but must be modeled as an input or output as chart connection.
- For the DCB library and its links, please select only names starting with letters in lower case.

Procedure

Model your DCC in the DCC editor.

Insert an **ADDER** and a PI controller (**PIC**), for example, in the chart and interconnect them. See Interconnection to chart connections (Page 46)

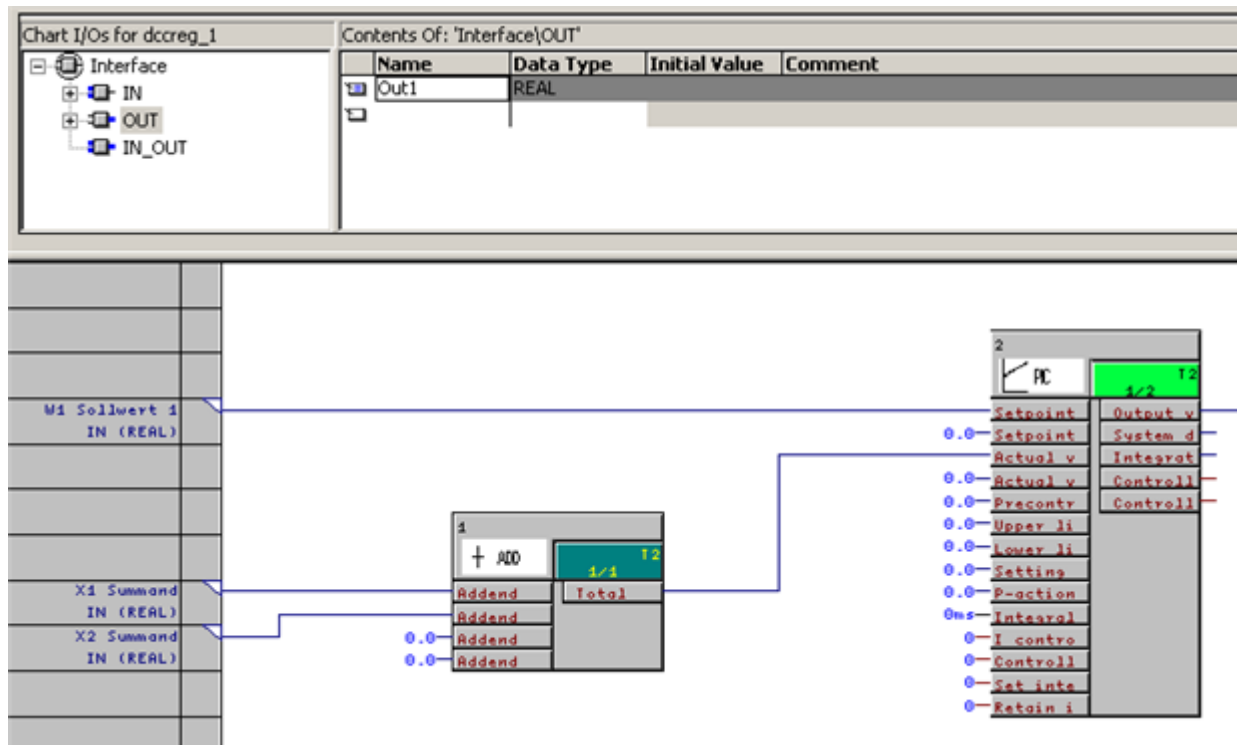


Figure 4-8 Example: Creating a chart as a program

Following successful configuration, the DCC can be compiled as a program. To do this, select **Chart > Compile > Chart as program** in the menu.

The newly created DCB library is available in the menu under **Options > Block types** in the DCC editor. The DCB library is not shown in the SIMOTION SCOUT; only native libraries are listed there.

See also

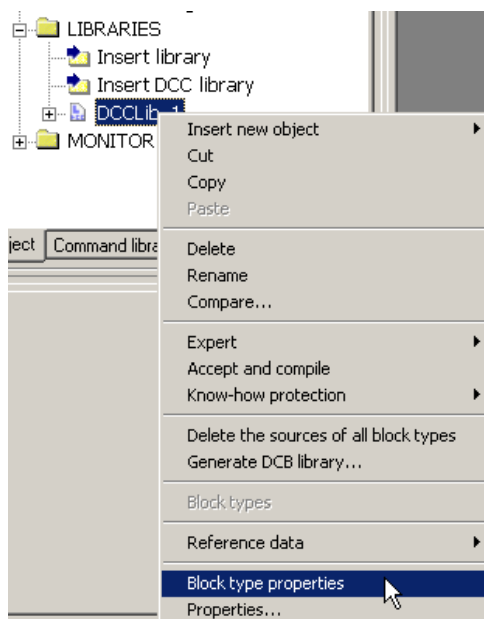
Creating an installable DCB library from DCC libraries (Page 160)

4.1.4.2 Creating comments for DCB libraries you have created yourself

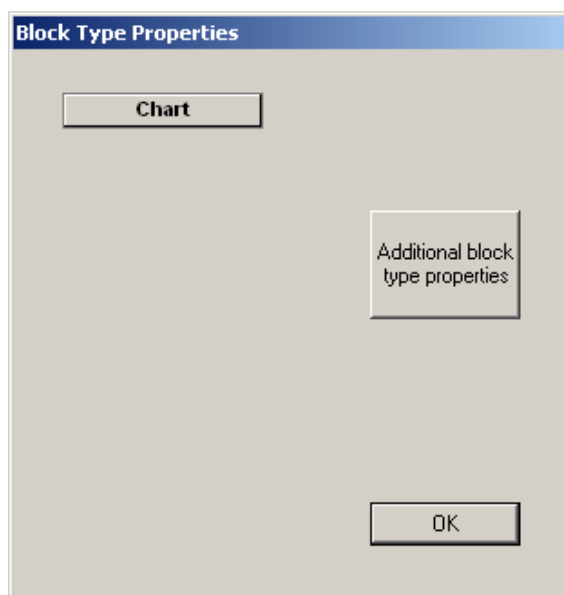
In the SIMOTION SCOUT engineering system, you can save additional information on individual blocks of a DCB library that you have created yourself.

Procedure

1. Select the block library you want to edit in the SIMOTION SCOUT engineering system.
2. In the context menu of the DCB library, select the **Block type properties** item.



3. In the **Block Type Properties** window, select the DCC to be edited and click the **Extra block type properties** button.



4. Now enter the information you require and then close the window with **OK**.

4.1.4.3 Special features of the 1:N interconnection of chart connection inputs

If you have defined an interconnection of a chart connection input with blocks for a chart, there are two different scenarios. First, there is the scenario of a 1:1 interconnection, which can be carried out without problems. However, the following must be noted for a multiple interconnection (1:N) of a chart connection input to several blocks.

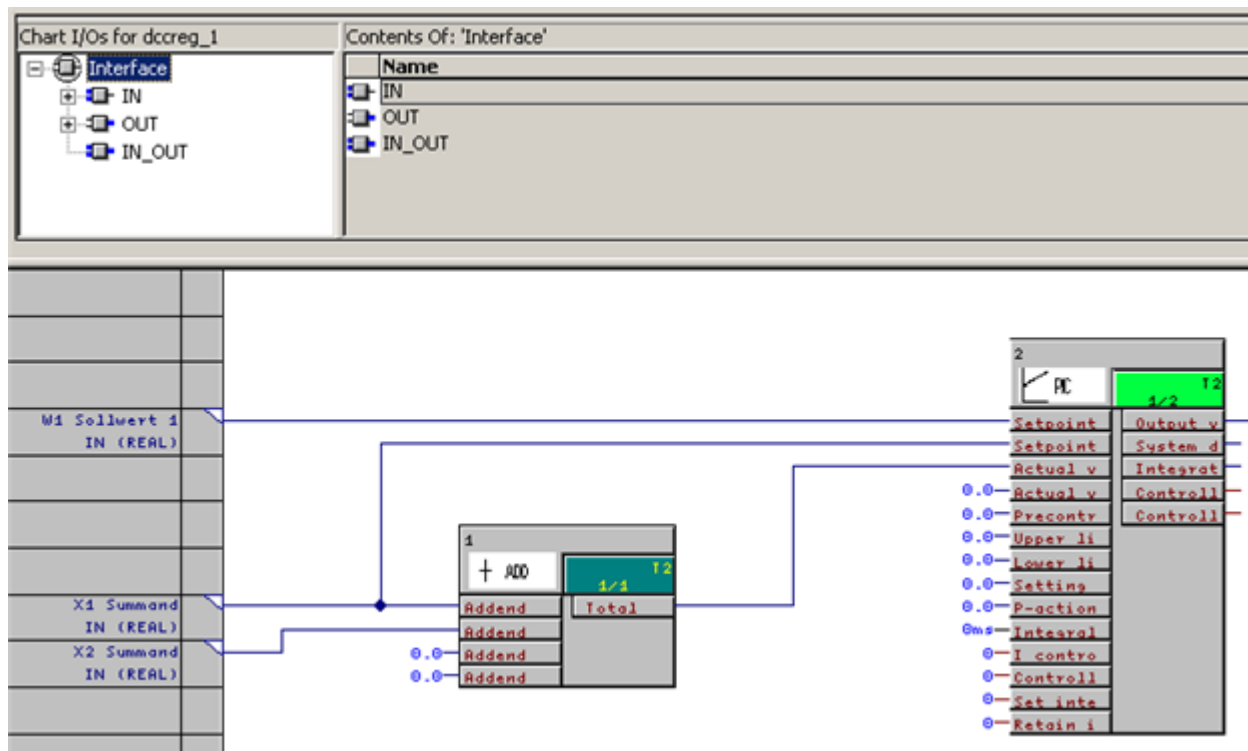


Figure 4-9 Connection example for a 1:N interconnection without NOP block

If there is a 1:N interconnection at an input within a hierarchical chart or block type, an **NOP_x** block must be used in the chart for consistency reasons - refer to the following example:

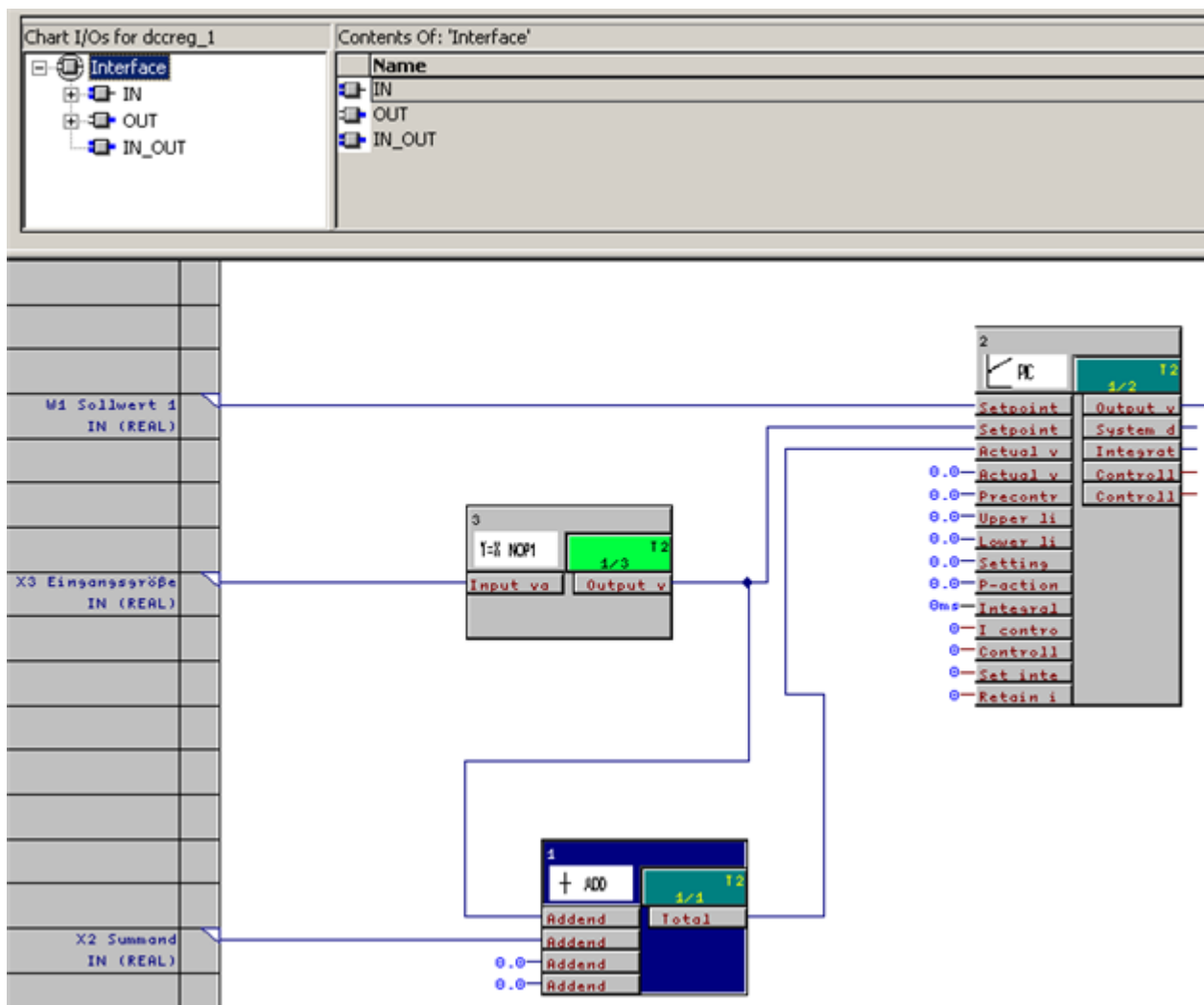


Figure 4-10 Connection example for a 1:N interconnection with NOP block

4.1.4.4 Reliance of the generated block on the underlying block libraries

A chart saved as a DCB library is created using blocks from other block libraries. Even after it has been converted, the chart contains a reference to the block libraries being used. This can lead to problems if the created DCB library is ported from the creation system to one or more target systems where the block libraries used are not available. Under these circumstances, the DCB library cannot be used.

Information on how to install DCB libraries that you have created can be found in Section Installation and uninstallation of DCB libraries (Page 164)

As of SIMOTION 4.1.2 / DCC 2.0.2, a DCB library with different versions of the basic DCC library can be used provided the interface of the called block types remains unchanged. See Creating an installable DCB library from DCC libraries (Page 160)

4.1.4.5 Creating an installable DCB library from DCC libraries

Up to SIMOTION V4.1.1, a DCC library had to be transported as project. As of SIMOTION V4.1.2, it is possible to create an installable DCB library from a DCC library in SCOUT.

To do this, select the entry **Create DCB library** in the context menu of the library; the **Create DCB Library** dialog box appears.

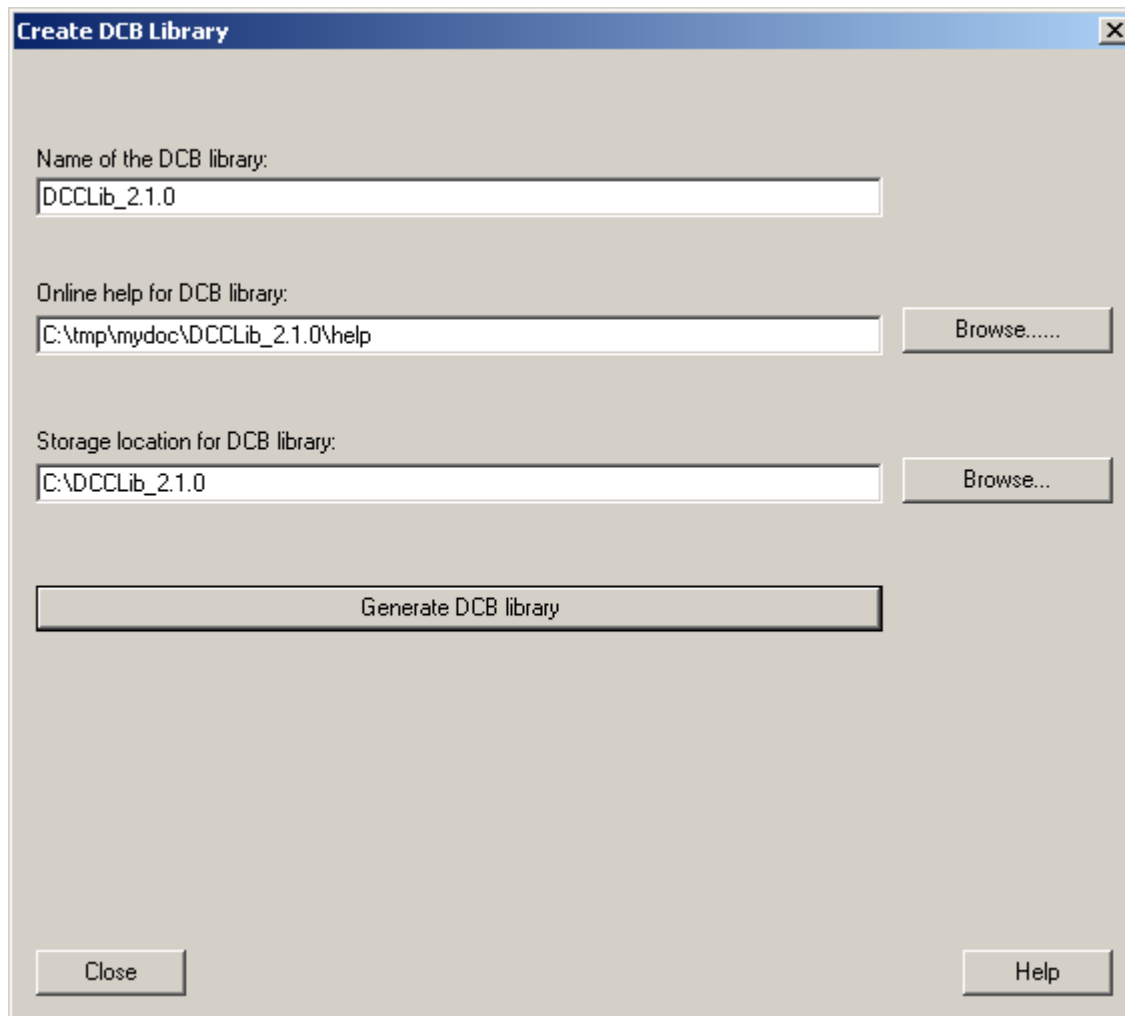


Figure 4-11 Create DCB library

In the dialog box, specify the online help directory for the library.

The name of the DCB library results from the name of the selected DCC library. You can freely select the storage location of the DCB library and then start the creation via the **Create DCB library** button. An error log appears in the output window in the lower part of the dialog box.

Note

The sources of the DCCs are no longer contained in the created DCB library. The contained functions can no longer be changed or monitored (know-how protection).

Information on how to create an online help for the created DCB library can be found in Section Creating the online help for block libraries (Page 161)

Note

It is possible to create C block libraries from exported DCC libraries automatically. This results in improvements to the memory utilization in the execution performance of the DCC charts as well as increased know-how protection.

4.1.4.6 Creating the online help for block libraries

You can create a separate online help for the created libraries with the aid of a supplied editor, the **DCB help editor**. You edit your help files in the same view in which they are later displayed.

The DCB help editor can be found at ...|Siemens|Step7|S7BIN|helpeditor DCB-Help-Editor.exe.

Storage structure for online help

Save your online help files under the name **<block>_doc.xml**, whereby <block> is the name of the written block library. Create a folder structure comprising a **help** folder, containing the **help_a**, **help_b**, **help_c**, **help_d** and **help_e** folders. This results in the following assignment:

Table 4- 2 Storage structure of the online help files

Folder	Subfolder	Language of the help file <block>_doc.xml
help	help_a	German
	help_b	English
	help_c	French
	help_d	Spanish
	help_e	Italian

A **<block>_doc.xml** file can be stored for each block in the help_x folder. For example, save your German language **<block>_doc.xml** file under **help > help_a**.

Procedure

1. Open the DCB help editor with a double-click on *helpeditor DCB-Help-Editor.exe*.
2. The STARTFILE_DOC.XML file, which is filled with elements and dummies, opens automatically.

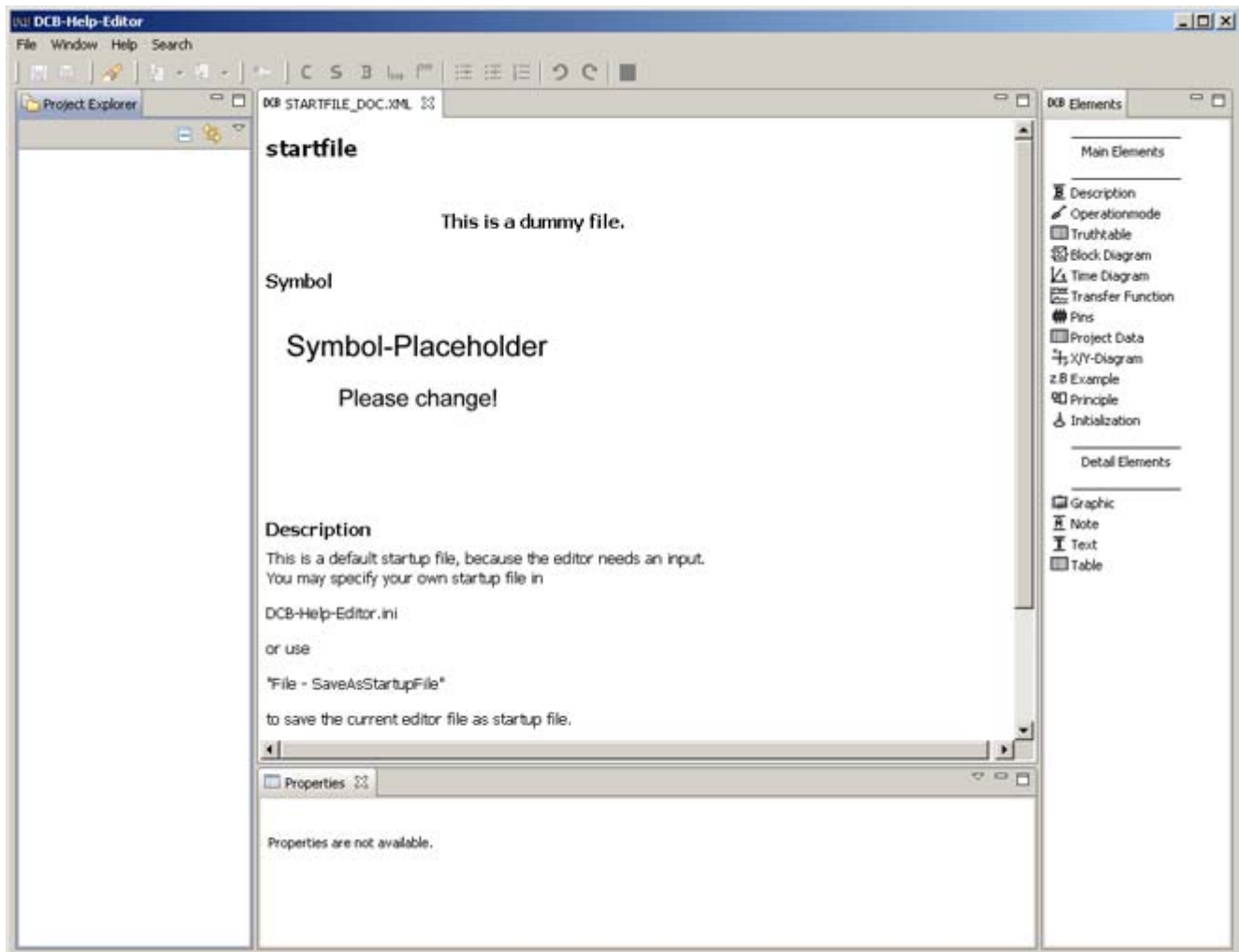


Figure 4-12 DCB help editor with STARTFILE

3. Replace these dummies with your own descriptions by selecting the dummies and entering or copying your text in the file.

4. Further elements for the description (e.g. *Truth table*, *Block diagram*, *Example*, etc), are available in the right-hand **Elements** pane, which you can drag to the desired position on your file using drag-and-drop and edit there.

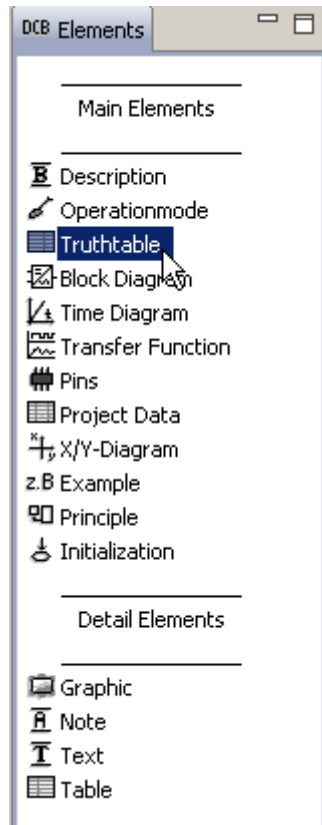


Figure 4-13 Elements for the description

5. Insert your own graphics by dragging the *Graphic* element to a suitable position in the file using drag-and-drop. In the following dialog box, enter the path under which the graphic is stored (use the **Browse** button).

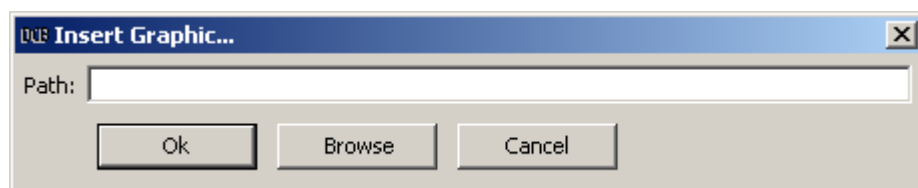


Figure 4-14 Inserting a graphic

6. Save the edited file with the  button or via the **File > Save as...** menu in the storage structure described above.

Creating further online help files

You can open existing files via the **File > Open** menu.

You can create a new block description via **File > New**.

Note

Detailed operating instructions for the **DCB help editor** can be found in the menu under **Help > Help contents**.

4.1.4.7 Installation and uninstallation of DCB libraries

From DCC Version 2.0.2 onward, there is the option of installing or uninstalling DCB libraries from SIMOTION SCOUT via a dialog box, in this case several libraries can be selected in parallel to the installation. You start the dialog box via the menu command **Options > Installation of libraries**.

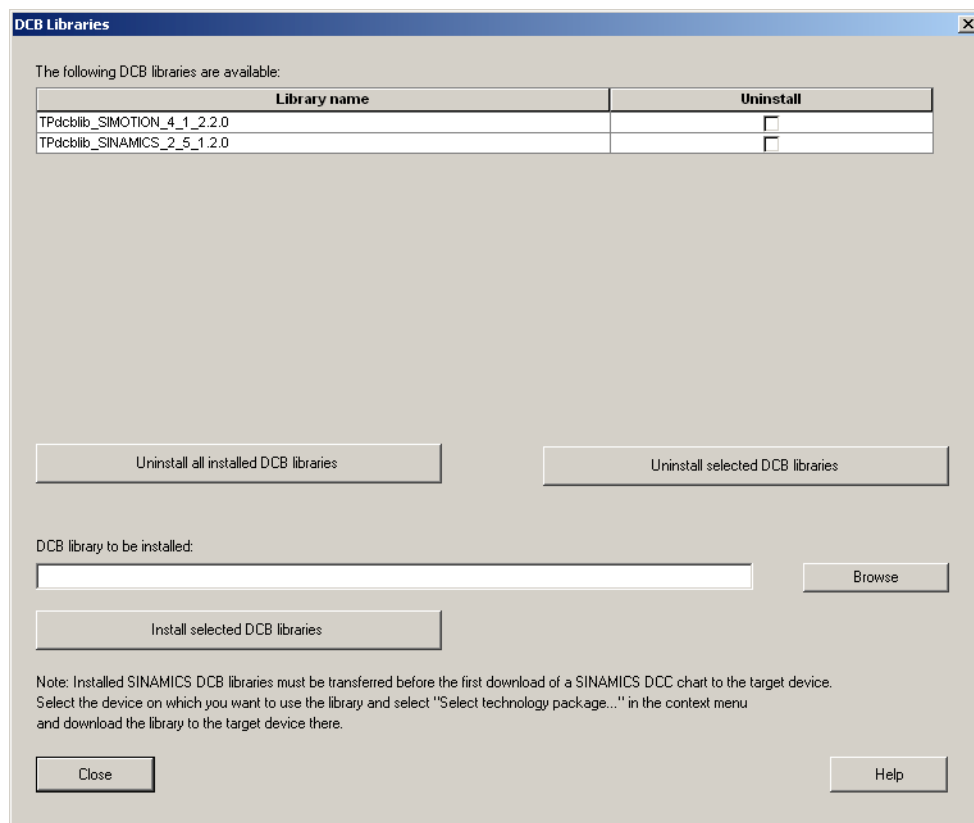


Figure 4-15 Installation and uninstallation of DCB libraries

Uninstallation of DCB libraries

A list of already installed libraries can be found in the selection list under **The following DCB libraries are available**.

- Activate the checkboxes behind the libraries to be deleted and select **Uninstall selected DCB libraries**.
- If you want to uninstall all available libraries, select **Uninstall all installed DCB libraries**. All libraries contained in the list are uninstalled.

Installation of DCB libraries

You can specify the path, under which the library that you want to install is stored, at **Install DCB library**: Enter the path directly or use the **Browse** button. Start the installation of the selected library by clicking **Install selected DCB libraries**.

Exit the **DCB Libraries** dialog box by clicking the **Close** button.

Note

The libraries are installed project-independently in SIMOTION SCOUT. The libraries are not transported when projects are archived or exported. If the project needs to be dearchived or exported on a different SCOUT, the libraries used there need to be reinstalled. After a SCOUT reinstallation, the libraries must also be reinstalled.

4.1.5 DCC and SIMOTION trace

As accustomed with SIMOTION-based devices, you can trace each block connection as a variable or use it in the trigger condition in SCOUT.

4.2 Working with DCC SIMOTION

4.2.1 Preliminary remarks on configuration

Preliminary remarks

The following is a brief explanation of what you will be configuring in this chart.

Configuration example

The configuration example deals with a straightforward oscillating circuit that creates sinusoidal oscillation at its output.

It will only take you a few minutes to create the chart; then you can execute it in test mode as a demonstration.

The following blocks are used:

- Two integrators (**INT**)
- One inverter (**SII**)

As indicated by the differential equation $\ddot{f}(x) = -f(x)$, the oscillating circuit is comprised of two integrator blocks that are linked by negation.

The frequency of the oscillating circuit is determined by the integrating time constant at the integrators.

The oscillation amplitude is specified by the initial value at the integrator output.

The DCC editor can be started via the SIMOTION SCOUT engineering system.

Configuration example structure

The configuration example is divided into the following steps:

1. Creating a new project
2. Inserting the DCC in the project
3. Inserting blocks in the DCC
4. Interconnecting blocks in the DCC
5. Compiling the DCC in the DCC editor
6. Downloading the compiled DCC program to a CPU
7. Starting the CPU
8. Switching between process and laboratory mode
9. Setting test mode and the test mode types
10. Monitoring values in laboratory and process modes
11. Logging on and logging off connections for monitoring
12. Using the "Enable" attribute for execution group control
13. Creating complete documentation for the example
14. Information on the chart reference data function

4.2.2 Creating a project

- Create a new project in the SIMOTION SCOUT engineering system, e.g. **dcc_ex**, see Creating a project (Page 20).
- Create a new device, e.g. a CPU C230-2 V4.1, using **Create new device**.
- Select the package **TPdcplib_SIMOTION_4_1.2.0 [x.y]** and close the window.
- You can now insert a chart.

4.2.3 Inserting a chart

- In the project navigator, double-click **Insert DCC** under **Programs**.
- Now give your chart another new name. The text field for the DCC name is selected automatically and the cursor is activated.
- Edit the new name here: **dcc_reg1**.
- If the **Open editor automatically** option is activated, the DCC will be started automatically. Alternatively, you can open the DCC by double-clicking **dcc_reg1**.
- If a library has not yet been imported, at this point you will be prompted to import one: Select **TPdcplib_SIMOTION_4_1.2.0 [x.y]** in the left-hand window, then click **>>** and finally **Close**.

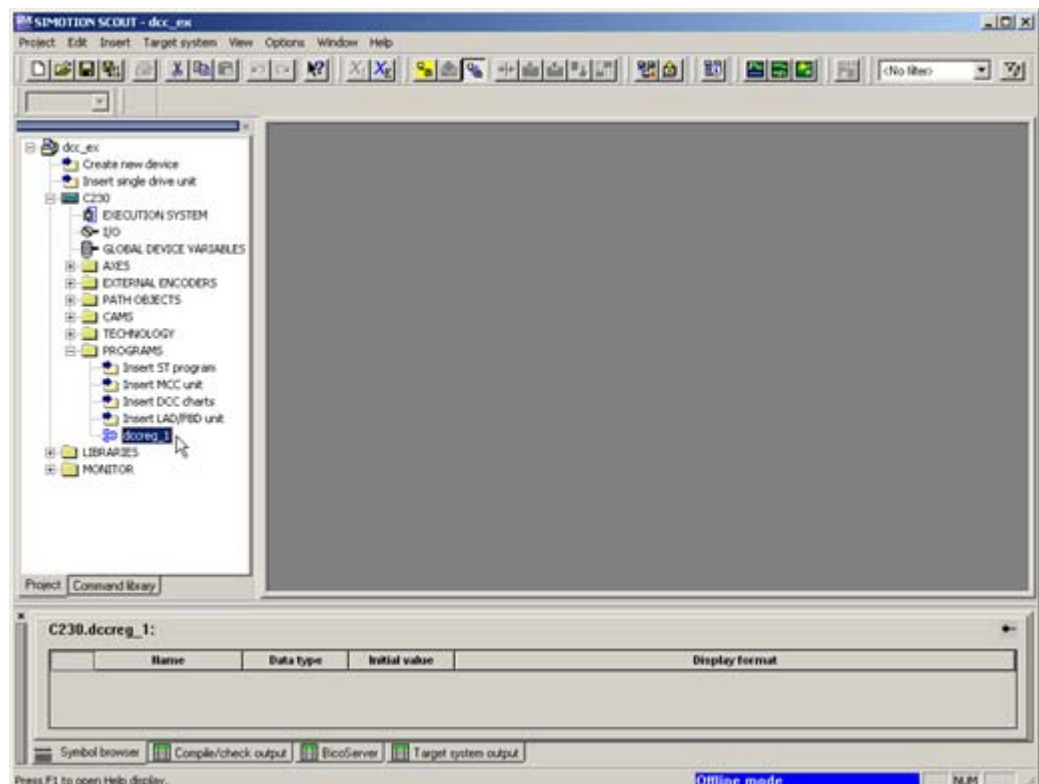


Figure 4-16 Inserting a chart

This creates the project structure and a chart. All that remains is to create some activity within the chart, i.e. by inserting blocks and interconnecting them.

4.2.4 Inserting blocks

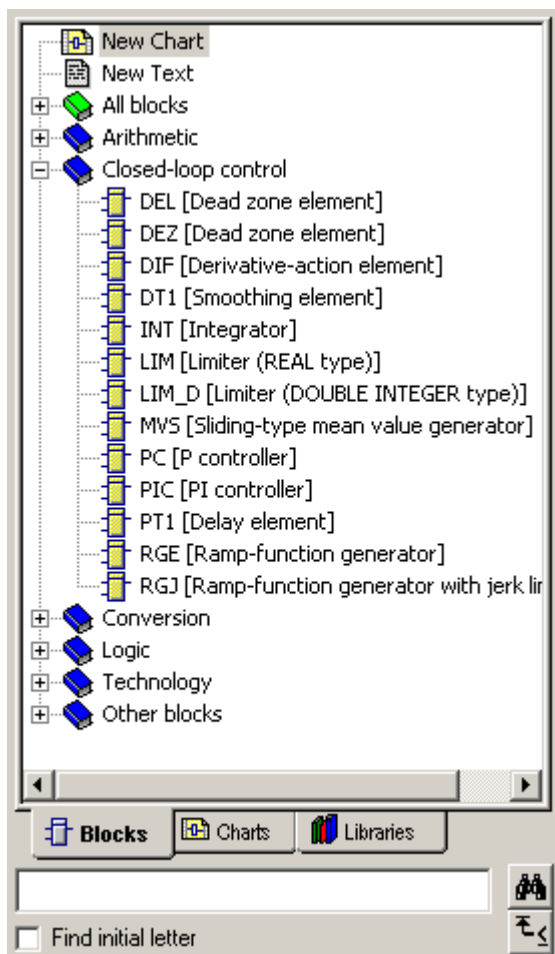


Figure 4-17 Inserting a block

- Open a block family in the **Closed-loop control** family with the closed-loop control blocks.
- Select the required block and insert it in the chart using drag-and-drop. Only the outline of the block in dashed lines is displayed during the copying procedure. Release the mouse button at the required point.
- To search for a block, enter its name in the input field of the block catalog and click the **binoculars** button. The search process begins. Once the block has been found, insert it in an empty space on the chart using drag-and-drop.

Note

If blocks are superimposed on the chart with other elements, such as other blocks or the sheet bar, the superimposed block will be displayed in gray and its connections will not be visible. You must reposition the blocks to ensure that all block information can be viewed.

Changing to the page view

To change to the page view from the overview representation, right-click an empty space in the chart and select **Display this page** in the context menu that appears. The names of the block connections are displayed in this enlarged view.

You can also switch to the page view and back to the overview again by double-clicking an empty area on a page.

Configuring the block display

You can change the display of the blocks. You can change the block width using **Options > Settings > Blocks / sheet bars width....** You can change the designation of the inputs and outputs of the blocks using **Options > Settings > Display** in the **Connections** submenu.

The block type can be displayed in the form of both text (name) and graphics (control symbol). This can be configured via **Options > Settings > Display** in the submenu **Block headers**.

4.2.5 Interconnecting blocks

Procedure

- Select the **Y** output of the first **INTegrator**, followed by the **X** input of the second **INTegrator**.
- Select the **Y** output of the second **INTegrator**, followed by the **X** input of the inverter (**SII**).
- Select the **Y** output of the inverter (**SII**), followed by the **X** input of the first **INTegrator**.

The autorouter creates the connecting lines from the outputs to the inputs and they are then interconnected.

4.2.6 Parameterizing block connections in the chart

For the first integrator INT 1/1, the following initial values should be assigned to the connections: LL = -10.0, LU = 10.0, SV = 2.0, Ti = 100 ms

For the second integrator INT 1/2, the following initial values should be assigned to the connections: LL = -10.0, LU = 10.0, Ti = 100 ms

To do this, open the **Properties - Connection** window of the associated block connection with a double-click. Enter the initial value for **Value** and click **OK** to close the window. Note that for the input of the values for Ti, the "ms" unit must follow the numeric value 100 without any spaces.

As the connections mentioned above are not interconnected, the entered values also remain valid during the cyclic processing of the chart.

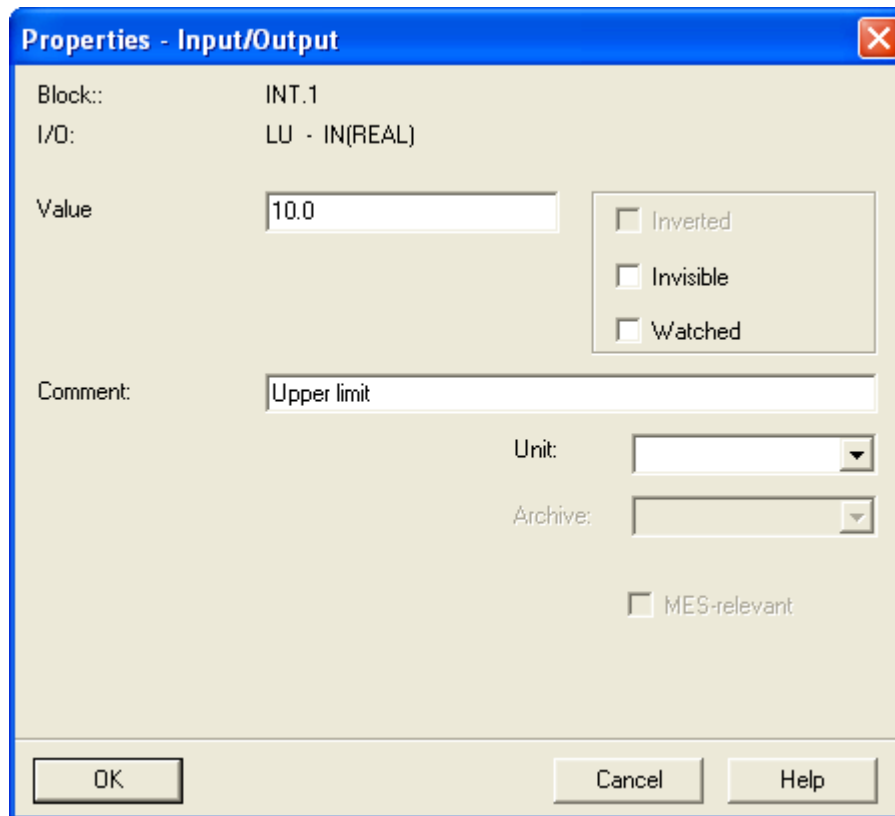


Figure 4-18 Properties window of the "LU" block connection

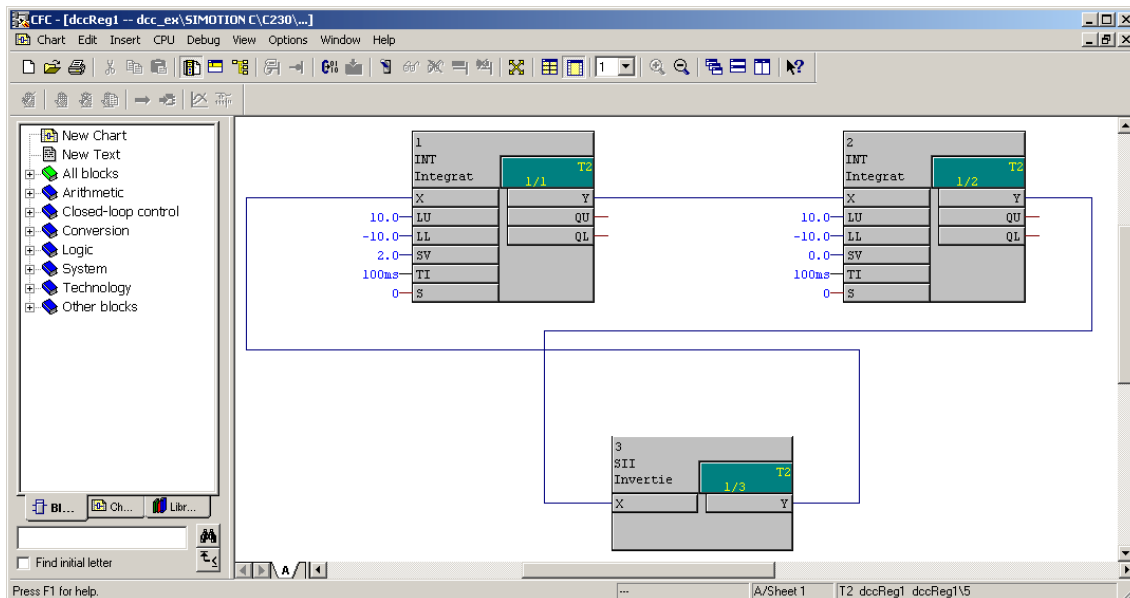


Figure 4-19 Chart "dccReg1" with interconnected blocks. Some connections have been assigned values that differ from the default values.

4.2.7 Default connection values for delta downloads

Changes made to the default values of the block connections at a later date are only transferred to SIMOTION if the option highlighted in the following image is activated:

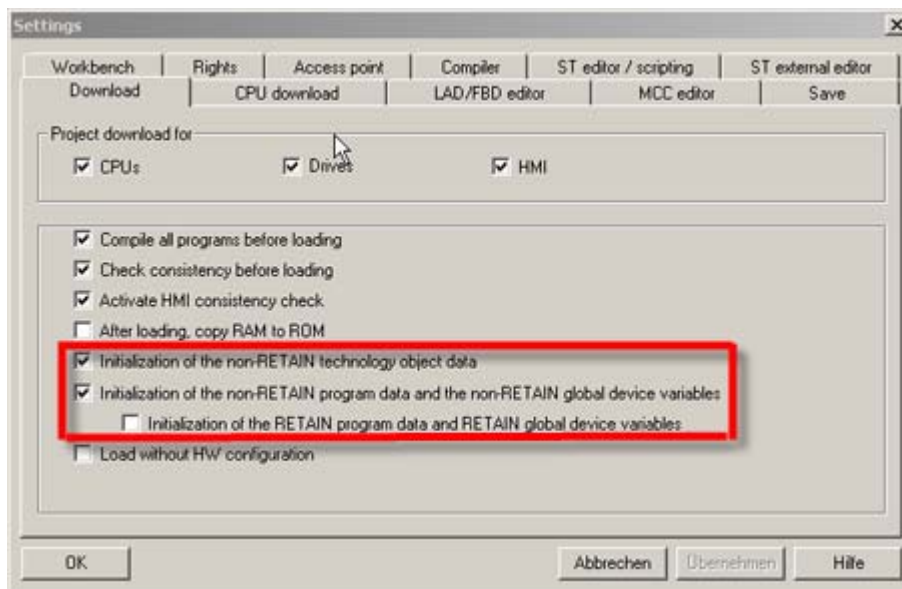


Figure 4-20 Settings for SIMOTION data transfer

4.2.8 Compiling the DCC in the DCC editor

Compiling

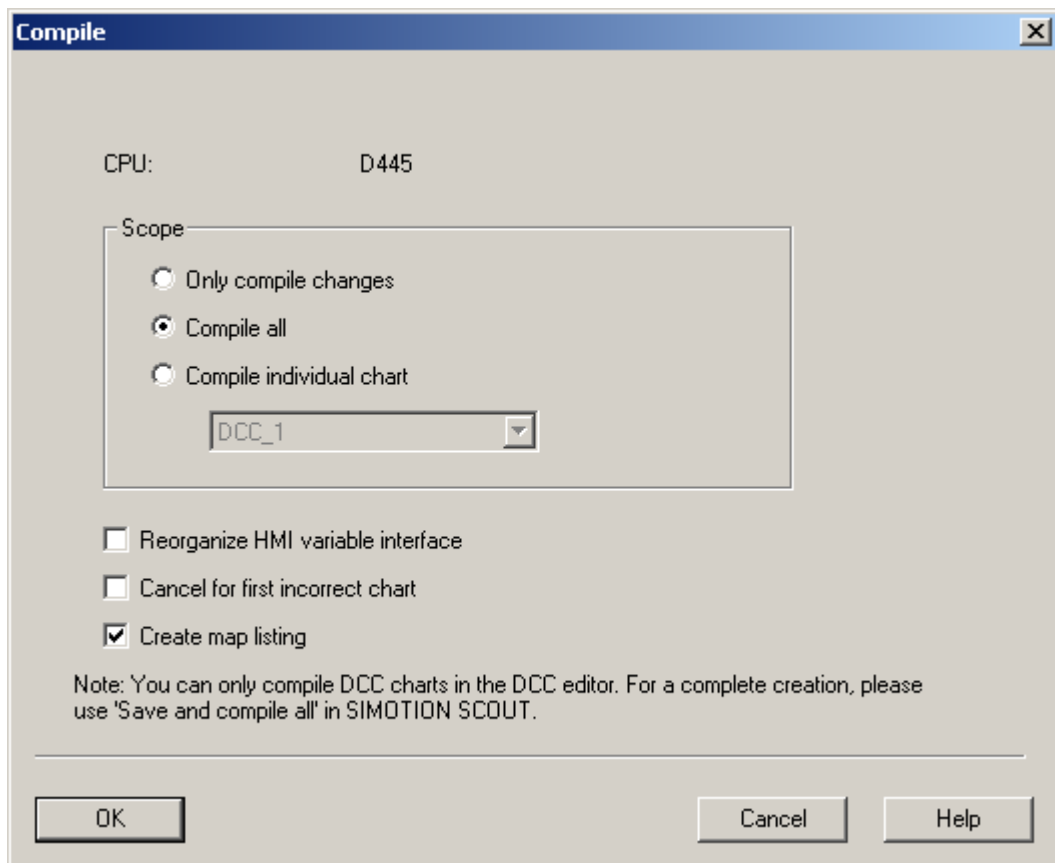


Figure 4-21 Compiling a dialog in the DCC editor

You can begin compiling with **Chart > Compile > Charts as program...** or via the  button. If errors occur during compilation, the **Logs** dialog box will automatically be displayed at the end of the procedure (just as in the case of the consistency check).

Note

You can check whether all the required block libraries and technology packages have been activated via the menu item **Options -> Block types**.

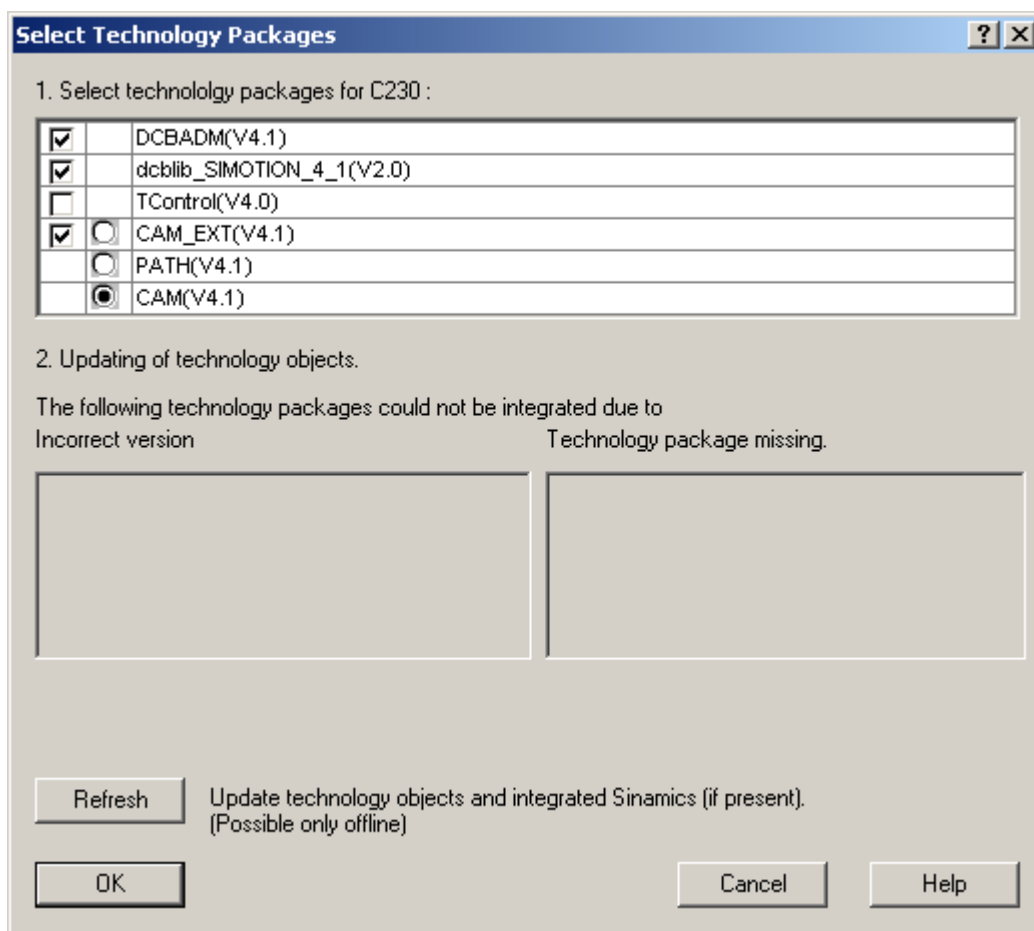


Figure 4-22 Window - Select technology packages

Compilation options

For detailed information about the compilation options, refer to Compiling (Page 48).

Note

The **Reorganize HMI variable interface** option enables you to reassign the addresses for all @ variables currently defined in the chart. @ variables that are still available in the interface, but are no longer used in the charts, are deleted.

After compilation

On completion of the compilation procedure, a detailed compilation log appears.

- To access the relevant block that is causing an error, select the error line in the log and click **Go to**.
- The log can be called again at a later point via the menu **Options > Logs** and, if necessary, can also be printed out.

4.2.9 Loading the compiled DCC

To be able to operate the DCC program on a CPU, it must have first been downloaded to the CPU.

Downloading can only take place in SIMOTION SCOUT.

Note

The DCC editor assigns the configured DCBs to tasks T1 to T5 in the execution system.

4.2.10 Starting the CPU

To execute the DCC program, the CPU must be switched to RUN.

4.2.11 Enable attribute, execution groups

The "Enable" attribute enables or disables an execution group (enable = 1; disable = 0).

The "Enable" attribute is set to 1 by default. However, it can also be set dynamically. The output value of a block determines whether the group is to be enabled or disabled. To this end, you can interconnect the digital output of a block with the execution group.

4.2.12 Creating documentation

Complete documentation

The example has been successfully configured and tested. You can now create the complete documentation for your example.

Chart reference data

In this case, the chart reference data is the **block type cross references** and the **execution sequence**. In the toolbar, click: 

You can open the **Chart Ref: Display Chart Reference Data** window via **Options > Reference data** in the menu. In this window, you can use the buttons in the toolbar to create and display the relevant lists. To begin printing, click: 

Then close the dialog box with **Reference data > Exit**.

Note

The **Chart Ref: Display Chart Reference Data** window can only be opened when the window of the execution editor is not open.

Printing a chart

You also want to print the chart. Since only one page has been used on this chart, a single page display is recommended for printing. The chart is now set to **Page view**. To begin printing, click: 

If your chart consists of several pages, we recommend that you print them individually in the page view. Click the  button to change to the page view.

Click  or  to customize the display so that all DCBs are shown on a single page on the PC monitor.

4.3 DCC SIMOTION specifications

4.3.1 Rules for assigning names in the DCC editor

Variables are used for data exchange between SCOUT and the DCC editor. The variable names used are subject to the same restrictions regarding names in SIMOTION.

Note

In the DCC editor, the length of the name is restricted to **a maximum of 49 characters**. In SIMOTION, however, longer names are supported. Where this arises, it is necessary to either use aliases or interpose ST variables.

The names in the DCC editor must therefore abide by the following rules:

- Basic chart:
 - No keyword or previously defined name permitted
 - Must start with a letter or underscore
 - May contain numbers, letters and "_"
 - "_" must be followed by a number
- Subchart:
 - Must start with a letter or underscore
 - May contain numbers and letters
 - "_" is not permitted
 - Keyword or previously defined name permitted (during compilation, a check is performed to determine whether this is unique)
 - During compilation in the DCC editor, the name is checked to determine whether it is unique
- Block instance:
 - Can start with a number
 - May contain numbers and letters
 - "_" is not permitted at the beginning
 - Keyword or previously defined name permitted
 - In the DCC editor, a check is performed to determine whether the name has previously been defined
- Execution group:
 - Keyword or previously defined name permitted
 - In the DCC editor, a check is performed to determine whether the name has previously been defined
- Chart connection:
 - No keyword or previously defined name permitted
 - In the DCC editor, a check is performed to determine whether the name has previously been defined

4.3.2 Field/name lengths and conventions

Field/name lengths and conventions

Object	Length	Remark
Chart name	22 *)	May not contain the following characters: \\ . : / * ? " < > # % () Use of the "_" character is subject to particular specifications.
Chart comment	255	All ANSI characters are permitted.
Execution group	22	Character set as for Chart , including "_"
Block type	6	Determined by DCC.
Block instance comment	80	All ANSI characters are permitted.
Parameter comment	80	All ANSI characters are permitted.
Block name (instance name)	16 *)	May not contain the following characters: \\ . : / * ? " < > # % Use of the "_" character is subject to particular specifications.
Global operand	Max. 49	If names from SIMOTION are too long, an ALIAS must be defined.

*) The chart name and block name must jointly consist of a maximum of 24 characters, including separating characters.

See also

Rules for assigning names in the DCC editor (Page 176)

4.3.3 Representation of the dynamic value display

The values are output next to the connections, according to their data type. They are displayed on the screen with a colored background.

Table 4- 3 Representation of the dynamic value display

Representation	Meaning
Blue on white	Representation of the values in edit mode (offline)
Black asterisks on yellow	Values during transfer to the dynamic display
Black value on yellow	Representation of the values read from the SIMOTION RT in test mode
#### on a red background	Representation of values while the dynamized values required from the SIMOTION RT are missing (fault, overload)

4.4 Faults and warnings

4.4.1 Notes on error message display

Notes on error message display

If an error message or warning appears in the DCC, there will be special characters that draw your attention to names or parts of names. You can double-click these characters to proceed directly to the relevant site of the error.

The characters have the following meanings:

- **Block instance or block input/block output**
 - Represented as:
"\" <instance name> ["\" <pin name>]
 - Example:
Interconnection of "_device\ST_1.i" with "_DCC_6_CFC1_1.x_1" cannot be achieved;
invalid type
- **Interconnection to chart connections**
 - Represented as:
"<" <variable name> ">"
 - Example:
The input connection DCC_5\<interconn_add_1_x_1> from the chart 'DCC_5' is not
interconnected with a sink.
- **Execution level/execution group**
 - Represented as:
"{" <group name> "}"
 - Example:
Empty execution group {EG}

Note

Error analysis when a DCB block crashes

If a DCB block crashes during operation, information about the cause and the block instance can be found in the device diagnostics buffer in SIMOTION. With this information, you can contact the creator of the block library, remove the relevant block instance or change the parameterization.

Appendix

A.1 List of abbreviations

Abbreviation	Description
(G)UI	(Grapical) User Interface
Connection	General term for block input or block output
BI parameter	Binector Input parameter. The parameter is used for the interconnection of a binector to a sink signal that can only have the states 0 or 1
BICO	Binector-Connector that designates an interconnectable parameter in the drive
BO parameter	Binector parameter (also Binector Output parameter). The parameter can be used as a binary signal source (0 or 1)
CFC	Drive Control Chart
CI parameter	Connector Input parameter. The parameter is used for the interconnection of a connector to a sink signal
CO parameter	Connector parameter (also Connector Output parameter). The parameter can be used as a signal source
CSV	Comma Separated Value, text format for column-oriented data
DCB	Drive Control Block
DCC	Drive Control Chart
DO	Drive Object
ELF file	File coded in Executable Linkable Format
FEAT	Feature document
ITCP	Instance, Time slice, Connection and Parameter/Variable
MBCS	Multi-Byte Character Set, corresponds to UTF-8
MDI	Multiple-Document Interface – an application with several windows
OEM	Original Equipment Manufacturer
TP	Technology Package

A.2 Glossary

Execution level	Interface to the execution system used for the execution of blocks.
Execution group	Execution groups are used for structuring or subdividing execution levels. The blocks and/or charts are integrated sequentially into the execution groups. SIMOTION only: Execution groups, for example, can be switched on and off separately through a connection to a block output (data type BOOL). When an execution group is switched off, all blocks/charts contained therein are no longer calculated.
Execution sequence	Sequence in which the blocks are to be calculated in an execution level or execution group.
User library	A user library which contains block libraries. As the blocks are based on charts, they can be handled individually in this case.
Output connections	The output connections can be interconnected to other inputs or assigned an initialization value. The value is then active at this connection when the block is calculated in the INIT mode for the first time. This is useful, for example, when a specific value is to be assigned to the output connection of a flip-flop block or a controller block.
HMI variables	You can declare block inputs and outputs as HMI variables and therefore generate a static interface for these for use in your system visualization.
Basic library	A library that has been created in C, e.g. with the aid of a DCB tool. Such a library is a closed unit, i.e. a DCB cannot, for example, be separated from a library in order to transfer it to another block. The supplied DCBLIB is an example.
Basic chart	The chart that is visible and can be managed in STEP 7 or SCOUT/STARTER. All other charts, i.e. chart partitions or subcharts can only be managed in the DCC editor. Only the term chart is used in the following.
Block help	Detailed information about a certain block type can be obtained by selecting the block and pressing the F1 key.
Block type library	Consists of the description of the block types contained in the library and the object file that implements the blocks, e.g. ELF.
INIT mode	Initialization, ramp-up
RUN mode	Cyclic operation
DCB library	A user-defined DCB is created from a chart.
DCB builder	Development environment for the programming of C block libraries.
DCB viewer	You can read the descriptions of the individual block types in the DCB viewer during the configuration in the DCC editor. The DCB viewer is opened by selecting a block and pressing the F1 key.

Input connections	The input connections can be parameterized with constants by the project engineer or interconnected to other block outputs. When the DCB blocks are called, the inputs and outputs have already been assigned default values that can be changed when required.
Creation mode	DCCs can be edited in creation mode.
Freely defined parameters	@ parameters and @ technology connectors. Interface defined for the parameterization by the user in the form of parameter numbers that can be defined as ALIAS for block inputs and block outputs.
Global operands	Global operands are connection partners for block inputs/outputs with which DCB blocks can read and write information from the system environment.
Hierarchical chart or also subchart	A chart (subchart) can be inserted in another chart (chart-in-chart-technique). Hierarchical structures can be formed here. Each inserted chart can be opened and, like every other chart, edited further and therefore changed individually. A chart can be encapsulated for further use, i.e. chart connections added. Which block connections are provided at the chart connections can also be specified individually.
Comments	Each block connection of the CFC page can have a comment added.
Connector	If no more lines can be drawn because the page is too full, the CFC inserts a connector at the block/chart connection and a number in the sheet bar. The corresponding connectors are assigned the same reference number. If several interconnections that cannot be displayed all start at one output, they are all assigned the same reference number. You can recognize the connection point location through the varying representation of the connector. For details, refer to the CFC help.
Laboratory mode	The blocks are automatically monitored in laboratory mode (display of the current values of the block/chart connections logged on for testing).
Offline	Project is processed without a connection to the device.
Online	Project is processed with a connection to the device.
Online RUN, or pulse enable at the drive	Project is processed with a connection to the device and the device is in open-loop or closed-loop control.
Parameterization	Instead of an interconnection, a constant different than the default can be parameterized at each input or output.
Process mode	The blocks are not automatically monitored in process mode (display of the current values of the block/chart connections logged on for testing) so that the additional load is very small.
Sheet bars	The sheet bars on the left and right sides of a CFC page contain the references to the interconnected objects, e.g. other blocks or execution groups, that are not on the current page. They also contain the number of the connector (breakpoint) when the connection line to the sheet bar cannot be drawn because the page is too full.

Technology package	A technology package (the DCBLIB) contains technology objects (DCB) that are capable of being instantiated.
Chart partition	Each chart comprises up to 26 chart partitions ("A".."Z"). Each chart partition consists of up to six DIN A4 pages. A newly created chart only contains chart partition "A".
Test mode	Test mode can be used to debug the DCCs.
Text interconnections	Are used to split projects into separate, configurable units and define an "open" interconnection between charts. A text interconnection can only be at a block/chart input and always references a block or chart output in CFC. The text interconnection is an "open" interconnection until it becomes a "real" interconnection upon closing. A closable text interconnection is the addressing of an input by means of a character string, which identifies a specific interconnection source (output) via the path specification (chart/block.connection).
Trace	The signals of the block outputs can be recorded with the trace function.
Trend display	In test mode with SIMOTION, you can use the value and trend display to analyze the input and output values of blocks.
Overflow pages	Overflow pages are created automatically when more sheet bar entries are generated than can be displayed on a page. An overflow page consists solely of the sheet bars and contains no further objects.
Interconnecting	The term interconnecting means: The connection of a block output to another DCB block input on the same device Interconnections of a block output to an execution group. SIMOTION only Interconnection of a block output to a global operand or a global operand to a DCB block input. A global operand can be the following: The name of a SIMOTION variable. The name of a SINAMICS parameter (can also be performed implicitly through the interconnection to a SB).
Workbench	The workbench is the navigation center for the individual engineering steps and is used for the centralized creation and management of projects. It provides a uniform, function-oriented, consistent, filterable view of all data and programs, even in distributed systems.

Index

"

"Enable" attribute, 174
Execution group, 174

@

@ parameters, 149

A

Abbreviations of the data types, 45
Alias identifier, 153
Array and name lengths, 177
SIMOTION, 177

B

Basic chart, 26
Block
Block connection units, 42
Configuring the display, 43
Delete, 47
Deleting online, 67
Execution properties, 40
Execution sequence, 40
Hidden connections, 42
Inserting online, 67
Interconnecting, 44
Pseudo comment, 42
Superimposition, 37
Block catalog
Binoculars, 37
Search, 37
Block library, 154
Changing the language, 32
Delete, 34
Importing, 29
Naming convention, 29
Updating, 30
Block type
Plant view, 36
Timing diagram, 36

C

Chart as block type, 154
Dependencies, 159
Multiple interconnection, 158
Chart partition, 26
Chart reference data
Block type cross references, 70
Execution group cross references, 70
Operand cross references, 70
Compiling, 49
Configuration example
SIMOTION, 166
Configuring the block display, 43
Connection
Publishing, 99
Creating a new device, 21
Creating a new project, 20

D

Data types
Abbreviations, 45
DCB library
Change, 52
DCC
Compiling, 49
Consistency Check, 48
Copy, 53
Copying block groups, 53
Creating an interconnection online, 66
Deleting a block online, 67
Deleting an interconnection online, 65
Insert block, 36
Inserting a block online, 67
Map listing, 50
Moving an interconnection online, 66
Reference data, 70
DCC chart
Exporting to WinCC, 150
DCC editor
Page view, 28
Rules for assigning names (SIMOTION), 176
Sheet bar, 47
Software requirements, 145
DCC Editor, 25

- Importing DCB library, 25
- DCC SINAMICS
 - Field of application, characteristics, 82
- DCC task, 146, 151

E

- Error log, 48
- Example of a SIMOTION configuration
 - Chart reference data, 174
 - Compiling a DCC, 172
 - Configuring the block display, 169
 - Creating a project, 167
 - Creating the complete documentation, 174
 - Data transfer in SIMOTION, 171
 - Laboratory mode, 58
 - Monitoring in test mode, 59
 - Page view, 169
 - Printing a chart, 175
 - Process mode, 58
 - Reorganising B&B variable interfaces, 173
 - Selecting Technology Packages, 172
- Execution group, 147
 - "Enable" attribute, 174
- Execution sequence, 147
 - change, 148
- Execution system
 - Change execution sequence, 148
 - Execution group, 147
 - Execution sequence, 147
 - Task, 146

G

- Global operand
 - Interconnection, 151
- Global operands, 47
 - Sheet bar, 47

H

- HMI variable, 149

I

- Importing DCB library, 25
- Inserting a DCC, 23
- Inserting a single drive unit, 21
- Interconnection
 - Array elements, 152
 - Creating online, 66

- Deleting online, 65
- Global operand, 151
- move online, 66
- ST program, 152
- Interconnection to array elements, 152
- Interconnection to global operands, 151

K

- Know-how protection, 73

M

- Map listing, 50

O

- OPC XML export, 150

P

- Parameter
 - Alias, 153
- Pseudo comment, 42
- Publishing
 - Connection, 99

R

- References, 3
- Representation of the dynamic value display, 178
- Rules for assigning names
 - Basic chart (SIMOTION), 176
 - Block instance (SIMOTION), 176
 - Chart connection (SIMOTION), 176
 - Execution group (SIMOTION), 176
 - SIMOTION, 176
 - Subchart (SIMOTION), 176

S

- Set system cycle clock, 147
- SIMOTION
 - Configuration example, 166
- SIMOTION SCOUT
 - Copy DCC, 53
 - Creating a project, 20
 - Inserting a DCC, 23
 - Inserting a device, 21
 - Know-how protection, 73

- Selecting technology packages, 31
- SIMOTION trace, 165
- Software requirements, 145
- Subchart, 26

T

- Task, 146
- Test mode, 57
 - Deactivating, 64
 - Laboratory mode, 57
 - Log on connection, 61
 - Monitoring, 61
 - Monitoring cycle, 57
 - Operating modes, 57

- Power up, 60
- Process mode, 57
- Trend display, 62
- Value display, 62
- Trace, 165
- Trend display, 62
- Typical
 - Dependencies, 159
 - Multiple interconnection, 158

V

- Value display, 62

